Working Draft

MEF W123.1 v0.1

LSO Cantata and LSO Sonata Product Order Management API - Developer Guide

This draft represents MEF work in progress and is subject to change.

December 2024

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List of Contributing Members

The following members of the MEF participated in the development of this document and have requested to be included in this list.

Member

Table 1. Contributing Members

1. Abstract

This standard is intended to assist the implementation of the Product Order functionality defined for the LSO Cantata and LSO Sonata Interface Reference Points (IRPs), for which requirements and use cases are defined in MEF 57.2 *Product Order Management Requirements and Use Cases* [MEF 57.2] and it's *Amendment* [MEF 57.2.1]. This standard consists of this document and complementary API definitions for Product Order Management and Product Order Notification.

This standard normatively incorporates the following files by reference as if they were part of this document, from the GitHub repository:

https://github.com/MEF-GIT/MEF-LSO-Sonata-SDK

- `productApi/order/productOrderManagement.api.yaml`
- `productApi/order/productOrderNotification.api.yaml`

https://github.com/MEF-GIT/MEF-LSO-Cantata-SDK

- `productApi/order/productOrderManagement.api.yaml`
- `productApi/order/productOrderNotification.api.yaml`

2. Terminology and Abbreviations

This section defines the terms used in this document. In many cases, the normative definitions of terms are found in other documents. In these cases, the third column is used to provide the reference that is controlling, in other MEF or external documents.

In addition, terms defined in the standards referenced below are included in this document by reference and are not repeated in the table below:

- MEF 55.1
- MEF 55.1.1
- MEF MEF 57.2
- MEF 79.1
- MEF 150

Term	Description	Reference
Application Program Interface	In the context of LSO, API describes one of the Management Interface Reference Points based on the requirements specified in an Interface Profile, along with a data model, the protocol that defines operations on the data and the encoding format used to encode data according to the data model. In this document, API is used synonymously with REST API.	[MEF 55.1]
Buyer	In the context of this document, denotes the organization or individual acting as the customer in a transaction over a Cantata (Customer <-> Service Provider) or Sonata (Service Provider <-> Partner) Interface.	This document; adapted from [MEF 55.1.1]
Cancellation Charge	A charge set by the Seller that results from the cancellation of a Product Order.	[MEF 57.2]
Connection Charge	A one-off charge set by the Seller to connect a Product Order Item to the Seller's network.	[MEF 57.2]
Construction Charge	A one-off charge set by the Seller resulting from special construction required to provide a Buyer requested Product Order Item.	[MEF 57.2]
Disconnect Charge	A one-off charge set by the Seller that results from a request by the Buyer to disconnect a Product.	[MEF 57.2]
Expedite Charge	A one-off charge set by the Seller resulting from a request by the Buyer to expedite the Product Order Item.	[MEF 57.2]
Telecommunication Service Priority	A US centric term used to assign a priority for restoration of a Product in the event of a natural or other disaster impacting multiple Products.	[MEF 57.2]
Requesting Entity	The business organization that is acting on behalf of one or more Buyers. In the most common case, the Requesting Entity represents only one Buyer and these terms are then synonymous.	[MEF 150]
Responding Entity	The business organization that is acting on behalf of one or more Sellers. In the most common case, the Responding	[MEF 150]

	Entity represents only one Seller and these terms are then synonymous.	
REST API	REST provides a set of architectural constraints that, when applied as a whole, emphasizes scalability of component interactions, generality of interfaces, independent deployment of components, and intermediary components to reduce interaction latency, enforce security, and encapsulate legacy systems.	[REST]
Seller	In the context of this document, denotes the organization acting as the supplier in a transaction over a Cantata (Customer <-> Service Provider) or Sonata (Service Provider <-> Partner) Interface.	This document; adapted from [MEF 55.1.1]

Table 2. Terminology

Term	Description	Reference
API	Application Program Interface	[MEF 55.1]
REST API	Representational State Transfer API	[REST]

Table 3. Abbreviations

3. Compliance Levels

The key words **"MUST"**, **"MUST NOT"**, **"REQUIRED"**, **"SHALL"**, **"SHALL NOT"**, **"SHOULD"**, **"SHOULD NOT"**, **"RECOMMENDED"**, **"NOT RECOMMENDED"**, **"MAY"**, and **"OPTIONAL"** in this document are to be interpreted as described in BCP 14 ([RFC 2119], [RFC 8174]) when, and only when, they appear in all capitals, as shown here. All key words must be in bold text.

Items that are **REQUIRED** (contain the words **MUST** or **MUST NOT**) are labeled as **[Rx]** for required. Items that are **RECOMMENDED** (contain the words **SHOULD** or **SHOULD NOT**) are labeled as **[Dx]** for desirable. Items that are **OPTIONAL** (contain the words MAY or OPTIONAL) are labeled as **[Ox]** for optional.

A paragraph preceded by **[CRa]<** specifies a conditional mandatory requirement that **MUST** be followed if the condition(s) following the "<" have been met. For example, **"[CR1]<[D38]"** indicates that Conditional Mandatory Requirement 1 must be followed if Desirable Requirement 38 has been met. A paragraph preceded by **[CDb]<** specifies a Conditional Desirable Requirement that **SHOULD** be followed if the condition(s) following the "<" have been met. A paragraph preceded by **[COc]<**specifies a Conditional Optional Requirement that **MAY** be followed if the condition(s) following the "<" have been met.

4. Introduction

This standard specification document describes the Application Programming Interface (API) for Product Order Management functionality of the LSO Cantata Interface Reference Point (IRP) and LSO Sonata IRP as defined in the *MEF 55.1 Lifecycle Service Orchestration (LSO): Reference Architecture and Framework* [MEF55.1]. The LSO Reference Architecture is shown in Figure 1 with both IRPs highlighted.

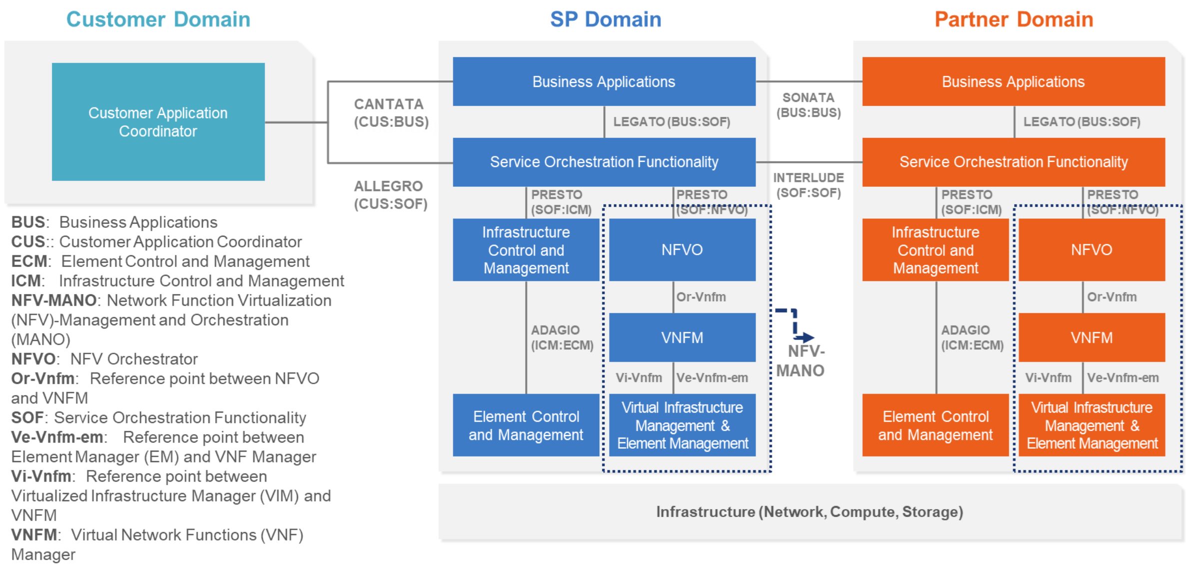

Figure 1. The LSO Reference Architecture

Cantata and Sonata IRPs define pre-ordering and ordering functionalities that allow an automated exchange of information between business applications of the Buyer (Customer or Service Provider) and Seller (Service Provider or Partner) Domains. Those are:

- Product Catalog
- Address Validation
- Site Retrieval
- Product Offering Qualification
- Product Quote
- Product Inventory
- Product Ordering
- Trouble Ticketing
- Billing

This document is structured as follows:

- Chapter 4 provides an introduction to Product Order Management and its description in a broader context of Cantata and Sonata and their corresponding SDKs.
- Chapter 5 gives an overview of endpoints, resource model, and design patterns.
- Use cases and flows are presented in Chapter 6.
- And finally, Chapter 7 complements previous sections with a detailed API description.

4.1. Description

The Product Order Management API allows the Buyer to submit a Product Order request containing one or more Product Order items. The Buyer may place a Product Order for an installation (`add`) of a new service, Change (`modify`) to an existing service, or a Disconnect (`delete`) of an existing service.

The API payloads exchanged between the Buyer and the Seller consist of product-independent and product-specific parts. The product-independent part is technically defined in this standard. The product-specific part is defined in the product specification standard of the concerned product. Both standards must be used in combination to validate the correctness of the payloads.

Section 5.4 explains how to use product specifications as the Product Order API payloads.

This document uses samples of Access E-Line Product specification definitions to construct API payload examples in Section 6.

4.2. Conventions in the Document

- Code samples are formatted using code blocks. When notation `<< some text >>` is used in the payload sample it indicates that a comment is provided instead of an example value and it might not comply with the OpenAPI definition.
- Model definitions are formatted as in-line code (e.g. `GeographicAddress`).
- In UML diagrams the default cardinality of associations is `0..1`. Other cardinality markers are compliant with the UML standard.
- In the API details tables and UML diagrams required attributes are marked with a `*` next to their names.
- In UML sequence diagrams `{{variable}}` notation is used to indicate a variable to be substituted with a correct value.

4.3. Relation to Other Documents

The requirements and use cases for Product Order Management are defined in MEF 57.2 [MEF 57.2] and it's *Amendment* [MEF 57.2.1]. Product specifications are defined using JSON Schema (draft 7) standard [JS], whereas Quote API is defined using OpenAPI 3.0 [OAS-V3]. The payloads exchanged through Quote endpoints must comply with respective Product specifications.

The API definition builds on *TMF622 Product Order Management API REST Specification R19.0.1* [TMF622]. Product Order Use Cases must support the use of any of MEF product specifications.

4.4. Approach

As presented in Figure 2. both Cantata and Sonata API frameworks consist of three structural components:

- Generic API framework
- Product-independent information (Function-specific information and Function-specific operations)
- Product-specific information (MEF product specification data model)

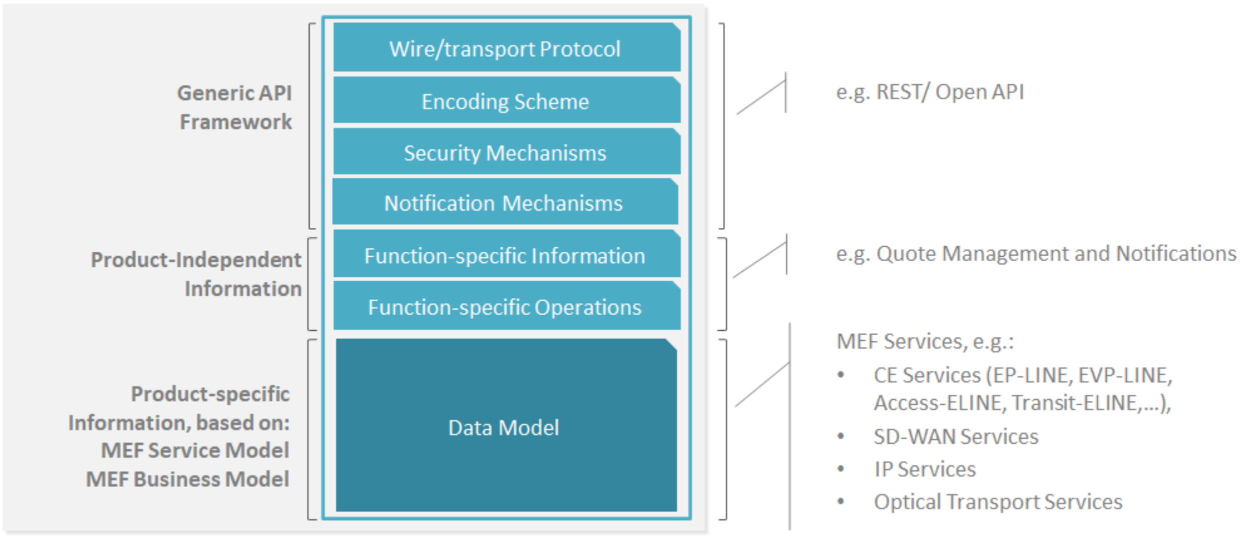

Figure 2. Cantata and Sonata API framework

The essential concept behind the framework is to decouple the common structure, information, and operations from the specific product information content.
Firstly, the Generic API Framework defines a set of design rules and patterns that are applied across all Cantata or Sonata APIs.
Secondly, the product-independent information of the framework focuses on a model of a particular Cantata or Sonata functionality and is agnostic to any of the product specifications. For example, this standard describes the Product Order model and operations that allow performing quoting of any product that is aligned with either MEF or custom product specifications.
Finally, the product-specific information part of the framework focuses on MEF product specifications that define business-relevant attributes and requirements for trading MEF subscriber and MEF operator services.

This Developer Guide does not define MEF product specifications but can be used in combination with any product specifications defined by or compliant with MEF.

4.5. High-Level Flow

Product Order Management is part of a broader Cantata and Sonata End-to-End flow. Figure 3. below shows a high-level diagram to get a good understanding of the whole process and Product Order Management's position within it.

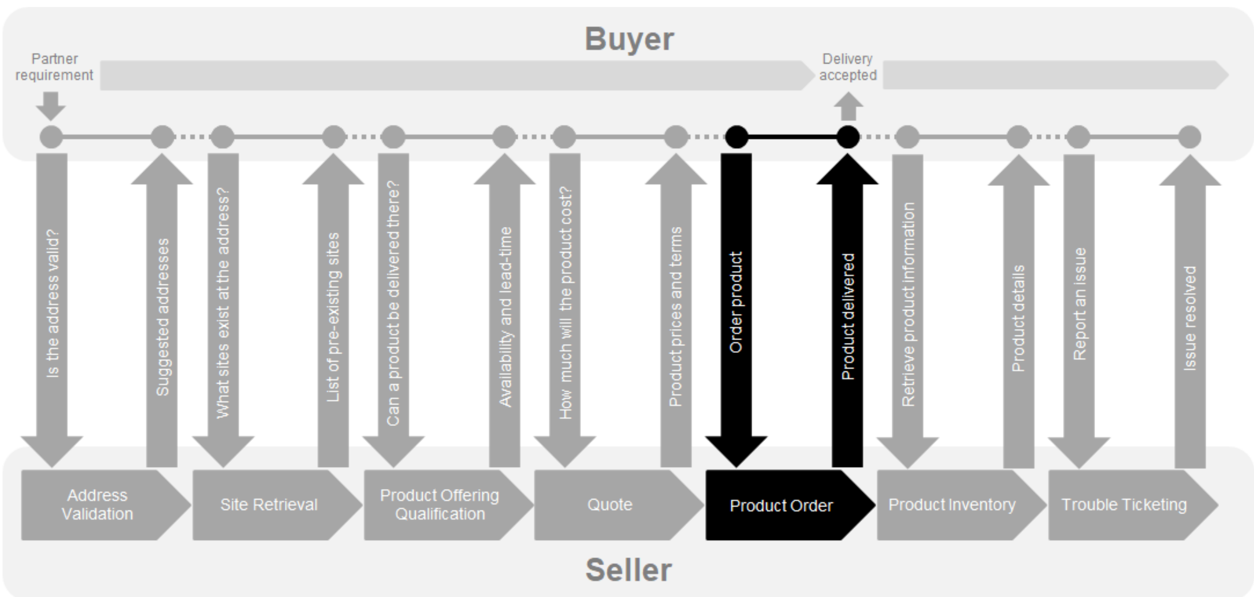

Figure 3. Cantata and Sonata End-to-End Function Flow

- Address Validation:
  - Allows the Buyer to retrieve address information from the Seller, including exact formats, for Geographic Addresses known to the Seller.
- Site Retrieval:
  - Allows the Buyer to retrieve Service Site information including exact formats for Service Sites known to the Seller.
- Product Offering Qualification (POQ):
  - Allows the Buyer to check whether the Seller can deliver a product or set of products from among their product offerings at the geographic address or a Geographic Site specified by the Buyer; or modify a previously purchased product.
- Quote:
  - Allows the Buyer to submit a request to find out how much the installation of an instance of a Product Offering, an update to an existing Product, or a disconnect of an existing Product will cost.
- Product Order:
  - Allows the Buyer to request the Seller to initiate and complete the fulfillment process of an installation of a Product Offering, an update to an existing Product, or a disconnect of an existing Product at the address defined by the Buyer.
- Product Inventory:
  - Allows the Buyer to retrieve the information about existing Product instances from Seller's Product Inventory.
- Trouble Ticketing:
  - Allows the Buyer to create, retrieve, and update Trouble Tickets as well as receive notifications about Incidents' and Trouble Tickets' updates. This allows managing issues and situations for a Product provided by the Seller.

5. API Description

This section presents the API structure and design patterns. It starts with the high-level use cases diagram. Then it describes the REST endpoints with use case mapping. Next, it gives an overview of the API resource model and an explanation of the design pattern that is used to combine product-agnostic and product-specific parts of API payloads. Finally, payload validation and API security aspects are discussed.

5.1. High-level Use Cases

Figure 4. presents a high-level use case diagram as specified in MEF 57.2 [MEF 57.2] in section 8.1. This picture aims to help understand endpoint mapping. Use cases are described extensively in chapter 6

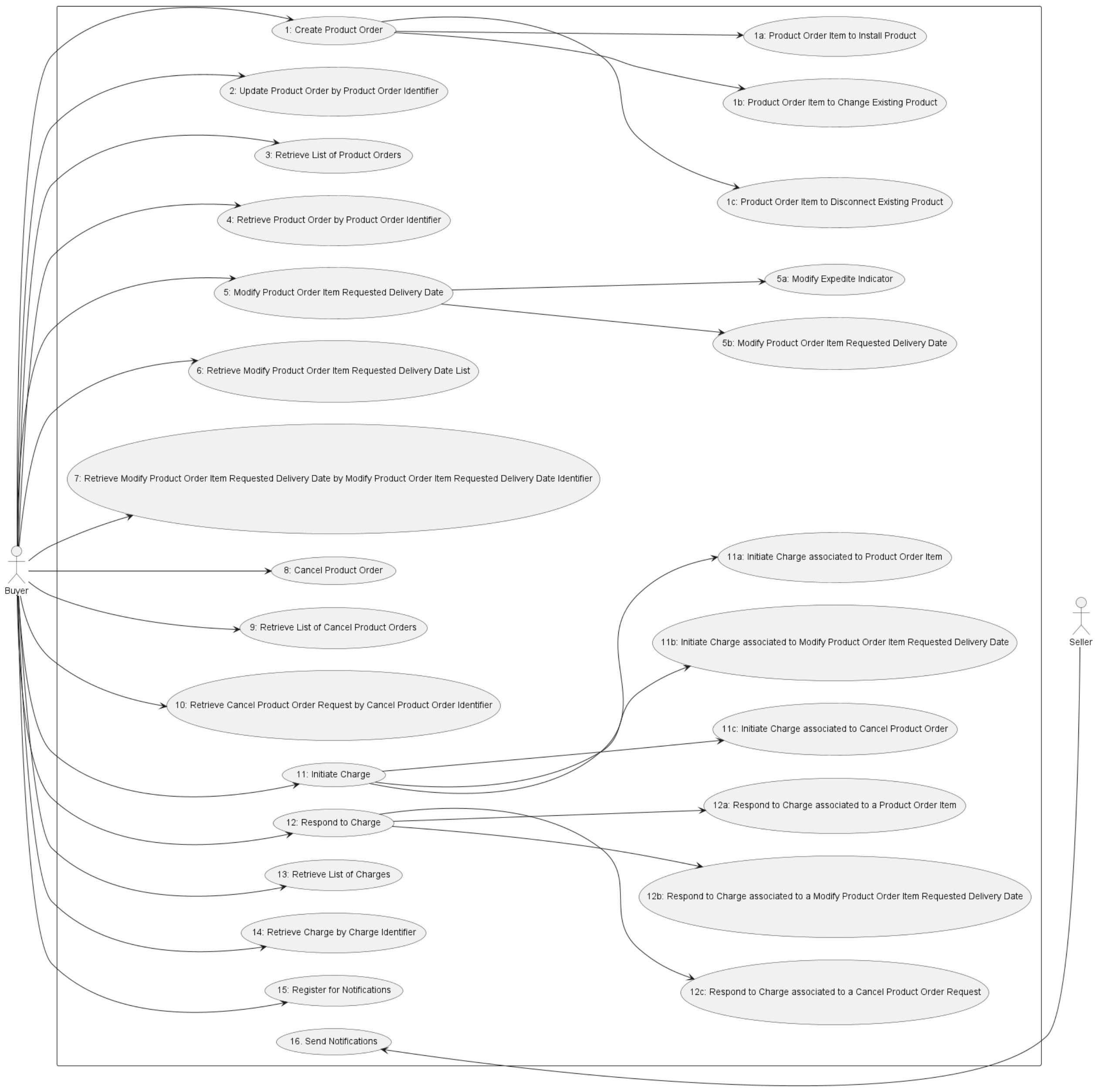

Figure 4. Use cases

5.2. API Endpoint and Operation Description

5.2.1. Seller side API Endpoints

Base URL for Cantata:

`https://{{serverBase}}:{{port}}{{?/seller_prefix}}/mefApi/cantata/productOrderingManagement/v6/`

Base URL for Sonata:

`https://{{serverBase}}:{{port}}{{?/seller_prefix}}/mefApi/sonata/productOrderingManagement/v11/`

The following API endpoints are implemented by the Seller and allow the Buyer to send Product Order requests, retrieve existing Product Orders or Product Order details, manage Charges and Notification registrations. The endpoints and corresponding data models are defined in `productApi/order/productOrderManagement.api.yaml`.

The first of the tables below lists the mandatory endpoints and the second one the optional ones.

API endpoint	Description	MEF 57.2 Use Case mapping
`POST /productOrder`	A request initiated by the Buyer to order a new product.	UC 1: Create Product Order UC 1a: Product Order Item to Install Product UC 1b: Product Order Item to Change Existing Product UC 1c: Product Order Item to Disconnect Existing Product
`GET /productOrder`	A request initiated by the Buyer to retrieve a list of Product Orders that match the provided filter criteria	UC 3: Retrieve List of Product Orders
`GET /productOrder/{{id}}`	A request initiated by the Buyer to retrieve the details associated with a specific Product Order with the given Product Order Identifier.	UC 4: Retrieve Product Order by Product Order Identifier

Table 4. Seller side mandatory API endpoints

[R1] The Seller **MUST** support API endpoints listed in Table 4. [MEF57.2 R1]

API endpoint	Description	MEF 57.2 Use Case mapping

API endpoint	Description	MEF 57.2 Use Case mapping
`PATCH /productOrder/{{id}}`	Allows the Buyer to update some Product Order and Product Order Item Attributes which have no impact on the fulfillment process of the Product Order	UC 2: Update Product Order by Product Order Identifier
`POST /cancelProductOrder`	A request initiated by the Buyer to cancel a Product Order.	UC 8: Cancel Product Order
`GET /cancelProductOrder`	A request initiated by the Buyer to retrieve a list of Cancel requests that match the provided filter criteria	UC 9: Retrieve List of Cancel Product Orders
`GET /cancelProductOrder/{{id}}`	A request initiated by the Buyer to retrieve the details associated with a specific Cancel Request with the given Cancel Product Order Identifier.	UC 10: Retrieve Cancel Product Order by Cancel Product Order Identifier
`GET /charge`	A request initiated by the Buyer to retrieve a list of Charges that match the provided filter criteria	UC 13: Retrieve List of Charges
`GET /charge/{{id}}`	A request initiated by the Buyer to retrieve the details associated with a specific Charge with the given Charge Identifier.	UC 14: Retrieve Charge by Charge Identifier

API endpoint	Description	MEF 57.2 Use Case mapping
PATCH /charge/{{id}}	A Buyer communicates to the Seller if they Accept or Decline Charge Items.	UC 12: Respond to Charge UC 12a: Respond to Charge Associated to a Product Order Item UC 12b: Respond to Charge Associated to a Modify Product Order Item Requested Delivery Date UC 12c: Respond to Charge Associated to a Cancel Product Order
POST /modifyProductOrderItemRequestedDeliveryDate	A request initiated by the Buyer to modify the requested delivery date of a Product Order Item.	UC 5: Modify Product Order Item Requested Delivery Date UC 5a: Modify Expedite Indicator UC 5b: Modify Product Order Item Requested Delivery Date RequestRequest
GET /modifyProductOrderItemRequestedDeliveryDate	A request initiated by the Buyer to retrieve a list of Modify Product Order Item Requested Delivery Date that matches the provided filter criteria	UC 6: Retrieve Modify Product Order Item Requested Delivery Date List

API endpoint	Description	MEF 57.2 Use Case mapping
`GET /modifyProductOrderItemRequestedDeliveryDate/{{id}}`	A request initiated by the Buyer to retrieve the details associated with a specific Modify Product Order Item Requested Delivery Date with the given Modify Product Order Item Requested Delivery Date Identifier.	UC 7: Retrieve Modify Product Order Item Requested Delivery Date by Modify Product Order Item Requested Delivery Date Identifier
`POST /hub`	The Buyer requests to subscribe to notifications.	UC 15: Register for Notifications
`GET /hub/{{id}}`	A request initiated by the Buyer to retrieve the details of the notification subscription.	UC 15: Register for Notifications
`DELETE /hub/{{id}}`	A request initiated by the Buyer to instruct the Seller to stop sending notifications.	UC 15: Register for Notifications

Table 5. Seller side optional API endpoints

[O1] The Seller **MAY** support API endpoints listed in Table 5. [MEF57.2 O1]

[CR1]<[O1] If any of the endpoints implementing Use Cases 5, 5a, 5b, 6, or 7 is supported then all endpoints implementing Use Cases 5, 5a, 5b, 6, and 7 **MUST** be supported. [MEF57.2 CR1<O1]

[CR2]<[O1] If any of the endpoints implementing Use Cases 8, 9, or 10 is supported, then all endpoints implementing Use Cases 8, 9, and 10 **MUST** be supported. [MEF57.2 CR2<O1]

[CR3]<[O1] If any of endpoints implementing Use Cases 11, 11a, 11b, 11c, 12, 12a, 12b, 12c, 13, or 14 is supported then all endpoints implementing Use Cases 11, 11a, 11b, 11c, 12, 12a, 12b, 12c, 13, 14, 15, and 16 **MUST** be supported. [MEF57.2 CR3<O1]

[CR4]<[O1] If either endpoints implementing Use Cases 15 or 16 are supported then both endpoints implementing Use Cases 15 and 16 **MUST** be supported. [MEF57.2 CR4<O1]

5.2.2. Buyer side API Endpoints

Base URL for Cantata: `https://{{serverBase}}:{{port}}{{?/buyer_prefix}}/mefApi/cantata/productOrderingNotification/v6/`

Base URL for Sonata: `https://{{serverBase}}:{{port}}{{?/buyer_prefix}}/mefApi/sonata/productOrderingNotification/v11/`

The following API Endpoints are used by the Seller to post notifications to registered listeners. The endpoints and corresponding data model are defined in `productApi/order/productOrderNotification.api.yaml`

All Buyer side endpoints are optional to implement. Please refer to the requirements stated in the previous chapter for more details.

API Endpoint	Description
`POST /listener/productOrderStateChangeEvent`	A request initiated by the Seller to nc `ProductOrder.state` change.
`POST /listener/productOrderItemStateChangeEvent`	A request initiated by the Seller to no `ProductOrderItem.state` change.
`POST /listener/productOrderItemExpectedCompletionDateSetEvent`	A request initiated by the Seller to no `productOrder.productOrderItem.expe` value set.
`POST /listener/productSpecificProductOrderItemMilestoneEvent`	A request initiated by the Seller to Product Specific Product Order Iter event.
`POST /listener/cancelProductOrderStateChangeEvent`	A request initiated by the Seller to nc `CancelProductOrder` state change.

API Endpoint	Description
`POST /listener/chargeCreateEvent`	A request initiated by the Seller to `Charge` create event to initiate the char
`POST /listener/chargeStateChangeEvent`	A request initiated by the Seller to `Charge` state change.

Table 6. Buyer side API endpoints

5.3. Specifying the Buyer ID and the Seller ID

A business Entity willing to represent multiple Buyers or multiple Sellers must follow requirements of [MEF 150] chapter 8.8, which states:

> For requests of all types, there is a business Entity that initiates an Operation (called a Requesting Entity) and a business Entity that is responding to this request (called the Responding Entity). In the simplest case, the Requesting Entity is the Buyer and the Responding Entity is the Seller. However, in some cases, the Requesting Entity may represent more than one Buyer and similarly, the Responding Entity may represent more than one Seller.

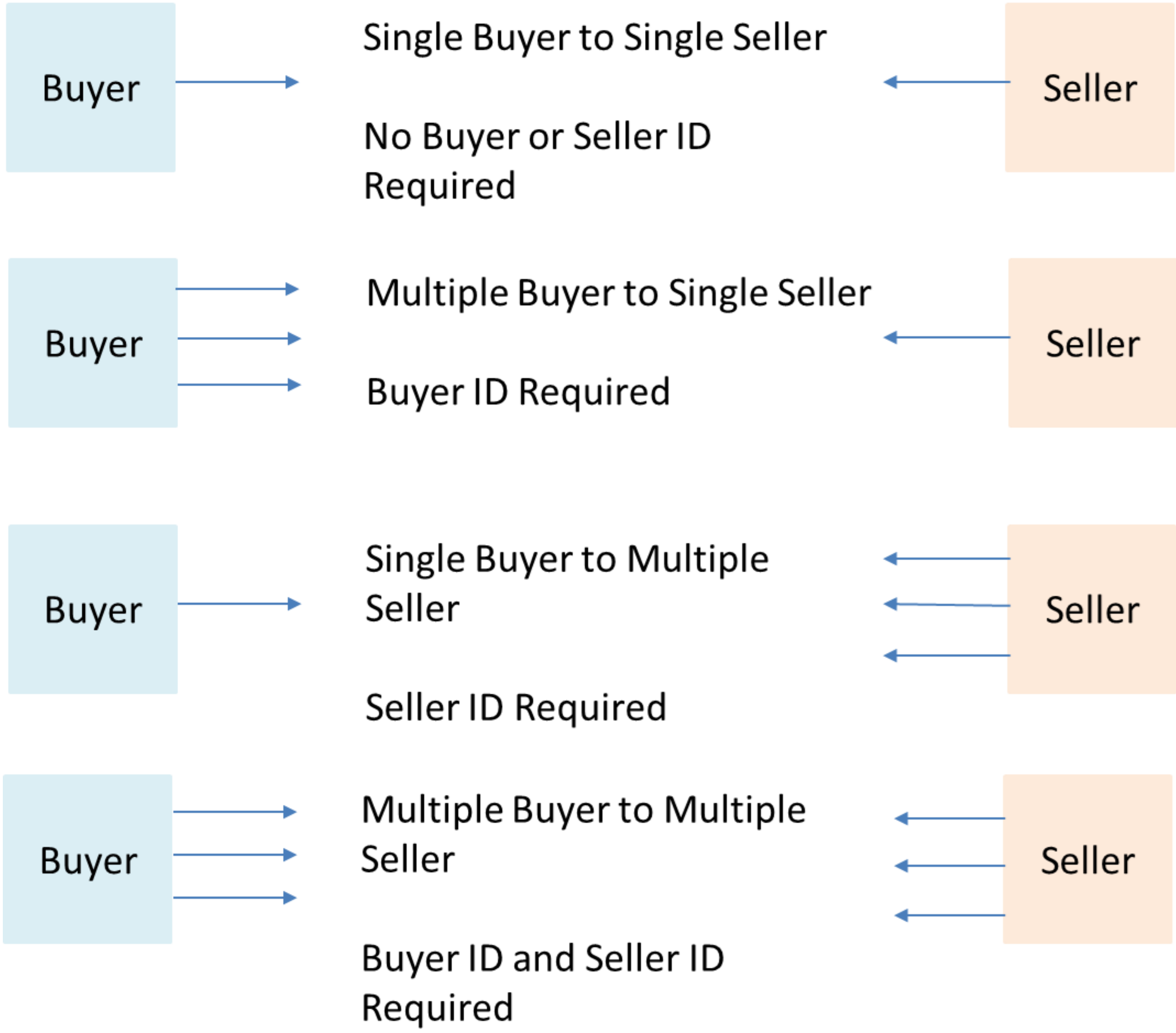

Figure 5. Buyer ID and Seller ID Examples

> As shown in Figure 5, if a Requesting Entity representing a single Buyer is doing business with a Responding Entity representing a single Seller, Buyer and Seller IDs are not required to be passed between the two Entities. If a Requesting Entity representing more than one Buyer is doing business with a Responding Entity representing a single Seller, the Buyer ID is required to be passed between the two Entities. If a Requesting Entity representing a single Buyer is doing business with a Responding Entity representing multiple Sellers, the Seller ID is required to be passed between the two Entities. If a Requesting Entity representing multiple Buyers is doing business with a Responding Entity representing multiple Sellers, both the Buyer ID and the Seller ID are required to be passed between the Entities.

> While it is outside the scope of this specification, it is assumed that the Requesting Entity and the Responding Entity are aware of each other and can authenticate requests initiated by the other party. It is further assumed that the Buying Entity knows:
>
> - the list of Buyers the Requesting Entity represents when interacting with this Responding Entity; and
> - the list of Sellers that this Responding Entity represents to this Requesting Entity.
>
> It is also assumed that the Responding Entity knows:
>
> - the list of Sellers that this Responding Entity represents to this Requesting Entity and

> - the list of Buyers the Requesting Entity represents when interacting with this Responding Entity.

In the API the `buyerId` and `sellerId` are represented as an optional query parameters in each operation defined.

[R2] If the Requesting Entity has the authority to represent more than one Buyer the request **MUST** include `buyerId` that identifies the Buyer being represented [MEF150 R62]

[R3] If the Responding Entity represents more than one Seller to this Buyer the request **MUST** include `sellerId` that identifies the Seller with whom this request is associated [MEF150 R63]

5.4. Integration of Product Specific Attributes

Product specifications are defined using JsonSchema format and are integrated into a Product Order payload using a standard TMF extension pattern.

The extension hosting type in the API data model is `MEFProductConfiguration`. The `@type` attribute of that type must be set to a value that uniquely identifies the product specification. A unique identifier for MEF standard product specifications is in URN format and is assigned by MEF. This identifier is provided as root schema `$id` and in product specification documentation. Use of non-MEF standard product definitions is allowed. In such a case, the schema identifier must be agreed between the Buyer and the Seller.

The example below shows a header of a Product Specification schema, where `"$id": urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all` is the above-mentioned URN:

```
'$schema': http://json-schema.org/draft-07/schema#
'$id': urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all
title: MEF LSO Sonata - Access Eline OVC Product Schema
```

Product specifications are provided as Json schemas without the `MEFProductConfiguration` context.

Product-specific attributes must be introduced into the `productConfiguration` attribute of the `MEFProductRefOrValueOrder`.

Implementations might choose to integrate selected product specifications into the data model during development. In such cases an integrated data model is built and product specifications are in an inheritance relationship with `MEFProductConfiguration` as described in OAS specification. This pattern is called **Static Binding**. The SDK is additionally shipped with a set of API definitions that statically bind all product-related APIs (POQ, Quote, Order, Inventory) with all corresponding product specifications available in the release. The snippets below present an example of a static binding of the Product Order API with a number of MEF product specifications, from both `MEFProductConfiguration` and product specification point of view:

```
MEFProductConfiguration:
  description:
    MEFProductConfiguration is used as an extension point for MEF-specific
    product/service payload.  The `@type` attribute is used as a discriminator
  discriminator:
    mapping:
                                                  urn:mef:lso:spec:sonata:carrier-ethernet-operator-uni:v5.0.0:all:
'#/components/schemas/CarrierEthernetOperatorUni'
        urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all: '#/components/schemas/AccessElineOvc'
    propertyName: '@type'
  properties:
    '@type':
```

```
        description:
          The name of the type, defined in the JSON schema specified above, for
          the product that is the subject of the Product Order Request. The named
          type must be a subclass of MEFProductConfiguration.
        type: string
```

```
AccessElineOvc:
  allOf:
    - $ref: '#/components/schemas/MEFProductConfiguration'
    - $ref: '#/components/schemas/AccessElineOvcCommon'
    - properties:
        uniEp:
          $ref: '#/components/schemas/AccessElineOvcEndPoint'
          description:
            MEF 26.2 sec. 16 - The OVC EP object for the OVC EP at the UNI. The
            UNI OVC End Point must be included in the Access E-Line Product.
        enniEp:
          $ref: '#/components/schemas/AccessElineOvcEndPoint'
          description:
            MEF 26.2 sec. 16 - The OVC EP object for the OVC EP at the ENNI. The
            ENNI OVC End Point must be included in the Access E-Line Product.
```

Alternatively, implementations might choose not to build an integrated model and choose different mechanisms allowing runtime validation of product-specific fragments of the payload. The system is able to validate a given product against a new schema without redeployment. This pattern is called **Dynamic Binding.**

Regardless of the chosen implementation pattern, the HTTP payload is exactly the same. Both implementation approaches must conform to the requirements specified below.

[R4] `MEFProductConfiguration` type is an extension point that **MUST** be used to integrate product specifications' properties into a request/response payload.

[R5] The `@type` property of `MEFProductConfiguration` **MUST** be used to specify the type of the extending entity.

[R6] Product attributes specified in the payload must conform to the product specification indicated by the `@type` property.

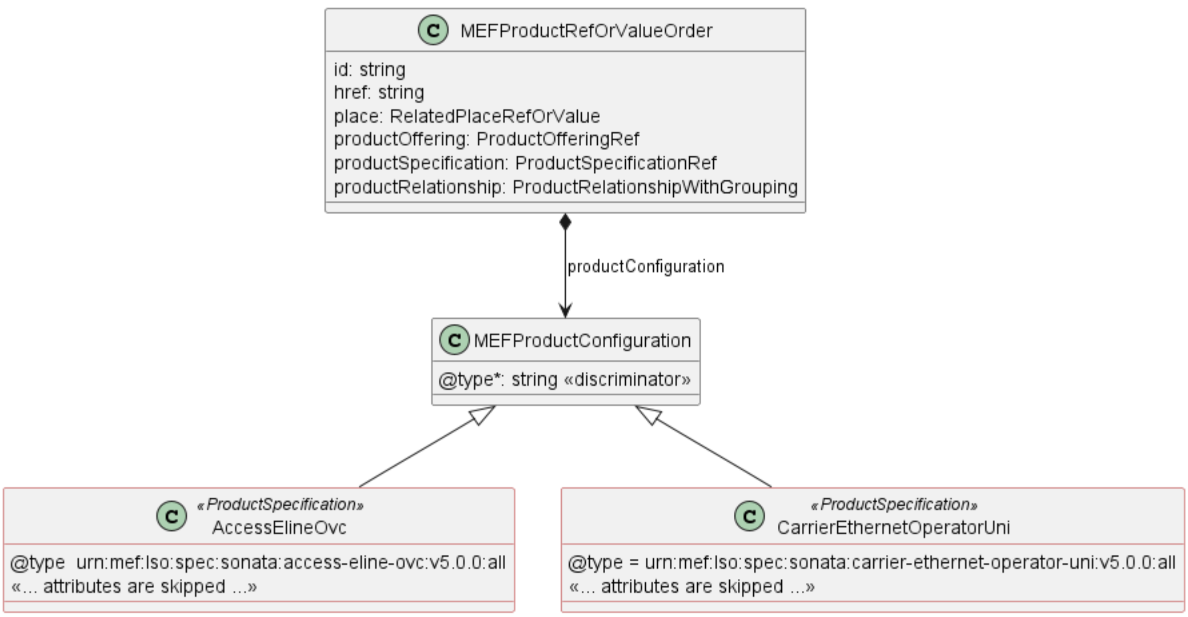

Figure 6. The Extension Pattern

Figure 6 depicts two MEF `<<ProductSpecifications>>` that represent Access E-Line and Operator UNI products. When these products are used in the Product Order payload the `@type` of `MEFProductConfiguration` takes `"urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all"` or `"urn:mef:lso:spec:sonata:carrier-ethernet-operator-uni:v5.0.0:all"` value to indicate which product specification should be used to interpret a set of product-specific attributes included in the payload.

The *all* suffix after the product type name in the URN comes from the approach that the product schemas may differ depending on the API they are used with. `all` means that this schema is applicable to all contexts.

This document uses samples of Access E-Line Product specification definitions to construct API payload examples in Section 6.

Note: The Access E-Line product is valid only in the Sonata context. It is used only for the explanation of the rules of combining the product-agnostic (envelope) and product-specific (payload) parts of the APIs. The examples do not represent full and consistent product configurations, they are not normative and are not kept up to date with their respective standards. It is out of the scope of this document to explain the details of any product.

5.5. Sample Product Specification

The Sonata SDK contains product specification definitions, from which Access E-Line [MEF 106] is used in the payload samples in this section. They are located in the SDK at:

```
\productSchema\carrierEthernet\operatorEthernet\accessEline\accessElineOvc.yaml
\productSchema\carrierEthernet\operatorEthernet\carrierEthernetOperatorUni\carrierEth
ernetOperatorUni.yaml
```

Figure 7 depicts a simplified view of the defined relationships with other products and places.

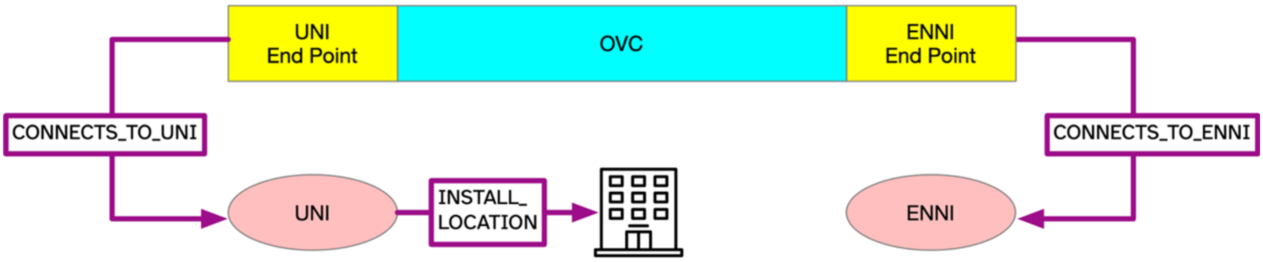

Figure 7. A Simplified View of Product and Place Relationships

Product specifications define a number of product-related and envelope-related requirements. Sample envelope-related requirements for Access E-Line:

- for an Access E-Line OVS product two mandatory relationship roles must be specified, one with the operator ENNI (`CONNECTS_TO_ENNI`) and a second with the operator UNI (`CONNECTS_TO_UNI`).
- in the case of a `modify` action, product relationships must have the same value as in the `add` action. They must not be changed
- for an operator UNI product a place relationship (`INSTALL_LOCATION`) must be specified
- in the case of a `modify` action, place relationships must have the same value as in the `add` action. They must not be changed

The product relationship (`product.productRelationship`) and the place relationship (`product.place`) are presented in Figure 7.

In case some of both product-related or envelope-related requirements are violated the Seller returns an error response to the Buyer which indicates specific functional errors. These errors are listed in the response body (a list of `Error422` entries) for HTTP `422` response.

5.6. Model Structural Validation

The structure of the HTTP payloads exchanged via Quote API endpoints is defined using:

- OpenAPI version 3.0 for the product-agnostic part of the payload
- JsonSchema (draft 7) for the product-specific part of the payload

[R7] Implementations **MUST** use payloads that conform to these definitions.

[R8] The Buyer and the Seller **MUST NOT** use any operation, entity or attribute that is not explicitly defined or allowed by this standard.

[R9] A product specification may define additional consistency rules and requirements that **MUST** be respected by implementations.

These are defined for:

- required relation type, multiplicity to other items in the same quote request
- required relation type, multiplicity to entities in the Seller's product inventory
- related contact information roles that are to be defined at the item level
- relations to places (locations) and their roles that are to be defined at the item level

5.7. Security Considerations

There must be an authentication mechanism whereby a Seller can be assured who a Buyer is and vice-versa. There must also be authorization mechanisms in place to control what a particular Buyer or Seller is allowed to do and what information may be obtained. However, the definition of the exact security mechanism and configuration is outside the scope of this document. Security considerations are standardized by *LSO API Security Profile* [MEF 128.1].

6. API Interactions and Flows

This section provides a detailed insight into the API functionality, use cases, and flows. It starts with Table 7 presenting a list and short description of all business use cases then presents the variants of end-to-end interaction flows, and in following subchapters describes the API usage flow and examples for each of the use cases.

Use Case #	Use Case Name	Use Case Description	Mandatory or optional
1	Create Product Order	A request initiated by the Buyer to order a new product or product component(s). A Product Order must contain at least one Product Order Item (Use Case # 1-a, 1-b, or 1-c) as shown below. A Product Order may contain more than one Product Order Item and Product Order Items within a Product Order are not required to have relationships between them.	Mandatory
1-a	Product Order Item to Install Product	Product Order Item installs a new Product.	Mandatory
1-b	Product Order Item to Change Existing Product	Product Order Item changes attributes of a specific active Product.	Mandatory
1-c	Product Order Item to Disconnect Existing Product	Product Order Item disconnects an active Product.	Mandatory
2	Update Product Order	Allows the Buyer to update some Product Order and Product Order Item Attributes	Optional
3	Retrieve List of Product Orders	A request initiated by the Buyer to retrieve a list of Product Orders that match the provided filter criteria	Mandatory
4	Retrieve Product Order by Product Order Identifier	A request initiated by the Buyer to retrieve the details associated with a specific Product Order with the given Product Order Identifier.	Mandatory
5	Modify Product Order Item Requested Delivery Date	A request initiated by the Buyer to modify either the Expedite Indicator or the Requested Completion Date of a Product Order Item.	Optional
6	Retrieve Modify Product Order Item Requested Delivery Date List	A request initiated by the Buyer to retrieve a list of Modify Product Order Item Requested Delivery Date that match the provided filter criteria	Optional

Use Case #	Use Case Name	Use Case Description	Mandatory or optional
7	Retrieve Modify Product Order Item Requested Delivery Date by Modify Product Order Item Requested Delivery Date Identifier	A request initiated by the Buyer to retrieve the details associated with a specific Modify Product Order Item Requested Delivery Date with the given Modify Product Order Item Requested Delivery Date Identifier.	Optional
8	Cancel Product Order	A request initiated by the Buyer to cancel a Product Order.	Optional
9	Retrieve List of Cancel Requests	A request initiated by the Buyer to retrieve a list of Cancel Product Order requests that match the provided filter criteria	Optional
10	Retrieve Cancel Product Order by Cancel Product Order Identifier	A request initiated by the Buyer to retrieve the details associated with a specific Cancel Product Order with the given Cancel Product Order Identifier.	Optional
11	Initiate Charge	Process to communicate charges from the Seller to Buyer	Optional
12	Respond to Charge	Process to communicate if the Buyer accepts or rejects the charges.	Optional
13	Retrieve List of Charges	A request initiated by the Buyer to retrieve a list of Charges that match the provided filter criteria	Optional
14	Retrieve Charge by Identifier	A request initiated by the Buyer to retrieve the details associated with a specific Charge with the given ChargeIdentifier.	Optional
15	Register for Notifications	The Buyer requests to subscribe to notifications for the Use Cases which the Seller supports including Create Product Order, Cancel Product Order, Charges, or Modify Product Order Item Requested Delivery Date.	Optional
16	Send Notification	A notification initiated by the Seller to the Buyer providing subsequent status information on Create Product Order, Cancel Product Order, Modify Product Order Item Requested Delivery Date, and Initiate Charge.	Optional

Table 7. Use cases description

The detailed business requirements of each of the use cases are described in sections 7 and 1 of [MEF 57.2] and in [MEF 57.2.1].

6.1. Use case 1: Create Product Order

This is the initial step for Product Order processing.

6.1.1. Interaction flow

The flow of this use case is very simple and is described in Figure 8.

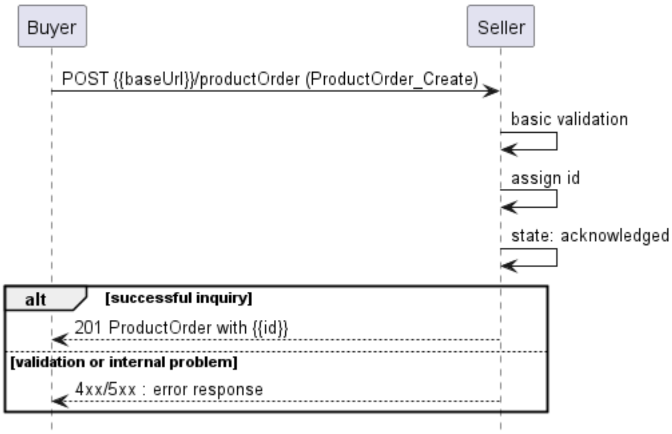

Figure 8. Use Case 1 - Product Order create request flow

The Buyer sends a request with a `ProductOrder_Create` type in the body. The Seller performs request validation, assigns an `id`, and returns `ProductOrder` type in the response body, with a `state` set to `acknowledged`. From this point, the Product Order is ready for further processing. The Buyer can track the progress of the process either by subscribing for notifications or by periodically polling the `ProductOrder`. The two patterns are presented in the following two diagrams.

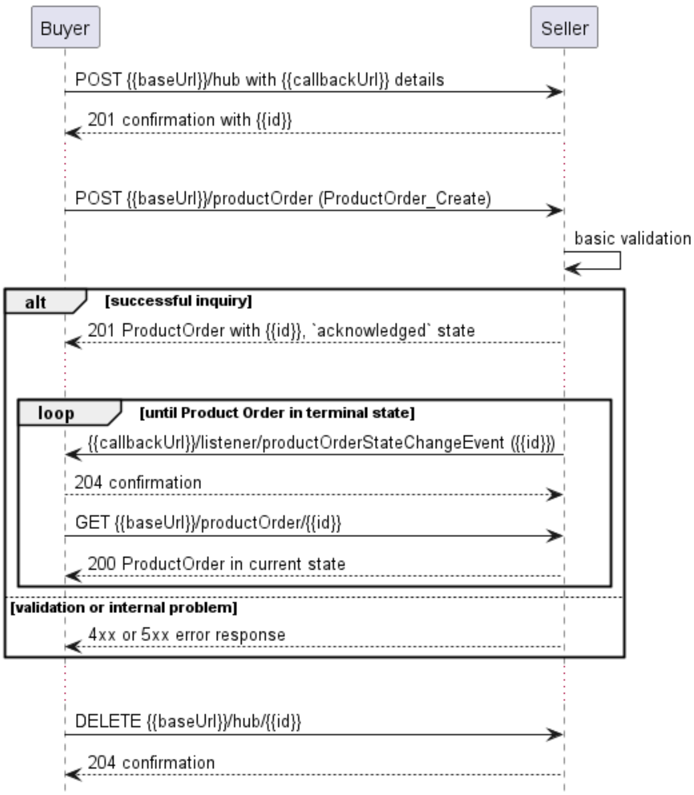

Figure 9. Product Order progress tracking - Notifications

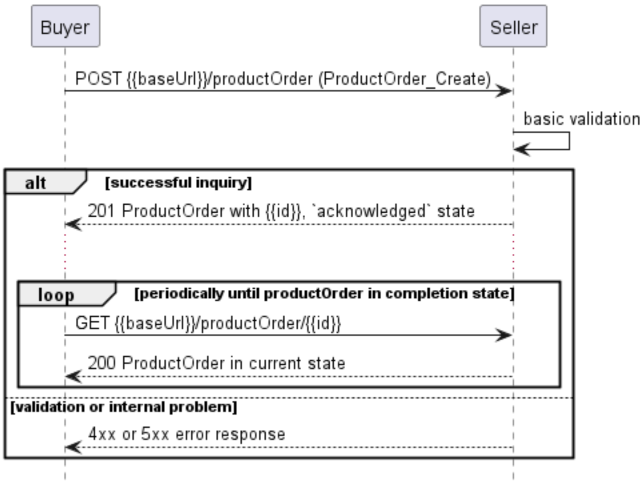

Figure 10. Product Order progress tracking - Polling

Note: The context of notifications is not a part of the considered use case itself. It is presented to show the big picture of end-to-end flow. This applies also to all further use case flow diagrams with notifications.

6.1.2. Key Entities - Request

Figure 11 presents the most important parts of the data model used during the Create Product Order request (`POST /productOrder`) that is sent by a Buyer (see Section 5.2.1 for details). The model of the request message is a subset of the `ProductOrder` model and contains only attributes that can (or must) be set by the Buyer. The Seller then enriches the entity in the response with additional information.

Note: `ProductOrder_Create` and `ProductOrderItem_Create` are entities used by the Buyer to make a request. `ProductOrder` and `ProductOrderItem` are entities used by the Seller to provide a response. The request entities have a subset of attributes of the response entities. Thus for visibility of these shared attributes `ProductOrder_Common` and `ProductOrderItem_Common` have been introduced. Though, these are not to be used directly in the exchange.

A `ProductOrderItem_Create` defines details of the product(s) being subject of the ordering (in `MEFProductRefOrValueOrder` structure) and allows for the definition of additional information like related parties (`RelatedContactInformation`) or relations to other items (`ProductOrderItemRelationship`).

`MEFProductRefOrValueOrder` allows for the introduction of MEF product-specific properties to the Product Order payload. The extension mechanism is described in detail in Section 5.4. `MEFProductRefOrValueOrder` may be also used to specify relations to places (using specializations of `RelatedPlaceRefOrQueryWithSubUnit`) and/or to a product that exists in the Seller's inventory (using `ProductRelationship`).

The full list of attributes is available in Section 7 and in the API specification which is an integral part of this standard.

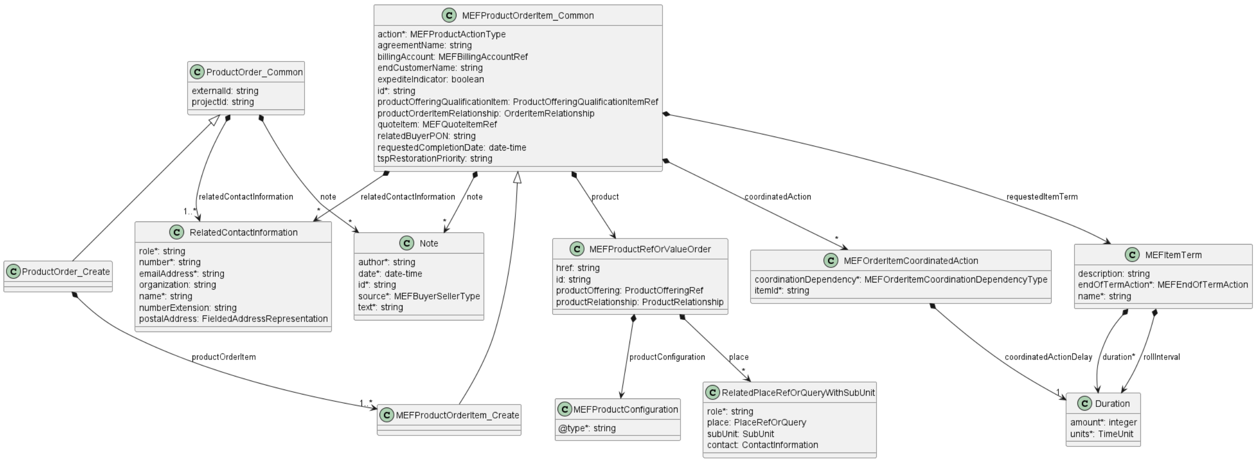

Figure 11. Key Entities - Create Request

6.1.3. Request Example

To send a Product Order request the Buyer uses the `createProductOrder` operation from the API: `POST /productOrder`. For clarity, some of the Product Order payload's attributes might be omitted to improve examples' readability. The `ProductOrder_Create` is a simple structure that is common for all types of requests (`add`, `modify`, `delete`), most of the information is in the `ProductOrderItem_Create`.

In the example below, the Buyer requests two Product Order Items with a specific requirement that the first one will be started one week after the completion of the second one. This is done with the use of the `coordinatedAction`. This action defines possible dependencies and the potential delay between the events.

- `startToStart` - Work on the Specified Product Order Item can only be started after the Coordinated Product Order Items are started
- `startToFinish` - The Coordinated Product Order Items must complete before work on the Specified Product Order Item begins
- `finishToStart` - Work on the Related Product Order Items begins after the completion of the Specified Product Order Item
- `finishToFinish` - Work on the Related Product Order Items completes at the same time as the Specified Product Order Item

`Product Order` Create

```
{
  "externalId": "buyerOrder-001",
  "projectId": "buyerProject-001",
  "relatedContactInformation": [
    {
      "emailAddress": "john.example@buyer.mef.com",
      "name": "John Example",
      "number": "12-345-6789",
      "numberExtension": "1234",
      "organization": "Example Co.",
      "role": "productOrderContact"
    }
  ],
  "productOrderItem": [
    {
      "id": "item-001",
      "action": "add",
      "endCustomerName": "End Customer Name",
      "expediteIndicator": false,
      "relatedBuyerPON": "PON-12-2021",
      "requestedCompletionDate": "2021-06-19T20:59:28.299Z",
      "agreementName": "Buyer-Seller General Agreement 03/2021",
```

```
        "billingAccount": {
          "id": "00000000-1111-0000-0000-000000000001"
        },
        "coordinatedAction": [
          {
            "itemId": "item-002",
            "coordinatedActionDelay": {
              "amount": 1,
              "units": "calendarWeeks"
            },
            "coordinationDependency": "startToFinish"
          }
        ],
        "product": {<< product specific attributes and configuration, see 6.1.6 >>
        },
        "productOfferingQualificationItem": {
          "id": "poqItem-001",
          "productOfferingQualificationId": "00000000-2222-0000-0000-000000000001"
        },
        "productOrderItemRelationship": [
          {
            "id": "item-002",
            "relationshipType": "CONNECTS_TO_UNI"
          }
        ],
        "quoteItem": {
          "id": "quoteItem-001",
          "quoteId": "00000000-4444-0000-0000-000000000001"
        },
        "relatedContactInformation": [
          {
            "emailAddress": "Buyer.ProductOrderItemContact@buyer.mef.com",
            "name": "Buyer Product Order Item Contact",
            "number": "+12-345-678-90",
            "role": "buyerProductOrderItemContact"
          },
          {
            "emailAddress": "Buyer.ImplementationContact@buyer.mef.com",
            "name": "Buyer Implementation Contact",
            "number": "+12-345-678-90",
            "role": "buyerImplementationContact"
          },
          {
            "emailAddress": "Buyer.TechnicalContact@buyer.mef.com",
            "name": "Buyer Technical Contact",
            "number": "+12-345-678-90",
            "role": "buyerTechnicalContact"
          },
          {
            "emailAddress": "bill.contact@buyer.mef.com",
            "name": "Bill Contact",
            "number": "+12-345-678-90",
            "organization": "Example Company",
            "role": "billingContact"
          }
        ],
        "requestedItemTerm": {
          "duration": {
            "amount": 12,
            "units": "calendarMonths"
          },
          "endOfTermAction": "autoRenew",
          "name": "Yearly Subscription"
        }
      },
      {
        "id": "item-002",
        "action": "add"
        ...
        << attributes skipped for readability >>
      }
    ]
  }
```

[R10] The Buyer's request **MUST** contain at least one `productOrderItem`. [MEF57.2 R14]

[R11] The Buyer's request **MUST** specify a `relatedContactInformation` item with a `role` set to `productOrderContact`. [MEF57.2 R61]

[O2] The Buyer and Seller **MAY** agree on using other contact `roles`. [MEF57.2 O3]

Note: During the onboarding the Seller may require to provide an additional contact `role`.

Note: It is up to Seller's discretion on how to react in case the Buyer provides a contact `role` that is not listed by this standard or agreed upon during the onboarding. Preferably the Seller should return an error with a message stating which `roles` are accepted. It may also be ignored.

For each `ProductOrderItem`:

[R12] The Buyer's Create Product Order request **MUST** contain: [MEF57.2 R22], [MEF57.2 R16]

- `id`
- `action`
- `requestedCompletionDate`
- `product`
- `relatedContactInformation` items with the following values of `role` set (only when `action` is `add` or `modify`):
  - `buyerProductOrderItemContact`
  - `buyerImplementationContact`
  - `buyerTechnicalContact`

[R13] The `ProductOrderItem` content **MUST** follow the product specification-related requirements when specifying values for `relatedContactInformation`, `productOrderItemRelationship`, `product.place`, and `product.productRelationship` attributes. [MEF57.2.1 A1-R2]

[R14] When specifying the `product.place`, the Buyer **MUST** provide following attributes`: [MEF57.2.1 A1-R2]

- `place`
- `role`

[R15] When specifying the `product.place`, with `GeographicSitesRef` the Buyer **MUST NOT** additionally provide `subUnit` to describe exactly where the Buyer wants the Product to be installed. [MEF57.2.1 A1-R3]

[O3] The Seller **MAY** require that the `billingAccount` attributes be the same for all Product Order Items in a Product Order. [MEF57.2 O8]

[O4] The Seller **MAY** require the Buyer to perform a POQ prior to submitting the Product Order. [MEF57.2 O7]

[CR5]<[O4] The Buyer's request **MUST** provide the `productOfferingQualificationItem` if required by the Seller. [MEF57.2 CR5<O7]

[O5] The Seller **MAY** require the Buyer to perform a Quote prior to submitting the Product Order. [MEF57.2 O8]

[CR6]<[O5] The Buyer's request **MUST** provide the `quoteItem` if required by the Seller. [MEF57.2 CR6<O8]

[R16] If the Buyer requires the `tspRestorationPriority` to be specified for the Product Order Item, the Buyer's Create Product Order request **MUST** provide it. [MEF57.2 R25]

6.1.4. Key Entities - Response

Figure 12 presents the most important data model parts used to provide a response to a Buyer's Create Product Order (`POST /productOrder`) or to retrieve a `ProductOrder` by identifier (`GET /productOrder/{{id}}`) request. Please note that the model differs only in the number of attributes for `ProductOrder` and `ProductOrderItem` entities.

`ProductOrder` is the root entity of a response and it contains the same number of `ProductOrderItems` as in the request.

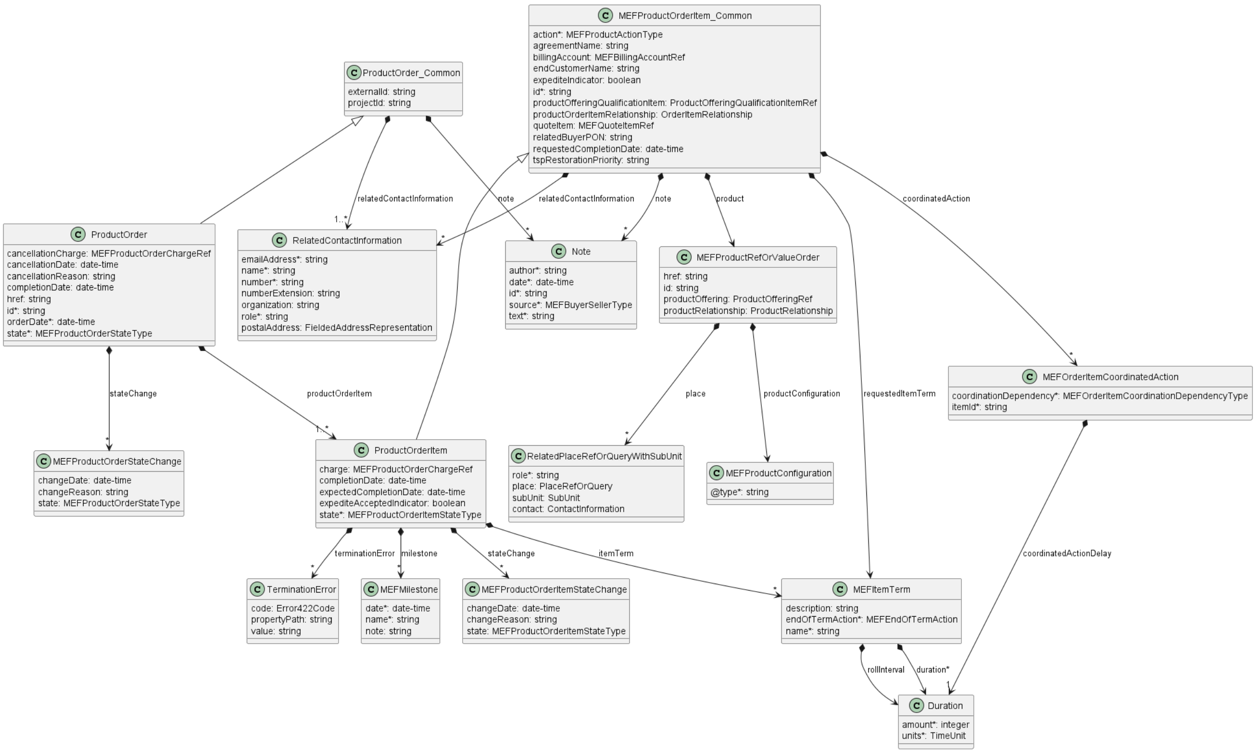

Figure 12. Key Entities - Response

Note: The term "Seller Response Code" used in the Business Requirements maps to HTTP response code, where `2xx` indicates *Success* and `4xx` or `5xx` indicate *Failure*.

6.1.5. Response Example

The following snippet presents the Seller's response. It has the same structure as in the retrieve by identifier operation.

```
{
  "id": "00000000-1111-2222-3333-000000000123",
  "href": "{{baseUrl}}/productOrder/00000000-1111-2222-3333-000000000123",
  "orderDate": "2021-05-19T07:01:02.983Z",
  "state": "acknowledged",
  "stateChange": [
    {
      "changeDate" : "2021-05-19T07:01:02.983Z",
      "state": "acknowledged"
    }
  ],
  "externalId": "buyerOrder-001", << as provided by the Buyer >>
  "projectId": "buyerProject-001", << as provided by the Buyer >>
  "relatedContactInformation": [
    { << as provided by the Buyer >>
```

```
        "emailAddress": "john.example@buyer.mef.com",
        "name": "John Example",
        "number": "12-345-6789",
        "numberExtension": "1234",
        "organization": "Buyer Example Co.",
        "role": "productOrderContact"
      },
      { << added by Seller >>
        "emailAddress": "kate.example@seller.mef.com",
        "name": "Kate Example",
        "number": "12-345-67890",
        "organization": "Seller Example Co.",
        "role": "sellerContact"
      }
    ],
    "productOrderItem": [
      {
        "id": "item-001", << as provided by the Buyer >>
        "action": "add", << as provided by the Buyer >>
        "endCustomerName": "End Customer Name", << as provided by the Buyer >>
        "expediteIndicator": false, << as provided by the Buyer >>
        "relatedBuyerPON": "PON-12-2021", << as provided by the Buyer >>
        "requestedCompletionDate": "2021-06-19T20:59:28.299Z", << as provided by the Buyer >>
        "expectedCompletionDate": "2021-05-31T00:00:00.000Z",
        "expediteAcceptedIndicator": false,
        "state": "acknowledged",
        "stateChange": [
          {
            "changeDate": "2021-05-19T07:01:02.983Z",
            "state": "acknowledged"
          }
        ],
        "agreementName": "Buyer-Seller General Agreement 03/2021", << as provided by the Buyer >>
        "billingAccount": { << as provided by the Buyer >>  },
        "coordinatedAction": [ << as provided by the Buyer >> ],
        "product": { << as provided by the Buyer >> },
        "productOfferingQualificationItem": { << as provided by the Buyer >> },
        "productOrderItemRelationship": [ << as provided by the Buyer >> ],
        "quoteItem": { << as provided by the Buyer >> },
        "relatedContactInformation": [
          {
            "emailAddress": "Buyer.ProductOrderItemContact@buyer.mef.com",
            "name": "Buyer Product Order Item Contact",
            "number": "+12-345-678-90",
            "role": "buyerProductOrderItemContact"
          },
          {
            "emailAddress": "Buyer.ImplementationContact@buyer.mef.com",
            "name": "Buyer Implementation Contact",
            "number": "+12-345-678-90",
            "role": "buyerImplementationContact"
          },
          {
            "emailAddress": "Buyer.TechnicalContact@buyer.mef.com",
            "name": "Buyer Technical Contact",
            "number": "+12-345-678-90",
            "role": "buyerTechnicalContact"
          },
          {<< added by Seller >>
            "emailAddress": "Seller.Contact@seller.mef.com",
            "name": "Seller Contact",
            "number": "+12-345-678-90",
            "role": "sellerContact"
          }

        ],
        "requestedItemTerm": {
          "duration": {
            "amount": 12,
            "units": "calendarMonths"
          },
          "endOfTermAction": "autoRenew",
          "name": "Yearly Subscription",
        },
        "itemTerm": [
          {
            "duration": {
              "amount": 12,
              "units": "calendarMonths"
            },
            "endOfTermAction": "autoRenew",
```

```
            "name": "Yearly Subscription",
          }
        ],
        "milestone": [
          {
            "date": "2021-05-19T07:01:02.983Z",
            "name": "EXAMPLE_MILESTONE_NAME",
            "note": "Additional comment when needed"
          }
        ]
      },
      {
        "id": "item-002",
        "action": "add"
        ...
        << attributes skipped for readability >>
      }
    ]
  }
```


The response to the create request does not contain all possible attributes. Some of them are valid only in the future lifecycle of the `Product Order` (e.g. `cancellationDate`, `cancellationReason`, `completionDate`).

[R17] The Seller's response **MUST** include all and unchanged attributes' values as provided in the request. [MEF57.2 R19], [MEF57.2 R33]

These attributes are indicated above with an appropriate comment: `<< as provided by the Buyer >>`.

The Seller might append related contact information if required, either at the item or Product Order level but cannot modify related contact information provided by the Buyer.

[R18] The Seller **MUST** specify the following attributes in a response: [MEF57.2 R16]

- `id`
- `orderDate`
- `productOrderItem`
- `state`
- `stateChange`
- `relatedContactInformation` item with a `role` set to `sellerContact`

[R19] The `id` **MUST** remain the same value for the life of the Product Order. [MEF57.2 R17]

[R20] The `stateChange` **MUST** contain a full history of the `productOrder.state`. [MEF57.2 R20], [MEF57.2 R57]

[O6] The Seller **MAY** add a `note` to any Product Order that is not in the `completed`, `partial`, `rejected`, `cancelled`, or `failed` states. [MEF57.2 O9]

[R21] The Seller **MUST** add a `note` only with `source=seller`. [MEF57.2 R9], [MEF57.2 R10]

[R22] Notes **MUST NOT** be able to be modified or deleted once entered. [MEF57.2 R13]

For each `productOrderItem`:

[R23] The response **MUST** have the `state` and `stateChange` attributes set. [MEF57.2 R32]

[R24] The `stateChange` **MUST** contain a full history of the `state`. [MEF57.2 R51], [MEF57.2 R60]

[R25] If in the request the `expediteIndicator` is `false`, the Seller's response **MUST NOT** have the `expediteAcceptedIndicator` attribute set to `true`. [MEF57.2 R34]

[R26] The response **MUST NOT** include the `expediteAcceptedIndicator` attribute set to `true` until the Charge process for any charges associated with the expedite is complete. [MEF57.2 R35]

[R27] If there are any additional costs associated with the Product Order Item and its `state` is `held.assessingCharge`, the Seller's response **MUST** have the `charge` attribute filled with these costs. [MEF57.2 R39]

[R28] If the Product Order Item `state` in the Seller's response is `cancelled` or `failed`, the `expectedCompletionDate` attribute **MUST NOT** be provided. [MEF57.2 R36]

[R29] If the Product Order Item `state` in the Seller's response is `inProgress`, the `expectedCompletionDate` attribute **MUST** be provided. [MEF57.2 R37]

[R30] If the Product Order Item `state` in the Seller's response is not `completed`, the response **MUST NOT** contain the `completionDate`. [MEF57.2 R38]

6.1.6. Use Case 1a: Product Order Item to Install Product

When requesting a new product installation (`action` equal to `add`) the Buyer needs to provide all of its configuration information. The example below shows a request for Access E-Line product (type `urn:mef:lso:spec:sonata:AccessElineOvc:1.0.0:order`). Assuming this is an extension of a previous example, the Product Order and less important attributes are omitted.

```
{
  <<ProductOrder attributes...>>
  "productOrderItem": [
    {
      "id": "item-001",
      "action": "add",
      ...
      "product": {
        "productConfiguration": {
          "@type": "urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all",
          "ceVlanIdPreservation": "PRESERVE",
          "maximumFrameSize": 1526,
          "listOfClassOfServiceNames": ["low"],
          "enniEp": {
            "identifier": "SP1_ENNI-EP1",
            "ingressClassOfServiceMap": {
              "mapType": "ENDPOINT",
              "map_M": "low",
              "l2cp_P": {
                "l2cpIdentifier": {
                  "l2cpProtocolType": "LLC",
                  "llcAddressOrEtherType": 66
                },
                "l2cpCosName": "low"
              }
            }
          },
          "uniEp": {
            "identifier": "NewYork_UNI-EP1",
            "ingressBandwidthProfilePerClassOfServiceName": [
              {
                "classOfServiceName": "low",
                "bwpFlow": {
                  "cir": {
                    "irValue": 0,
                    "irUnits": "MBPS"
                  },
                  "cirMax": {
                    "irValue": 0,
                    "irUnits": "MBPS"
                  },
```

```
                  "eir": {
                    "irValue": 10,
                    "irUnits": "GBPS"
                  },
                  "eirMax": {
                    "irValue": 10,
                    "irUnits": "GBPS"
                  }
                }
              }
            ],
            "ingressClassOfServiceMap": {
              "mapType": "ENDPOINT",
              "map_M": "low",
              "l2cp_P": {
                "l2cpIdentifier": {
                  "l2cpProtocolType": "LLC",
                  "llcAddressOrEtherType": 66
                },
                "l2cpCosName": "low"
              }
            }
          }
        },
        "productOffering": {
          "id": "000073"
        },
        "productRelationship": [
          {
            "relationshipType": "CONNECTS_TO_ENNI",
            "id": "SP1_ENNI"
          }
        ]
      },
      "productOfferingQualificationItem": {
        "id": "poqItem-001",
        "productOfferingQualificationId": "00000000-2222-0000-0000-000000000001"
      },
      "productOrderItemRelationship": [
        {
          "id": "item-002",
          "relationshipType": "CONNECTS_TO_UNI"
        }
      ],
      "quoteItem": {
        "id": "quoteItem-001",
        "quoteId": "00000000-4444-0000-0000-000000000001"
      },
      "relatedContactInformation": [
        {
          "emailAddress": "Buyer.ProductOrderItemContact@buyer.mef.com",
          "name": "Buyer Product Order Item Contact",
          "number": "+12-345-678-90",
          "role": "buyerProductOrderItemContact"
        },
        {
          "emailAddress": "Buyer.ImplementationContact@buyer.mef.com",
          "name": "Buyer Implementation Contact",
          "number": "+12-345-678-90",
          "role": "buyerImplementationContact"
        },
        {
          "emailAddress": "Buyer.TechnicalContact@buyer.mef.com",
          "name": "Buyer Technical Contact",
          "number": "+12-345-678-90",
          "role": "buyerTechnicalContact"
        }
      ],
      "requestedItemTerm": {
        "duration": {
          "amount": 12,
          "units": "calendarMonths"
        },
        "endOfTermAction": "autoRenew",
        "name": "Yearly Subscription",
      }
    },
{
      "id": "item-002",
      "action": "add",
      ...
```

```
        "product": {
          "productOffering": {
            "id": "000074"
          },
          "place": [
            {
              "place": {
                "@type": "GeographicAddressRef",
                "id": "NewYorkAddress-id-1"
              },
              "role": "INSTALL_LOCATION",
              "contact": [
                {
                  "number": "+12-345-678-90",
                  "emailAddress": "LocationContact@buyer.mef.com",
                  "name": "Location Contact"
                }
              ]
            }
          ],
          "productConfiguration": {
            "@type": "urn:mef:lso:spec:sonata:carrier-ethernet-operator-uni:v5.0.0:all",
            "defaultCeVlanId": 4094,
            "maximumNumberOfEndPoints": 6,
            "lagLinkMeg": "DISABLED",
            "linkAggregation": "NONE",
            "tokenShare": "ENABLED",
            "maximumServiceFrameSize": 1522,
            "listOfPhysicalLinks": [
              {
                "id": "01",
                "physicalLink": "10GBASE_SR",
                "uniConnectorGender": "SOCKET",
                "synchronousEthernet": "ENABLED",
                "uniConnectorType": "SC",
                "precisionTiming": "DISABLED"
              }
            ]
          }
        }
      }
    ]
  }
```

The following requirements apply when `productOrderItem.action` is `add`:

[R31] The Buyer **MUST** provide the `product.productConfiguration`. [MEF57.2 R24]

[R32] If there is a relationship with another Product Order Item within the same Product Order, the `productOrderItemRelationship` **MUST** be specified. [MEF57.2 R45]

[R33] `product.productOffering` **MUST** be provided. [MEF57.2 R45]

[R34] The Buyer **MUST** provide the `billingAccount` even if the presumed price is zero. [MEF57.2 R47]

[R35] The Buyer **MUST** provide the `requestedItemTerm`. [MEF57.2 R43]

[R36] If the `requestedItemTerm.endOfTermAction` is `roll`, the Buyer **MUST** provide the `requestedItemTerm.rollInterval`. [MEF57.2 R44]

[R37] The Buyer **MUST NOT** specify the `productOrderItem.product.id` in the request. It is the Seller who assigns this id.

The following requirements apply to a Seller's lifecycle response when `productOrderItem.action` is `add`:

[R38] If the Seller does not support the `requestedItemTerm`, the Seller **MUST** reject the Product Order Item and move the Product Order Item to the `rejected` state. [MEF57.2 R49]

[R39] If the `requestedItemTerm` does not match the term from a referenced Quote, the Seller **MUST** reject the Product Order Item and move the Product Order Item to the `rejected` state. [MEF57.2 R50]

An Access E-Line product specification defines two mandatory relationship types that have to be specified in case of an `add` action: `CONNECTS_TO_ENNI` and `CONNECTS_TO_UNI`.
The reference to an operator UNI product might use another Product Order Item or an existing product from the Seller's inventory. In this example, the UNI product is another item of the request with a unique identifier `item-002`. This Access E-Line product references an existing ENNI product which is uniquely identified with id `SP1_ENNI` in the Seller's inventory.

The place is not provided as the Access E-Line product specification does not allow for a place description to be part of the request. Values for some of the available product attributes are provided under the `productConfiguration` node. This example uses only a tiny subset of available Access E-Line attributes. It aims to explain the Product definition and relation patterns, not to focus on the product configurations themselves.

This specification describes the structure and requirements defined for this product with which the payload should be validated. Product specification is a subject of MEF standardization. It is published as a dedicated MEF standard. It is built of:

- the JSON Schemas for technical specifications. Those can be found in the SDK in the `\productSchema\` directory.
- a document with a textual description of the product and a list of the requirements (not all of them can be technically included in the JSON schema). Such documents can be found in the `\documentation\productSchema\` directory of the SDK package.

The product offering is a business representation of a product specification version offered by the Seller for purchase. Product offering associates commercial attributes to a product specification. The product offering model is not part of the standardization and is up to the Seller to define their offering.

Both product specifications and product offerings are not negotiated and exchanged within Cantata and Sonata. They are agreed between the Buyer and the Seller during the onboarding process. After that, they are only referenced as in the example above.

6.1.7. Use case 1b: Product Order Item to Change Existing Product

The following example shows a request for an order of an existing Access E-Line Product modification (`action` equal to `modify`). In particular, changes to `cir` (Committed Information Rate) and `cirMax` (Maximum Committed Information Rate) values for `uniEp` bandwidth profiles are introduced.
The Access E-Line product exists in Seller's inventory and is identified as `AccessElineOVC-0001`.

The following requirements apply to `productOrderItem` when `action` is `modify`:

[R40] If `action=modify` the Buyer **MUST** provide following attributes of `productOrderItem`: [MEF57.2 R24], [MEF57.2 R56]

- `product.id`
- `product.productOffering`
- `product.productConfiguration`

[R41] The modify request **MUST** provide a full state of the `product` attributes, including values of (specified or empty) of `product.productRelationship` and `product.place`.

The Product Specification defines if the relationships to products or places can be changed.

[R42] If there is a relationship with another Product Order Item within the same Product Order, the `productOrderItemRelationship` **MUST** be specified. [MEF57.2 R55]

[R43] The Buyer **MUST** provide the `requestedItemTerm` if required by the Seller for the requested change. [MEF57.2 R54]

[O7] The Seller **MAY** allow the Buyer to specify a different `product.productOffering` than the one of the existing Product.

[R44] If `product.productOffering` changes, it **MUST** be based on the same Product Specification` as the existing Product.

[O8] The Buyer **MAY** include the `billingAccount`. [MEF57.2 O12]

[O9] The Seller **MAY** require that the `billingAccount` attributes be the same for all Product Order Items in a Product Order. [MEF57.2 O10]

There is no possibility to send an update to single attributes. The Buyer must send a full product description (the whole `product.productConfiguration` section and if set previously or to be set: `product.productRelationship` and `product.place`), which means all attributes that represent the desired state, even if some of them do not change.
If the Seller does not allow for some of the attributes to change an appropriate error response (`422`) must be returned to the Buyer.

Please also note, that in the `add` case, a reference to the UNI product used the `productOrderItemRelationship` pointing to another `productOrderItem` in the same Product Order Request. This is because the UNI did not exist at that moment and was also a part of the Order. In the case of ordering the update of an existing Access E-Line, the UNI is also existing and it must be referenced with the use of `productRelationship`. This example assumes that the UNI product is available in Seller's Inventory with the `id=SP1_UNI`.

The references to `quoteItem` and `productOfferingQualificationItem`, if provided, would point to a different POQ and Quote items than the ones provided in the `add` request, for the `modify` case also the POQ and Quote have to be performed explicitly for the `modify` action.

```
{
  <<ProductOrder attributes...>>
  "productOrderItem": [
    {
      "id": "item-001",
      "action": "modify",
      ...
      "product": {
        "id": "AccessElineOVC-0001",
        "productConfiguration": {
          "@type": "urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all",
          "ceVlanIdPreservation": "PRESERVE",
          "maximumFrameSize": 1526,
          "listOfClassOfServiceNames": ["low"],
          "enniEp": {
            "identifier": "SP1_ENNI-EP1",
            "ingressClassOfServiceMap": {
              "mapType": "ENDPOINT",
              "map_M": "low",
              "l2cp_P": {
                "l2cpIdentifier": {
                  "l2cpProtocolType": "LLC",
                  "llcAddressOrEtherType": 66
                },
                "l2cpCosName": "low"
              }
            }
```

```
          },
          "uniEp": {
            "identifier": "NewYork_UNI-EP1",
            "ingressBandwidthProfilePerClassOfServiceName": [
              {
                "classOfServiceName": "low",
                "bwpFlow": {
                  "cir": {
                    "irValue": 1,
                    "irUnits": "GBPS"
                  },
                  "cirMax": {
                    "irValue": 1,
                    "irUnits": "GBPS"
                  },
                  "eir": {
                    "irValue": 10,
                    "irUnits": "GBPS"
                  },
                  "eirMax": {
                    "irValue": 10,
                    "irUnits": "GBPS"
                  }
                }
              }
            ],
            "ingressClassOfServiceMap": {
              "mapType": "ENDPOINT",
              "map_M": "low",
              "l2cp_P": {
                "l2cpIdentifier": {
                  "l2cpProtocolType": "LLC",
                  "llcAddressOrEtherType": 66
                },
                "l2cpCosName": "low"
              }
            }
          }
        },
        "productOffering": {
          "id": "000073"
        },
        "productRelationship": [
          {
            "relationshipType": "CONNECTS_TO_ENNI",
            "id": "SP1_ENNI"
          },
          {
            "relationshipType": "CONNECTS_TO_UNI",
            "id": "SP1_UNI"
          }
        ]
      },
      "relatedContactInformation": [
        {
          "emailAddress": "Buyer.ProductOrderItemContact@buyer.mef.com",
          "name": "Buyer Product Order Item Contact",
          "number": "+12-345-678-90",
          "role": "buyerProductOrderItemContact"
        },
        {
          "emailAddress": "Buyer.ImplementationContact@buyer.mef.com",
          "name": "Buyer Implementation Contact",
          "number": "+12-345-678-90",
          "role": "buyerImplementationContact"
        },
        {
          "emailAddress": "Buyer.TechnicalContact@buyer.mef.com",
          "name": "Buyer Technical Contact",
          "number": "+12-345-678-90",
          "role": "buyerTechnicalContact"
        }
      ]
    }
  ]
}
```


6.1.8. Use case 1c: Product Order Item to Disconnect Existing Product

The example below represents a single Product Order request for deletion (`action` equals `delete`) of an existing Access E-Line product.

```
{
  <<ProductOrder attributes...>>
  "productOrderItem": [
    {
      "id": "item-001",
      "action": "delete",
      "requestedCompletionDate": "2022-06-19T20:59:28.299Z",
      "product": {
        "id": "01494079-6c79-4a25-83f7-48284196d44d"
      }
    }
  ]
}
```

The following requirements apply to `productOrderItem` when `action` is `delete`:

[R45] In the `delete` request the Buyer **MUST** provide the `productOrderItem.product.id` attribute. [MEF57.2 R58]

[R46] In the `delete` request the Buyer **MUST NOT** provide any of the following attributes: [MEF57.2 R59]

- `productOrderItem.productOfferingQualificationItem`
- `productOrderItem.requestedItemTerm`
- `productOrderItem.relatedContactInformation`
- `productOrderItem.product.*` - product attributes other than `id`

6.1.9. Product Order State Machine

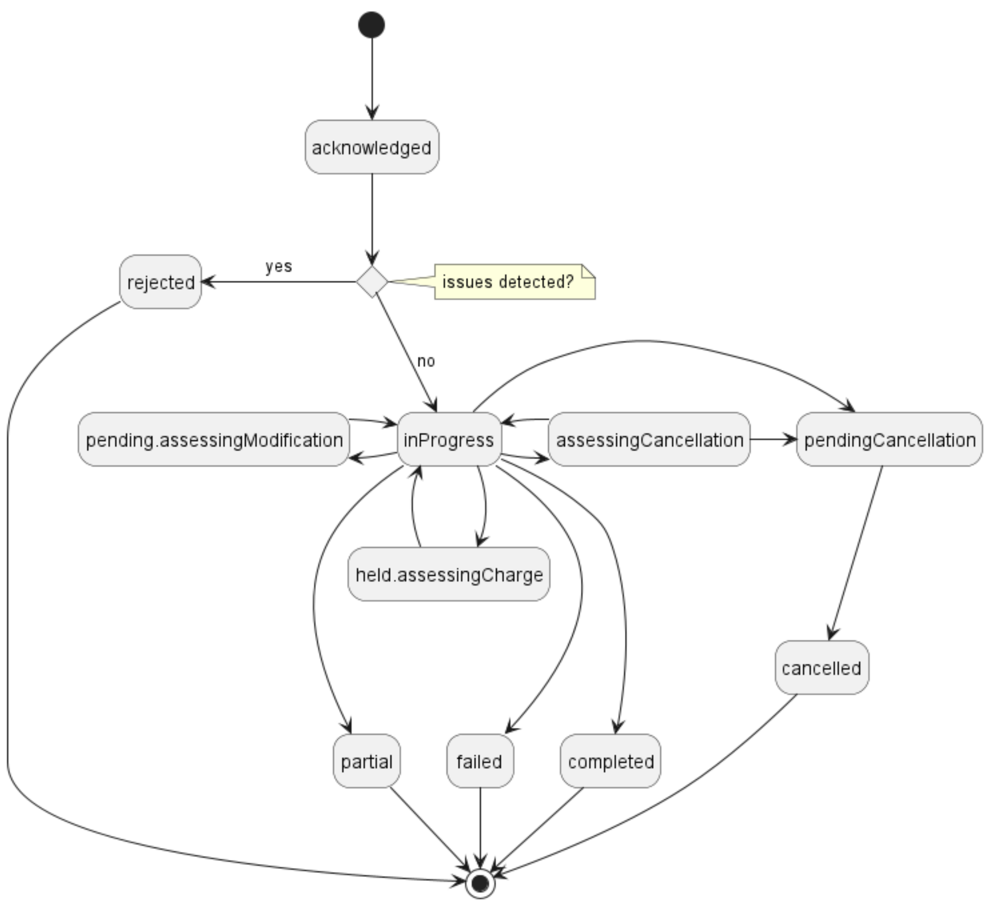

Figure 13. Product Order State Machine

Figure 13 presents the state machine for the Product Order. After receiving the request, the Seller performs basic checks of the message. If any problem is found an Error response is provided. If the validation passes a response is provided with `ProductOrder` and all `ProductOrderItems` in `acknowledged` state. Before moving the order to the `inProgress` state, the Buyer performs all the remaining business and time-consuming validations. At this point, an Error response cannot be provided anymore so the order moves to a `rejected` state if some issues are found. The `productOrderItem.terminationError` acts as a placeholder to provide a detailed description of what caused the problem.

Table 8 presents the mapping between the API `state` names (aligned with TMF) and the MEF 57.2 naming, together with the states' description.

state	MEF 57.2 name	Description
`acknowledged`	ACKNOWLEDGED	A Product Order has been received by the Seller and has passed basic validation. A `productOrder.id` is assigned in the `acknowledged` state and a response is returned to the Buyer. The Product Order remains in the `acknowledged` state while validations of Product Order and Product Order Item(s) attributes as applicable is completed. If the Product Order and Product Order Item attributes are validated the Product Order moves to the `inProgress` state. If not validated, the Product Order moves to the `rejected` state.

state	MEF 57.2 name	Description
`assessingCancellation`	ASSESSING_CANCELLATION	A Cancel Product Order request has been received by the Seller. The Product Order is being assessed to determine if the Product Order can be cancelled. If there are charges associated with cancelling the Product Order, these are communicated to the Buyer using the Charge process. The Product Order remains in the `assessingCancellation` state until any relevant Charge is completed or withdrawn by the Seller. Once the Buyer's request has been validated and any associated Charges completed, the Product Order moves to the `pendingCancellation` state. If the request is not validated or if any associated Charges are not completed, the Product Order moves to the `inProgress` state and the Product Order is not cancelled.
`held.assessingCharge`	ASSESSING_CHARGE	A Charge has been initiated by the Seller that is not the result of a Modify Product Order Item Requested Delivery Date or Cancel Product Order request and the Seller is awaiting a Buyer response to the Charge. If a blocking or non-blocking charge is accepted by the Buyer, the Product Order moves to `inProgress`. If a non-blocking charge is declined by the Buyer, the Product Order moves to `inProgress`. If a blocking charge is declined by the Buyer and there are no unrelated Product Order Items in the Product Order, the Product Order moves to the `inProgress` and then to the `failed` state. If a blocking charge is declined by the Buyer and there are unrelated Product Order Items in the Product Order, the Product Order moves to the `inProgress` state.

state	MEF 57.2 name	Description
`pending.assessingModification`	ASSESSING_MODIFICATION	A request has been made by the Buyer to modify either the `expediteIndicator` or the `requestedCompletionDate` of a Product Order Item. The Product Order Item is currently being assessed to determine whether the Modify Product Order Item Requested Delivery Date is valid. If there is a charge associated with the Modify Product Order Item Requested Delivery Date, the Product Order remains in the `pending.assessingModification` state until the Charge is completed or withdrawn by the Seller. Once the Buyer's request has been validated and any associated Charges completed, the Product Order returns to the `inProgress` state.
`cancelled`	CANCELLED	The Product Order has been successfully cancelled. This is a terminal state.
`pendingCancellation`	CANCELLING	The Buyer's Cancel Request has been assessed and it has been determined that it is feasible to proceed with the cancellation. This state can also result from a Seller cancelling the Product Order within their systems without a request from the Buyer.
`completed`	COMPLETED	The Product Order has completed fulfillment. This is a terminal state
`failed`	FAILED	All Product Order Items have failed which results in the entire Product Order failing. This is a terminal state.
`inProgress`	IN_PROGRESS	The Product Order has been successfully validated, and fulfillment has started.
`partial`	PARTIAL	At least one Product Order Item is `failed` or `rejected`, and fulfillment of at least one Product Order Item has been successful. This is a terminal state

state	MEF 57.2 name	Description
`rejected`	REJECTED	A Product Order was submitted, and it has failed at least one of the validation checks the Seller performs after it reached the `acknowledged` state

Table 8. Product Order states

6.1.10 Product Order Item State Machine

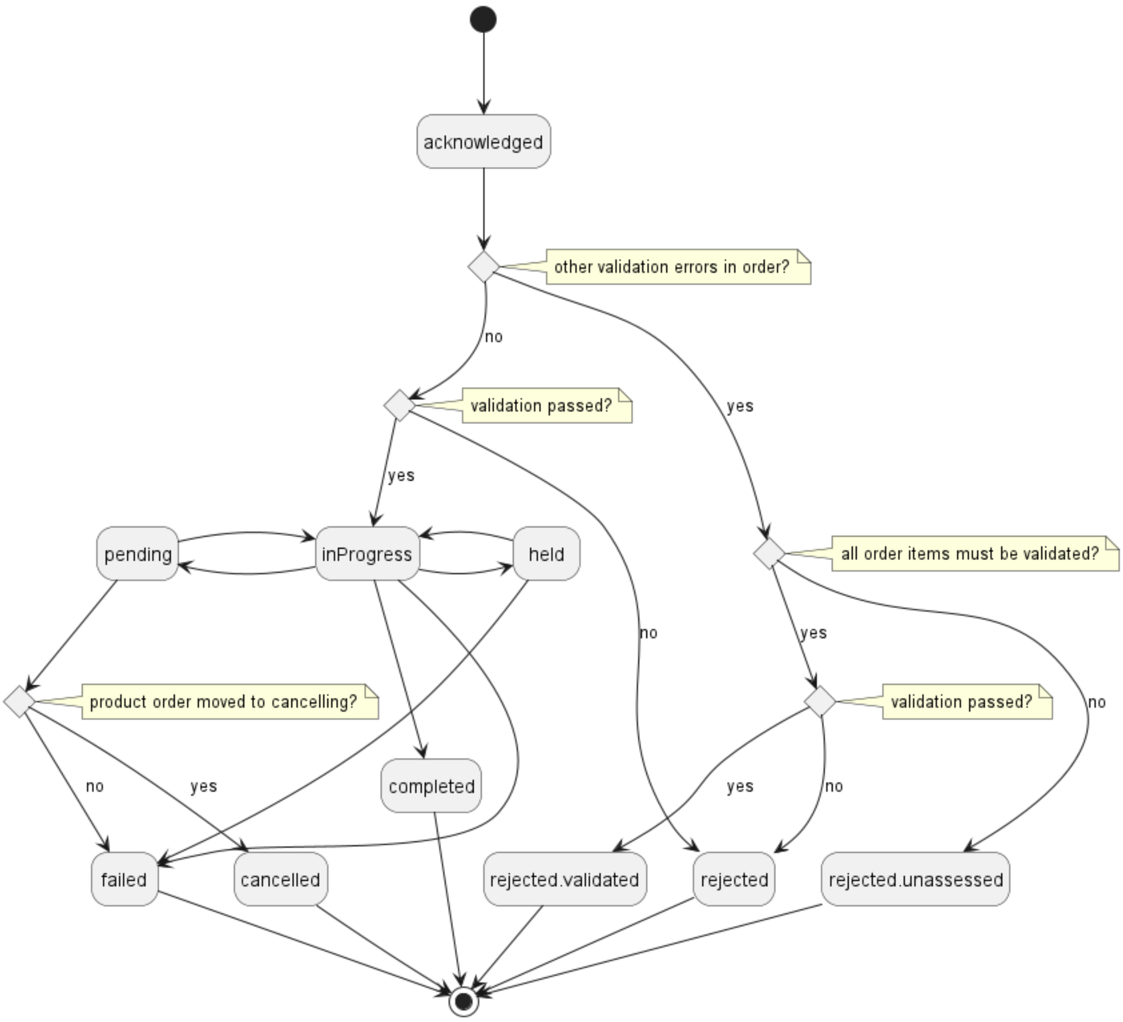

Figure 14. Product Order Item State Machine

Table 9 presents the mapping between the API `state` names (aligned with TMF) and the MEF 57.2 naming, together with the corresponding descriptions.

state	MEF 57.2 name	Description
`acknowledged`	ACKNOWLEDGED	A Product Order Item has been received and has passed basic business validations. From the `acknowledged` state the Product Order Item is further validated and depending on the results of the validation and if other Product Order Items in the Product Order are also validated the Product Order Item moves to `inProgress`, `rejected.validated`, or `rejected.unassessed`.

state	MEF 57.2 name	Description
`cancelled`	CANCELLED	The Product Order has moved to the `pendingCancellation` state. All Product Order Items move to `cancelled`.
`completed`	COMPLETED	The Product Order Item has completed provisioning. This is an end state
`failed`	FAILED	The fulfillment of a Product Order Item has failed. A Product Order Item may fail because the Buyer declined a Blocking charge identified via the Charge, the Buyer failed to respond to a Charge Item included in a Charge, or the Seller is unable to fulfill the Product Order Item. A Product Order Item moving to `failed` state results in the Product Order State being `failed` or `partial`. This is a terminal state.
`held`	HELD	The Product Order Item cannot be progressed due to Charge the Seller awaiting a response from the Buyer on a Charge. The Seller stops work on the Product Order Item until the Charge has completed. Upon acceptance by the Buyer of all Blocking charges, the Product Order Item returns to `inProgress` state If the Buyer rejects a Blocking charge, the Product Order Item moves to the `failed` state.
`inProgress`	IN_PROGRESS	The Product Order Item has been successfully validated and fulfillment has started. If the Seller's system links validation between Product Order Items in a Product Order, a Product Order Item in this state also indicates that the other Product Order Items passed validation.
`pending`	PENDING	The Product Order Item cannot be progressed due to the Seller assessing a Cancel Product Order or Modify Product Order Item Requested Delivery Date request. The Seller stops work on the Product Order Item until either the Cancel Product Order has been accepted and the Product Order state moves to `pendingCancellation` and the Product Order Item state moves to `cancelled`, the Cancel Product Order has been rejected and the Product Order Item State moves to `inProgress`, the Modify Product Order Item Requested Delivery Date has been accepted and the Product Order Item State moves to `inProgress`, or the Modify Product Order Item Requested Delivery Date moves to `done.declined` and the Product Order Item state moves to `inProgress` with original delivery dates.

state	MEF 57.2 name	Description
rejected	REJECTED	A Product Order Item was submitted, and it has failed at least one validation checks the Seller performs during the acknowledged state. Other Product Order Items may continue to be processed.
rejected.unassessed	UNASSESSED	A Product Order was submitted and all validation checks the Seller performs during the acknowledged state have not been completed, but another Product Order Item in the Product Order has moved to the rejected state.
rejected.validated	VALIDATED	A Product Order was submitted, and it has passed all validation checks the Seller performs during the acknowledged state, but another Product Order Item in the Product Order has moved to the rejected state

Table 9. Product Order Item states

6.1.11. Requirements for Product Order and Product Order Item Lifecycle

Requirements below are applied to a Product Order processing lifecycle - after providing an initial response where the Product Order was `acknowledged`. It assumes a Seller's response to a GET by `id` request.

[R47] If the Product Order `state` in the Seller's response is `completed`, the response **MUST** contain the `completionDate` attribute. [MEF57.2 R21]

[R48] If the Product Order Item `state` in the Seller's response is `inProgress`, the `expectedCompletionDate` attribute **MUST** be provided. [MEF57.2 R40]

[R49] If the Product Order Item `state` in the Seller's response is `cancelled`, the `expectedCompletionDate` attribute **MUST NOT** be provided. [MEF57.2 R42]

[R50] If the Product Order Item `state` in the Seller's response is `completed`, the response **MUST** contain the `completionDate` attribute. [MEF57.2 R41]

[R51] If the Product Order Item `state` in the Seller's response is not `completed`, the response **MUST NOT** contain the `completionDate` attribute. [MEF57.2 R52]

[R52] If the Seller revises the `expectedCompletionDate` for any Product Order Item, they **MUST** include a `note` that indicates that the date has been revised and the reason for the revision. [MEF57.2 R53]

6.1.12. Providing the place information

When required by product specification, the Buyer must point to the place where the Product is to be provided. This is done with the use of the `ProductOrderItem's` attribute: `product.place` of type `RelatedPlaceRefOrQueryWithSubUnit`, which is presented in Figure 15.

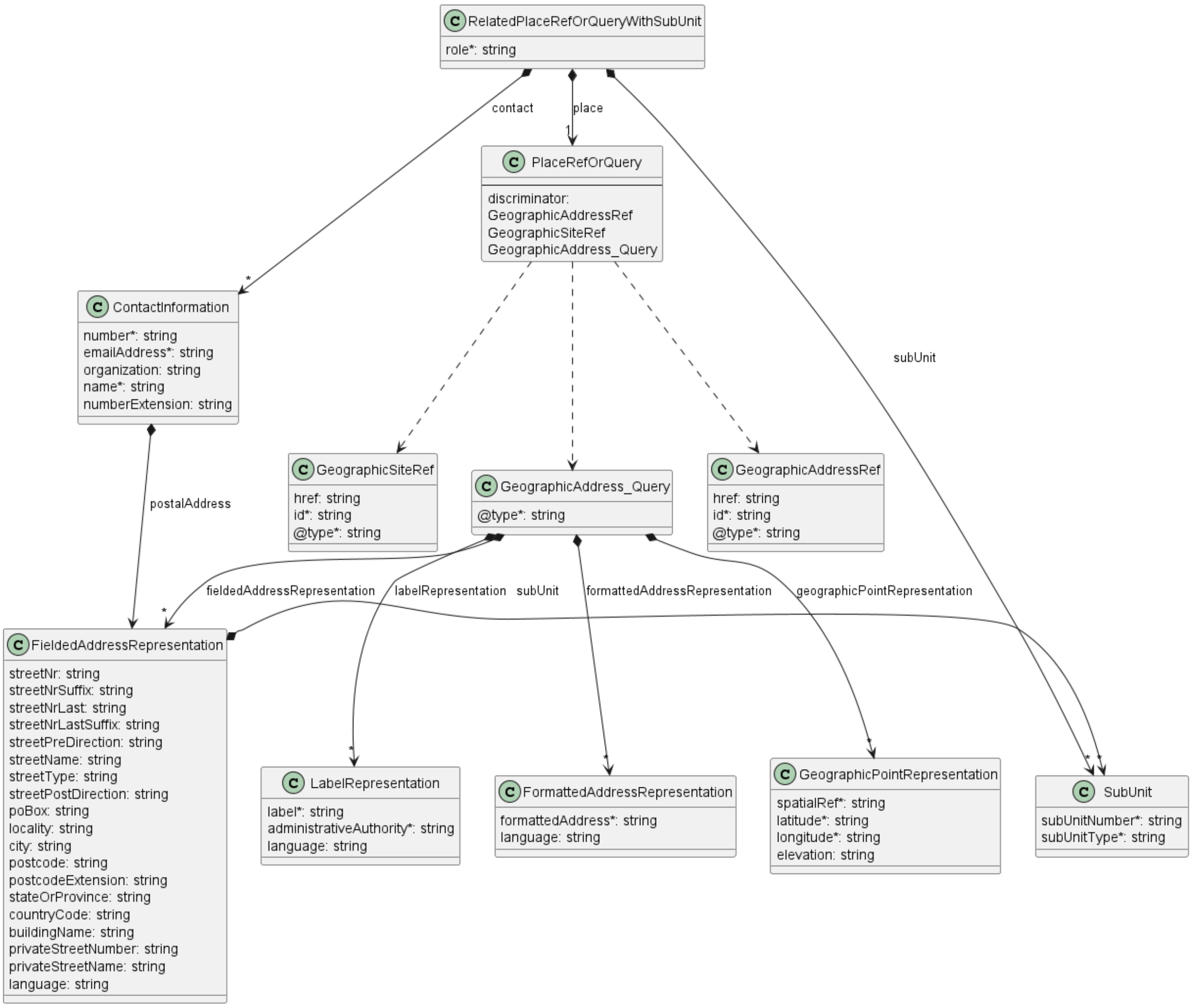

Figure 15. Data model - referring to a place

The `role` defines the function that the place plays for a given Product. The name of the role to be provided is strictly defined by the product specification. Usually, it is `INSTALL_LOCATION`.

`contact` provides additional information about the person to contact to get access to this place in case such access is required to complete the evaluation of this Product Order Item.

`place` is where the actual place is pointed. The attribute is of type `PlaceRefOrQuery` which is an abstract class that can be of one of three types: `GeographicAddressRef`, `GeographicSiteRef`, or `GeographicAddress_Query`. The first two are simple identifiers to reference a `GeographicAddress` or `GeographicSite` respectively. The Buyer usually first validates the `GeographicAddress` and gets its identifier from the Seller and then optionally retrieves `GeographicSite` information for that address. In the unlikely case that the Seller does not provide the Address Validation API and the Buyer is not able to obtain the address identifier in any other way, the `GeographicAddressQuery` type might be used. It contains lists of Geographic Address Representations to provide the address information by value. There are four types of Geographic Address Representations:

- `FieldedAddressRepresentation`
- `FormattedAddressRepresentation`
- `LabelRepresentation`
- `GeographicPointRepresentation`

The Buyer may use one or more of these representations to describe a single desired place. The Buyer must provide sufficient clarity that allows the Seller to match to precisely one place. For this reason, the success rate of Product Order inquiries is significantly better when identifiers are used.

In case when there is no desired `GeographicSite` object in the Seller's system, or `GeographicAddress` precision is not sufficient, the Buyer may use the `subUnit` attribute to provide more detailed information about the precise location of the installation. This information may be used by the Seller to create an instance of a `GeographicSite` with the same `subUnit` attribute value.

The `GeographicAddress` model together with its above-mentioned representations and respective requirements are defined by MEF 121.1 (chapter 5.3). That standard is the owner of those definitions. This API specification contains a model of `GeographicAddress` but does not define it. Any further changes of these types will update the API specification, but will not be reflected in this document.

The mandatory `@type` attribute of `GeographicSiteRef`, `GeographicAddressRef` and `GeographicAddress_Query` is used as a discriminator to unambiguously identify the intended type when using in the context of the `oneOf` section of `PlaceRefOrQuery` type.

[D1] For all Addresses that have not been validated, a Buyer **SHOULD** initiate a Validate Address request, to get the Seller's full details associated with this Address prior to submitting a Create Product Order request.

[D2] For all Addresses that have been validated, the Buyer **SHOULD** use the Seller's Address Identifier to describe the location when submitting a Create Product Order request and when a Service Site does not exist.

[D3] After validating an Address, a Buyer **SHOULD** initiate a Retrieve Service Site List request and obtain the Seller's Service Site Identifiers for all Service Sites at the validated Address prior to submitting a Create Product Order request.

[D4] Once the Buyer has obtained the Service Site Identifier of the Service Site, then for all subsequent operations related to this Service Site, the Buyer **SHOULD** use the Seller's Service Site Identifier to reference this Service Site when submitting a Create Product Order request.

6.2. Use Case 2: Update Product Order

The update is realized with the use of the `PATCH /productOrder/{{id}}` operation. For that purpose specialized types `ProductOrder_Update` and `ProductOrderItem_Update` are provided. Their lists of attributes are limited to a subset that includes only the Buyer settable and not Product Order processing affecting attributes. If the Buyer wishes to update any attribute not listed in the abovementioned types (e.g. product-related) then the pending Product Order must be canceled and a new one must be resubmitted.

The PATCH usage recommendation follows TMF 622 json/merge (https://tools.ietf.org/html/rfc7386).

Figure 16 presents the model used in the PATCH request. The Seller responds with a `ProductOrder` type.

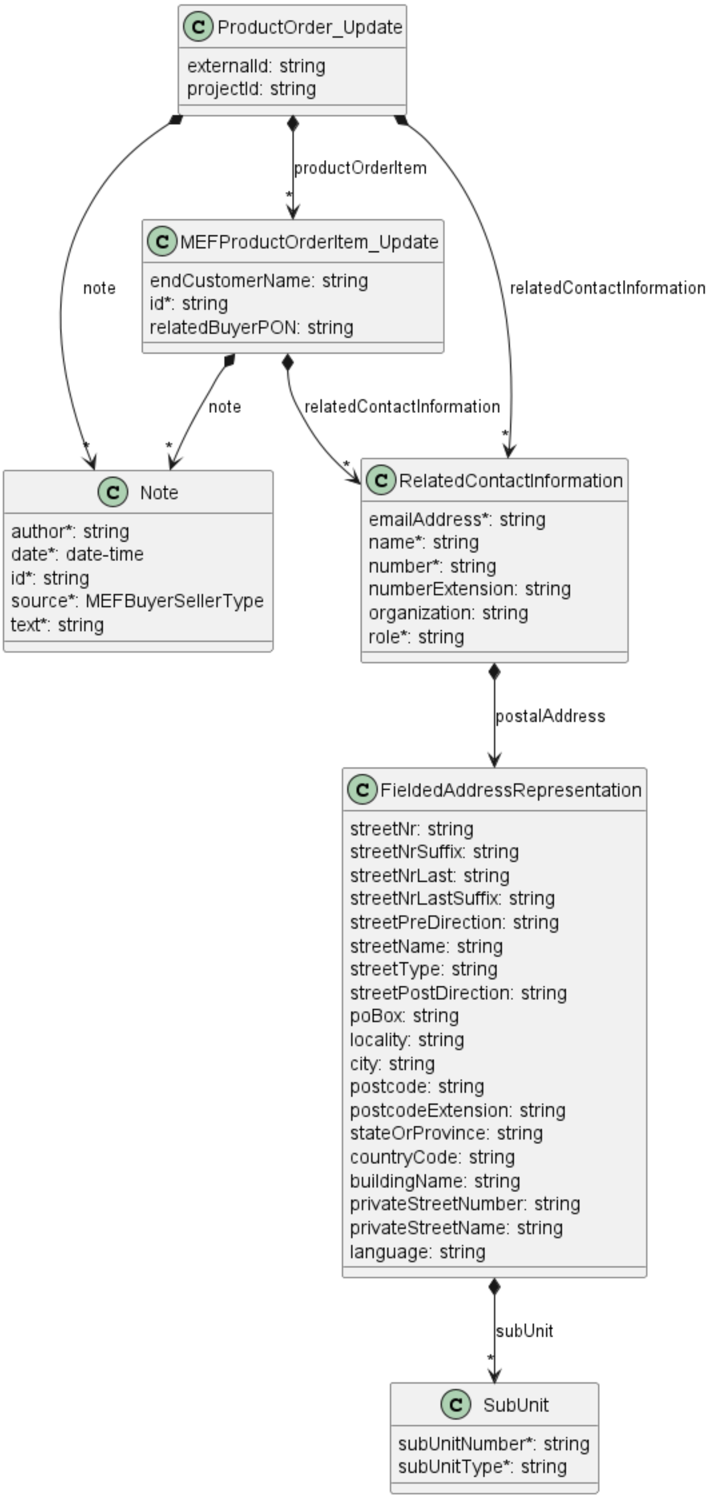

Figure 16. Patch request Model

The example below shows a request to change the Product Order Contact (`relatedContactInformation` with `role` set to `productOrderContact`), and the `endCustomerName` of the first Product Order Item.

```
{
  "relatedContactInformation": [
    { << updated contact >>
      "emailAddress": "Richard.example@buyer.mef.com",
      "name": "Richard Example",
      "number": "98-765-4321",
      "organization": "Buyer Example Co.",
      "role": "productOrderContact",
    },
    { << not changed >>
```

```
        "emailAddress": "kate.example@buyer.mef.com",
        "name": "Kate Example",
        "number": "12-345-67890",
        "organization": "Seller Example Co.",
        "role": "sellerContact"
      }
    ],
    "productOrderItem": [
      {
        "id": "item-001",
        "endCustomerName": "Updated End Customer Name"
      },
      {
        "id": "item-002"
      }
    ]
  }
```


[R53] A Buyer's Patch request **MUST** contain one or more of the `ProductOrder` updatable attributes. [MEF57.2 R62]

[R54] If a Buyer's Patch request contains a Product Order Item, it **MUST** provide one or more of the Product Order Item's updatable attributes (apart from `id`). [MEF57.2 R69]

If the Buyer wishes to update a Product Order Item:

[R55] A Buyer's Patch request **MUST** contain `productOrderItem.id` [MEF57.2 R68]

Note: The `productOrderItem.id` attribute cannot be updated. It is used only to refer to identify and items to be updated.

[R56] If a Buyer wants to update a Product Order Item, the Patch request **MUST** contain one or more of the `ProductOrderItem` updatable attributes. [MEF57.2 R69]

[R57] The Buyer **MUST** only be able to modify the Buyer-related contact information. [MEF57.2 R4]

The Buyer can update a Buyer-related contact by providing a full list of existing `relatedContactInformation` items, and updating the value of those with Buyer-related `roles`.

The Buyer can not update a Buyer-related note. New notes can only be appended to the existing list of `note` items.

[R58] The Buyer **MUST** add a `note` only with `source=note`. [MEF57.2 R11], [MEF57.2 R12]

[R59] When the Buyer is adding a Note, the `note` list **MUST** be appended with the new `note` item. [MEF57.2 R14]

6.3. Use Case 3: Retrieve List of Product Orders

The Buyer can retrieve a list of `ProductOrders` by using a `GET /productOrder` operation with desired filtering criteria.

[O10] The Buyer's request **MAY** contain none or more of the following attributes: [MEF57.2 O16], [MEF57.2 O17]

- `state`
- `externalId`
- `projectId`
- `orderDate.gt`
- `orderDate.lt`

- `completionDate.gt`
- `completionDate.lt`
- `cancellationDate.gt`
- `cancellationDate.lt`
- `itemRequestedCompletionDate.gt`
- `itemRequestedCompletionDate.lt`
- `itemExpectedCompletionDate.gt`
- `itemExpectedCompletionDate.lt`

The Buyer may also ask for pagination of the response when the number of results is too big. The following query attributes related to pagination can be provided:

- `limit` - number of expected list items
- `offset` - offset of the first element in the result list.

```
https://serverRoot/mefApi/sonata/productOrderingManagement/v11/productOrder?
state=completed&projectId=myProject&limit=10&offset=0
```

The example above shows a Buyer's request to get all `ProductOrders` that are in the `completed` state and are part of `myProject`. Additionally, the Buyer asks only for a first (`offset=0`) pack of 10 results (`limit=10`) to be returned. The correct response (HTTP code `200`) in the response body contains a list of `ProductOrder_Find` objects matching the criteria. To get more details (e.g. the item level information), the Buyer has to query a specific `ProductOrder` by `id`.

The Seller returns a list of elements that comply with the requested `limit`. If the requested `limit` is higher than the supported list size then the smaller list of results is returned. In that case, the size of the result is returned in the header attribute `X-Result-Count`. The Seller can indicate that there are additional results available using:

- `X-Total-Count` header attribute with the total number of available results
- `X-Pagination-Throttled` header set to `true`

[D5] The Seller **SHOULD** support the pagination mechanism.

[CR7]<[D5] Seller **MUST** use either `X-Total-Count` or `X-Pagination-Throttled` to indicate that the page was truncated and additional results are available.

[R60] The Seller **MUST** put the following attributes (if set) into the `ProductOrder_Find` object in the response: [MEF57.2 R99]

- `id`
- `cancellationDate`
- `completionDate`
- `externalId`
- `orderDate`
- `projectId`
- `state`

[R61] In case no items matching the criteria are found, the Seller **MUST** return a valid response with an empty list. [MEF57.2 R101]

[R62] In case of too many matching items are found (the definition of 'too many' is up to Seller's discretion), the Seller **MUST** return an `Error422` with `code=tooManyRecords`.

In that case, the Buyer can change the filter criteria and/or repeat the query with pagination in order to avoid receiving the tooManyRecords error.

6.4. Use Case 4: Retrieve Product Order by Product Order Identifier

The Buyer can get detailed information about the Product Order from the Seller by using a GET /productOrder/{{id}} operation. In case id does not allow to find a ProductOrder in Seller's system, an error response Error404 must be returned. The payload returned in the response includes all attributes the Buyer has provided while sending a Product Order create request. The attributes provided by the Seller depend on the status of the ProductOrder and may require some time to be set.

[R63] Once the product identifier (productOrder.productOrderItem.product.id) is assigned, it **MUST** be provided in the Seller's response.

[R64] The Seller's response **MUST** comply with the states and attributes detailed in Table 10 and Table 11. [MEF57.2 R103]

Please note that for readability purposes following tables do not show attributes specified by the Buyer that must be only echoed back ("E") by the Seller without any change. Attributes required to be provided by the Seller are shown by an "R", Required if Populated by the Seller shown by a "PR", or Optional to be provided by the Seller or the Buyer shown by an "O". If there are two values in a cell (e.g. E / PR) the first one relates to values set by the Buyer, the second for the Values set by the Seller.

	acknowledged	assessingCancellation	held.assessingCharge	cancelle
id	R	R	R	R
orderDate	R	R	R	R
state	R	R	R	R
stateChange	R	R	R	R
relatedContactInformation	R	R	R	R
cancellationReason				E / R
cancellationDate				R
cancellationCharge				PR
completionDate				R
note	E / PR	E / PR	E / PR	E / PR

Table 10. Seller Response Product Order Attributes Based on Product Order State

	acknowledged	cancelled	completed	failed	held	inProgress	penc
note	E / PR	E / PR	E / PR	E / PR	E / PR	E / PR	E / P
expediteAcceptedIndicator	PR	PR	PR	PR	PR	PR	PR
charge		PR	PR	PR	PR	PR	PR
state	R	R	R	R	R	R	R
stateChange	R	R	R	R	R	R	R
expectedCompletionDate		R	R	R	R	R	R

	acknowledged	cancelled	completed	failed	held	inProgress	penc
completionDate			R				
relatedContactInformation	E / R	E / R	E / R	E / R	E / R	E / R	E / R
itemTerm		PR	PR	PR	PR	PR	PR
terminationError				R			

Table 11. Seller Response Product Order Item Attributes Based on Product Order Item State

6.5. Use case 5: Modify Product Order Item Requested Delivery Date

The Product Order PATCH operation is limited to a subset of attributes that includes only the Buyer settable and not Product Order processing affecting ones (Section 6.2). Modification of `requestedCompletionDate` or `expediteIndicator` may bring a significant processing and business impact hence it is extracted to a separate dedicated process.

The Buyer issues the request by using a dedicated endpoint: `POST /modifyProductOrderItemRequestedDeliveryDate` and providing a `MEFModifyProductOrderItemRequestedDeliveryDate_Create` in the request body.

There are two functions supported by the Modify Product Order Item Requested Delivery Date request:

- changing the `expediteIndicator`
- changing the `requestedCompletionDate` of the Product Order Item.

Figure 17 presents entity types that take part in the Modify Product Order Item Requested Delivery Date use cases:

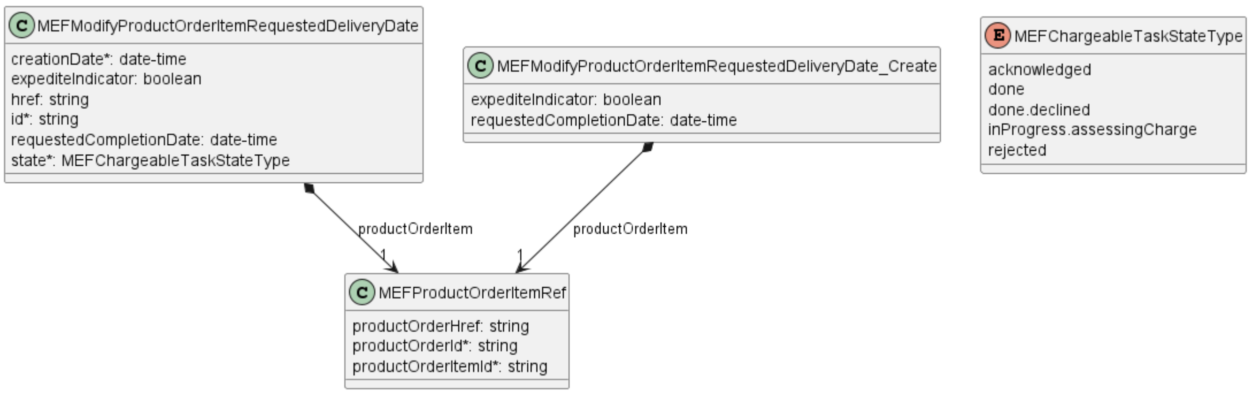

Figure 17. Modify Product Order Item Requested Delivery Date Model

The state transition and detailed description are presented in Figure 18 and Table 12:

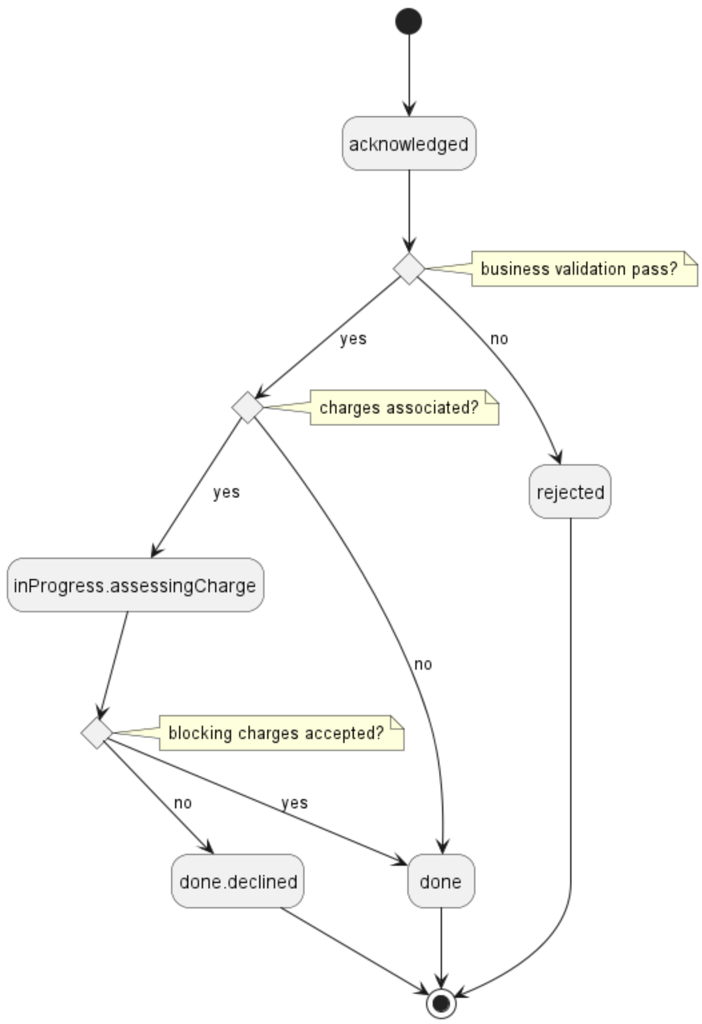

Figure 18. Modify Product Order Item Requested Delivery Date State Machine

Name	MEF 57.2 Name	Description
`inProgress.assessingCharge`	ASSESSING_CHARGE	The Modify Product Order Item Requested Delivery Date request results in a Charge being initiated by the Seller. The Modify Product Order Item Requested Delivery Date remains in this state until the Charge is completed or withdrawn by the Seller. All charges within a Charge that was initiated due to a Modify Product Order Item Requested Delivery Date are considered Blocking charges. If any charge is not accepted by the Buyer, the Modify Product Order Item Requested Delivery Date moves from the `inProgress.assessingCharge` state to the `done.declined` state.

Name	MEF 57.2 Name	Description
acknowledged	ACKNOWLEDGED	A Modify Product Order Item Requested Delivery Date request has been received and has passed basic validation. The Modify Product Order Item Requested Delivery Date Identifier is assigned in the acknowledged state. Validation of Modify Product Order Item Requested Delivery Date attributes as applicable is completed in the acknowledged state.
done	COMPLETED	A Modify Product Order Item Requested Delivery Date request has been received, passed all validations, if a Charge is associated all Charge Items have been accepted by the Buyer, and the Product Order Item Completion Date has been updated as requested.
done.declined	DECLINED	Blocking charges associated with a Modify Product Order Item Requested Delivery Date have been declined by the Buyer. No updates are made to the Product Order Item.
rejected	REJECTED	A Modify Product Order Item Requested Delivery Date request was submitted by the Buyer, and it has failed any validation checks the Seller performs during the acknowledged state. No updates are made to the referenced Product Order Item.

Table 12. Modify Product Order Item Requested Delivery Date States

Example of a Buyer's request (modifyProductOrderItemRequestedDeliveryDate_Create):


```
{
  "expediteIndicator": true,
  "requestedCompletionDate": "2021-05-25T21:32:28.826Z",
  "productOrderItem": {
    "productOrderId": "00000000-1111-2222-3333-000000000123",
    "productOrderItemId": "item-001"
  }
}
```


Example of a Seller's response (modifyProductOrderItemRequestedDeliveryDate):


```
{
  "id": "00000000-8888-0000-0000-000000000001",
  "expediteIndicator": true,
  "requestedCompletionDate": "2021-05-25T21:32:28.826Z",
  "productOrderItem": {
    "productOrderId": "00000000-1111-2222-3333-000000000123",
```

```
      "productOrderItemId": "item-001"
    },
    "state": "acknowledged"
  }
```

Below you can find a flow of this use case when there are no additional charges identified. A case with additional charges handling is presented in Section 6.11.2

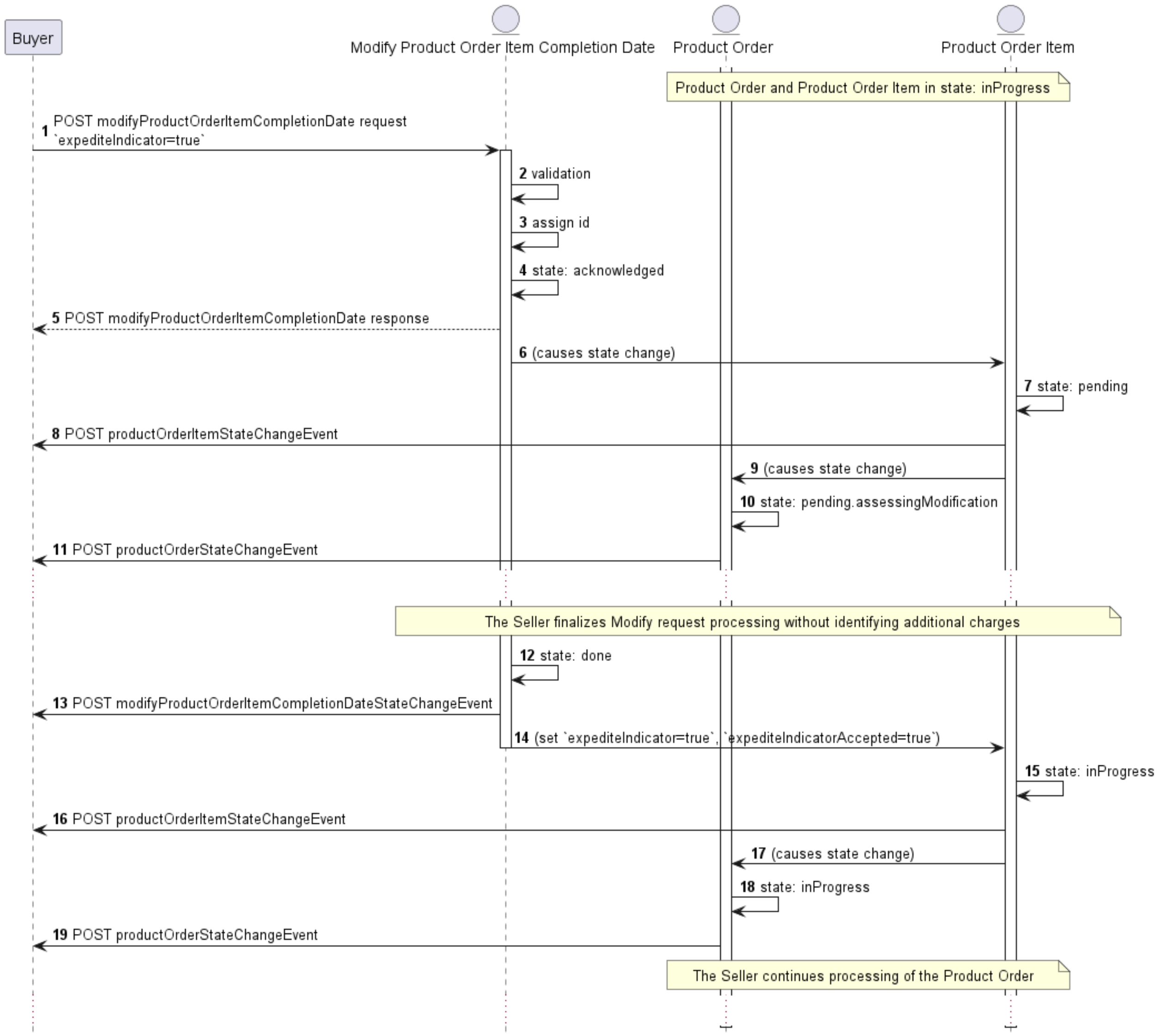

Figure 19. Modify Product Order Item Requested Delivery Date Flow

- The Buyer sends a `modifyProductOrderItemRequestedDeliveryDate` request with the `expediteIndicator` set to `true` (and/or `requestedCompletionDate`) set to new value **(1)**.
- The Seller validates the request **(2)**.
- The Seller initiates the Modify Date process, assigns a unique `id` **(3)**, then sets the `modifyProductOrderItemRequestedDeliveryDate.state`to`acknowledged`**(4)**, changes the `state` of the referenced `ProductOrderItem` to `pending` **(6,7)**, and the `state` of `ProductOrder` to `pending.assessingModification` **(9,10)**.
- The Seller notifies the Buyer of any charges resulting from the request while the `modifyProductOrderItemRequestedDeliveryDate` is in the `acknowledged` state (The details of the Charge process are not present here for clarity. They are provided in Section 6.11.2 for details).
- The Seller accepts the requested change. The `modifyProductOrderItemRequestedDeliveryDate` is set to `done` **(12)** and the Seller updates the

expediteIndicator and the expediteAcceptedIndicator (and/or requestedCompletionDate) **(14)**.

- The Seller sets the referenced ProductOrderItem.state back to inProgress **(15)** and ProductOrder.state to inProgress **(17,18)**.
- The Seller continues the work to fulfill the Product Order.

Note: The are places where the sequence of the state changes is performed "at the same time" (e.g. **4,7,10**, **12,14,15,18**). The diagrams in this document show additional "causes ..." steps for explanation purposes. The actual order of those state transitions is not mandated and may depend on Seller's implementation.

6.5.1. Use case 5a: Modify Expedite Indicator

In this case, the Buyer requests to expedite a Product Order.

[R65] The Buyer's sent modifyProductOrderItemRequestedDeliveryDate_Create **MUST** contain the following attributes: [MEF57.2 R74]

- productOrderItem
- expediteIndicator

The Buyer sets the expediteIndicator to true if they want the Seller to fulfill the Product Order Item in a shorter period than the installationInterval (provided in product offering qualification and/or quote step).

[O11] The Buyer's sent modifyProductOrderItemRequestedDeliveryDate_Create **MAY** contain the requestedCompletionDate. [MEF57.2 O13]

If the Buyer sets the expediteIndicator to true and sets a requestedCompletionDate then they are requesting that the Product Order Item be fulfilled in a shorter time period than the installationInterval and have provided a date they would like it fulfilled by. The requestedCompletionDate must indicate a shorter time period than the installationInterval. The Seller may try to honor the date or may ignore it.

[R66] The Seller's response **MUST** specify the id, state, and creationDate attributes. [MEF57.2 R75]

[R67] The Seller's response **MUST** echo back all attributes and values in the Buyer's request. [MEF57.2 R76]

6.5.2. Use case 5b: Modify Product Order Item Requested Delivery Date

In this case, the Buyer requests to change the expectedCompletionDate of a Product Order Item.

[R68] The Buyer's sent modifyProductOrderItemRequestedDeliveryDate_Create **MUST** contain the following attributes: [MEF57.2 R78]

- productOrderItem
- requestedCompletionDate

If the Buyer wants to push out or delay the fulfillment of the Product Order Item, they set a new requestedCompletionDate and the expediteIndicator to false (or just not specify it all as the default value for expediteIndicator is false).

[R69] The Seller's response **MUST** specify the id, state, and creationDate attributes. [MEF57.2 R79]

[R70] The Seller's response **MUST** echo back all attributes and values in the Buyer's request. [MEF57.2 R80]

6.6. Use case 6: Retrieve Modify Product Order Item Requested Delivery Date List

The Buyer can retrieve a list of `modifyProductOrderItemRequestedDeliveryDate` by using a `GET /modifyProductOrderItemRequestedDeliveryDate` operation with desired filtering criteria.

[O12] The Buyer's request **MAY** contain none or any of the following attributes: [MEF57.2 O18]

- `state`
- `expediteIndicator`
- `productOrderId`
- `requestedCompletionDate.gt`
- `requestedCompletionDate.lt`
- `creationDate.gt`
- `creationDate.lt`

The rules of using pagination and an example request are provided in section 6.3. Please refer to it as the rules also apply to this case.

[R71] The Seller **MUST** put the following attributes (if set) into the response: [MEF57.2 R104]

- `id`
- `creationDate`
- `expediteIndicator`
- `productOrderItem`
- `requestedCompletionDate`
- `state`

[R72] In case no items matching the criteria are found, the Seller **MUST** return a valid response with an empty list. [MEF57.2 R105]

6.7. Use case 7: Retrieve Modify Product Order Item Requested Delivery Date by Modify Product Order Item Requested Delivery Date Identifier

The Buyer can get detailed information about the Modify Product Order Item Requested Delivery Date from the Seller by using a `GET /modifyProductOrderItemRequestedDeliveryDate/{{id}}` operation.

[R73] The Seller **MUST** put the following attributes (if set) into the response: [MEF57.2 R108]

- `id`
- `creationDate`
- `expediteIndicator`
- `productOrderItem`
- `requestedCompletionDate`
- `state`

[R74] In case `id` does not allow to find a `modifyProductOrderItemRequestedDeliveryDate` in Seller's Inventory, an error response `404` must be returned. [MEF57.2 R109]

6.8. Use case 8: Cancel Product Order

The Buyer may request to Cancel a Product Order by using `POST /cancelProductOrder` and providing a `CancelProductOrder_Create` in the request body.

The following Figures present the use case's model and flow diagrams.

Figure 20. Cancel Product Order Model

The state transition and detailed description are presented in Figure 21 and Table 13:

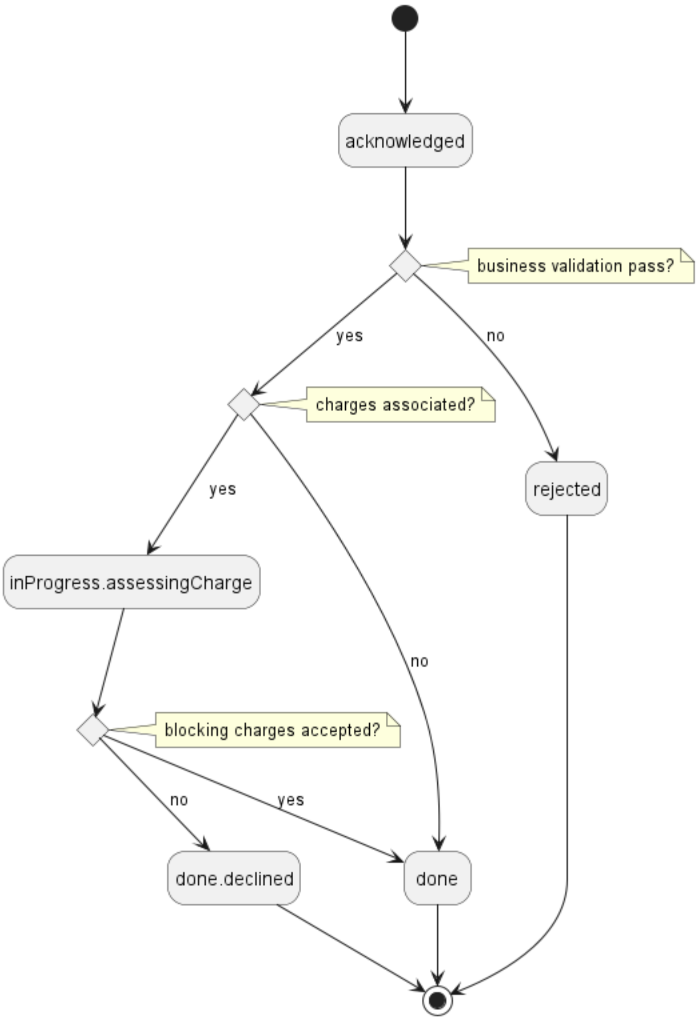

Figure 21. Cancel Product OrderState Machine

Name	MEF 57.2 Name	Description
`acknowledged`	ACKNOWLEDGED	A Cancel Request has been received and has passed basic validation. Seller `id` is assigned in the `acknowledged` state. Validation of Cancel attributes as applicable is completed in the `acknowledged` state.
`inProgress.assessingCharge`	ASSESSING_CHARGE	The Cancel Request results in a Charge being initiated by the Seller. The Cancel Request remains in this state until the Charge is completed or withdrawn by the Seller.
`done`	COMPLETED	A Cancel Request has been received, and passed all validations, if a Charge is associated all Charge Items have been accepted by the Buyer, and the Product Order has been canceled as requested.
`done.declined`	DECLINED	Blocking charges associated with a Cancel Product Order have been declined by the Buyer. No updates are made to the Product Order.

Name	MEF 57.2 Name	Description
`rejected`	REJECTED	A Cancel Request was submitted, and it has failed any validation checks the Seller performs during the `acknowledged` state e.g. the Product Order being in an incorrect state. No updates are made to the referenced Product Order.

Table 13. Cancel Product Order States

Example of a Buyer's request (`CancelProductOrder_Create`):


```
{
  "cancellationReasonType": "technical",
   "cancellationReason": "We have an equipment swap and the requirements will change. Will issue another Product
Order once done.",
  "relatedContactInformation": [
    {
      "emailAddress": "Cancel.ProductOrderContact@buyer.mef.com",
      "name": "Cancel Product Order Contact",
      "number": "+12-345-678-90",
      "organization": "Buyer",
      "role": "cancelProductOrderContact"
    }
  ],
  "productOrder": {
    "id": "00000000-1111-2222-3333-000000000123"
  }
}
```


[R75] A Buyer **MUST** have submitted the Product Order Request to be able to submit a Cancel Request on the Product Order. [MEF57.2 R93]

[R76] The Buyer's Cancel Product Order request **MUST** contain the following attributes: [MEF57.2 R94]

- `productOrder`
- `relatedContactInformation` (`role=cancelProductOrderContact`)

  Example of a Seller's response (`CancelProductOrder`):


```
{
  "id": "00000000-9999-0000-0000-000000000003",
  "state": "acknowledged",
  "cancellationReasonType": "technical",
   "cancellationReason": "We have an equipment swap and the requirements will change. Will issue another Product
Order once done.",
  "relatedContactInformation": [
    {
      "emailAddress": "Cancel.ProductOrderContact@buyer.mef.com",
      "name": "Cancel Product Order Contact",
      "number": "+12-345-678-90",
      "organization": "Buyer",
      "role": "cancelProductOrderContact"
    },
    {
      "emailAddress": "Seller.Contact@seller.mef.com",
      "name": "Seller Contact",
      "number": "+12-345-678-90",
      "organization": "Seller",
      "role": "sellerContact"
    }
  ],
```

```
    "productOrder": {
      "id": "00000000-1111-2222-3333-000000000123"
    }
  }
```

[R77] The Seller **MUST** echo back all Buyer specified attributes in the Buyer's Cancel Product Order request. [MEF57.2 R95]

[R78] The Seller **MUST** specify the following attributes in the response: [MEF57.2 R96]

- `id`
- `state`
- `relatedContactInformation` (add item with `role=cancelProductOrderSellerContact`)

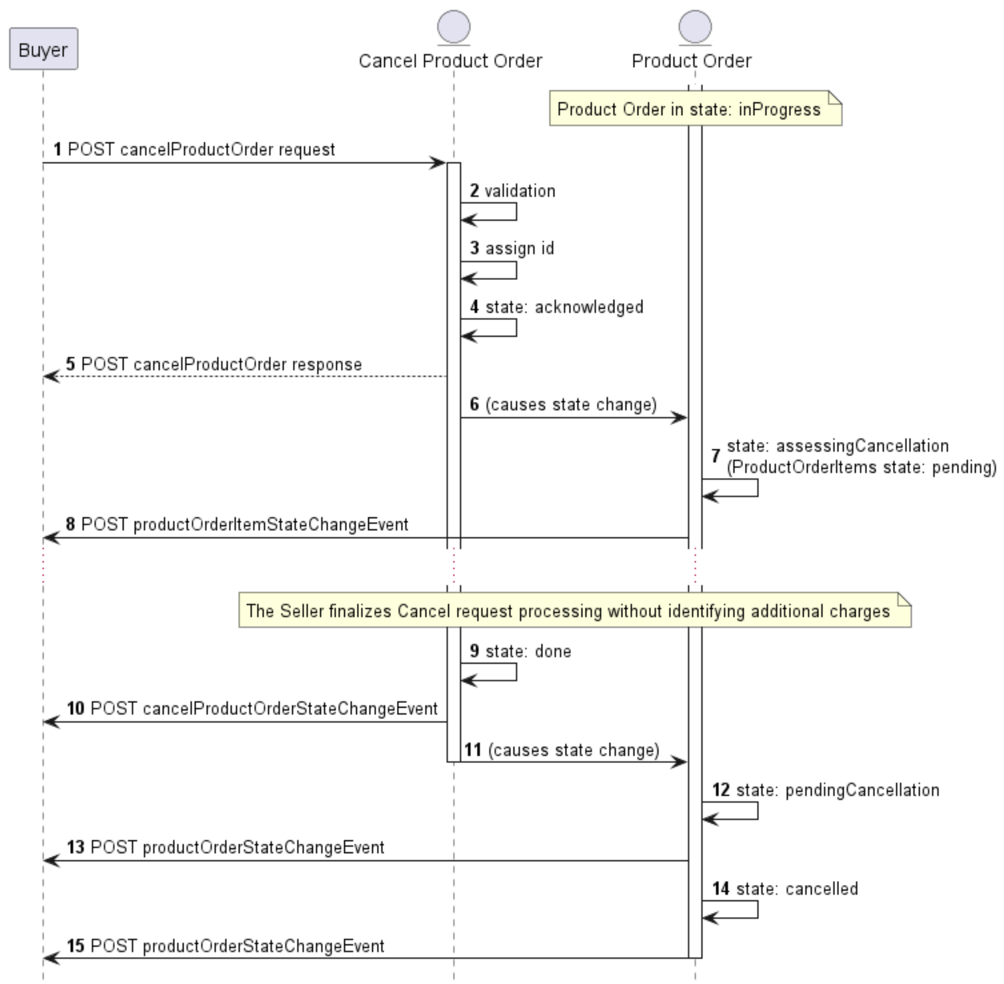

Figure 22. Cancel Product Order Flow

- The Buyer sends a Cancel Product Order request with `CancelProductOrder_Create` **(1)**.
- The Seller validates the request **(2)**.
- The Seller initiates the Cancel process, assigns a `CancelProductOrder.id` **(3)**, sets the `CancelProductOrder.state` to `acknowledged` **(4)**, and changes the referenced `ProductOrder.state` to `assessingCancellation` **(6,7)**.
- The Seller notifies the Buyer of any charges resulting from cancelling the referenced Product Order while the Cancel Request is in the `acknowledged` state (The details of the Charge process are not present here for clarity. They are provided in Section 6.11.3 for details).
- The Seller accepts the Cancel Request. The `CancelProductOrder.state` is set to `done` **(9)** and the referenced `ProductOrder.state` is set to `pendingCancellation` **(11,12)**.
- Once the Seller has completed the cancellation process, the referenced `ProductOrder.state` is changed to `cancelled` **(14)**.

6.9. Use case 9: Retrieve List of Cancel Product Orders

The Buyer can retrieve a list of `CancelProductOrder` by using a `GET /cancelProductOrder` operation with desired filtering criteria.

[O13] The Buyer's request **MAY** contain none or any of the following attributes: [MEF57.2 O19]

- `productOrderId`
- `state`
- `cancellationReasonType`

The rules of using pagination and an example request are provided in section 6.3. Please refer to it as the rules also apply to this case.

[R79] The Seller **MUST** put the following attributes (if set) into the response: [MEF57.2 R110]

- `id`
- `cancellationReasonType`
- `productOrder`
- `state`

[R80] In case no items matching the criteria are found, the Seller **MUST** return a valid response with an empty list. [MEF57.2 R111]

6.10. Use case 10: Retrieve Cancel Product Order by Cancel Product Order Identifier

The Buyer can get detailed information about the Cancel Product Order request from the Seller by using a `GET /cancelProductOrder/{{id}}` operation.

[R81] The Seller's response **MUST** echo back all attributes provided by the Buyer in the request and provide the following attributes (if set): [MEF57.2 R114]

- `id`
- `cancellationDeniedReason`
- `cancellationReason`
- `cancellationReasonType`
- `productOrder`
- `relatedContactInformation` (items with `role=cancelProductOrderSellerContact` and `role=cancelProductOrderContact`)
- `state`

6.11. Use case 11: Initiate Charge

When new or changes to existing charges are identified by the Seller during processing of a Product Order the Seller must communicate them to the Buyer and the Buyer must respond if they accept or reject each charge.

Within the Charge, the Seller indicates for each Charge Item, if the Charge Item is Blocking or non-blocking. After sending a `ChargeCreateEvent` the Seller puts the associated Product Order in `held.assessingCharge` and/or Product Order Item in `held` state and waits for the response. If the Buyer rejects a Blocking Charge, the Seller will cancel the processing of the related entity (depending on the sub-case - as described below).

The seller may identify Charges during:

- standard processing Product Order Item,

- processing of Buyer's Cancel Product Order,
- processing of Buyer's Modify Product Order Item Requested Delivery Date Request.

The variants are described as separate use cases and are explained in the next sections.

Figure 23 presents the model taking part in the use case. It is common for all sub-use cases:

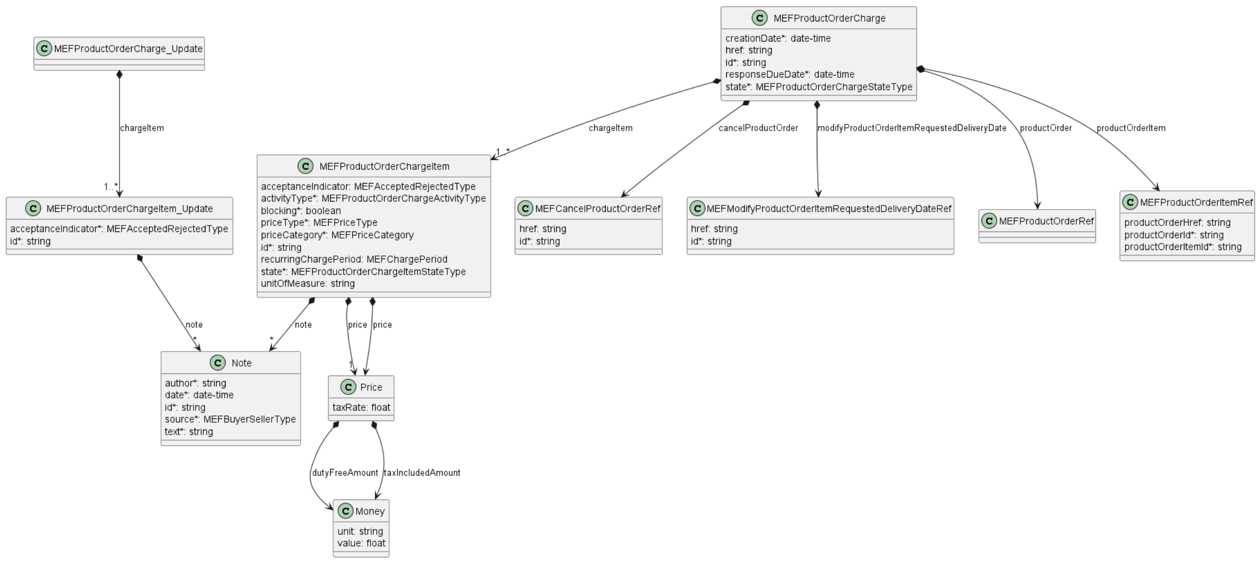

Figure 23. Charge Model

The Figures and Tables below present the Charge and Charge Item states.

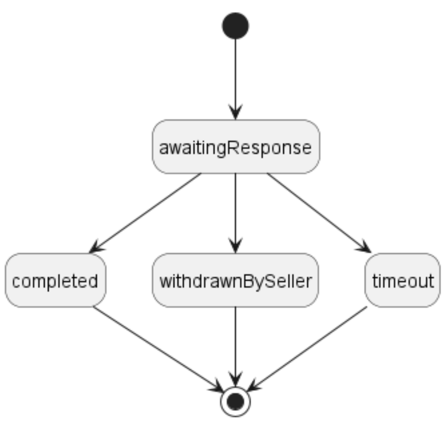

Figure 24. Charge State Machine

State	MEF 57.2 Name	Description
`awaitingResponse`	AWAITING_RESPONSE	A Charge has been initiated by the Seller. The charge includes one or more charges related to a Product Order or Product Order Item. Buyer has not indicated whether they accept or reject the charges via a Respond to Charge request.
`completed`	COMPLETED	All Charge Items included in the Charge for a given Product Order Item have moved to either the `acceptedByBuyer` state, the `declinedByBuyer` state, or the `withdrawnBySeller` state.

State	MEF 57.2 Name	Description
timeout	TIMEOUT	A Charge Item has been declined by the Buyer. The referenced Product Order and Product Order Items are updated. If a Blocking charge is declined, the Seller may cancel the referenced Product Order Item and any related Product Order Items, the related Cancel Product Order, or the related Modify Product Order Item Requested Delivery Date.
withdrawnBySeller	WITHDRAWN_BY_SELLER	The Seller determines that the Charge Item is incorrect. They withdraw the Charge Item and initiate a new Charge with the required correction(s) if needed.

Table 14. Charge States

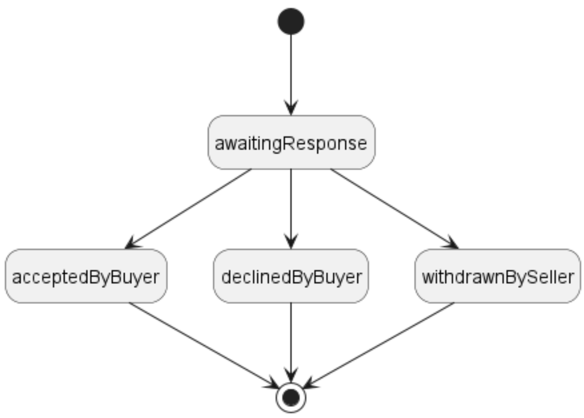

Figure 25. Charge Item State Machine

State	Description
acceptedByBuyer	A Charge Item identified in the Charge has been accepted by the Buyer.
awaitingResponse	A Charge Item has been identified by the Seller and awaits the Buyer's acceptance.
declinedByBuyer	A Charge Item identified in the Charge has been declined by the Buyer. The referenced Product Order and Product Order Items are updated.
withdrawnBySeller	The Seller determines that the Charge Item is incorrect. They withdraw the Charge Item and initiate a new Charge with the required correction(s).

Table 15. Charge Item States

[R82] The Charge **MUST** contain only Charge Items related to the same Product Order Item (or Product Order, depending on the Use Case).

[R83] A Product Order **MUST NOT** have more than one Charge in `state=awaitingResponse` at the same time. [MEF57.2 R88]

[R84] A Product Order Item **MUST NOT** have more than one Charge in `state=awaitingResponse` at the same time. [MEF57.2 R87]

[R85] A Product Order and a Product Order Item within the Product Order **MUST NOT** have Charges `state=awaitingResponse` at both the Product Order and Product Order Item at the same time. [MEF57.2 R89]

[R86] A Charge **MUST** be initiated for a Product Order only in one of the following states: `assessingCancellation` or `pending.assessingModification`.

[R87] A Charge **MUST** be initiated for a Product Order Item only in one of the following states: `inProgress`, or `pending` state.

[R88] The Seller **MUST** support the `chargeCreateEvent` notification if the Seller supports Charge use cases. [MEF57.2 R8]

[O14] The Seller **MAY** support the `chargeStateChangeEvent` and `chargeTimeoutEvent` notifications if the Seller supports Charge use cases. [MEF57.2 O5]

[R89] The Buyer **MUST** register for `chargeCreateEvent` notifications if the Seller supports charge use cases. [MEF57.2 R7]

[O15] The Buyer **MAY** register for other Charge related notifications if they are supported by the Seller [MEF57.2 O4]

[R90] When the Seller creates a Charge, the following attributes **MUST** be set: [MEF57.2 R82]

- `id`
- one of `productOrder` or `productOrderItem`
- `chargeItem`
- `creationDate`
- `responseDueDate`
- `state`

[R91] When the Seller initiates the Charge associated with Product Order Item, the `productOrderItem` attribute **MUST** be provided. [MEF57.2 R82], [MEF57.2 R83]

[R92] When the Charge was identified as an effect of a Modify Product Order Item Requested Delivery Date request the Seller **MUST** provide the `productOrderItem` and `modifyProductOrderItemRequestedDeliveryDate` attributes. [MEF57.2 R83]

[R93] When the Charge was identified as an effect of a Cancel Product Order request the Seller **MUST** provide the `productOrder` and `cancelProductOrder` attributes. [MEF57.2 R84]

[R94] For each Charge Item included in the Charge, the Seller **MUST** include the following attributes: [MEF57.2 R85]

- `id`
- `activityType`
- `priceType`
- `priceCategory`
- `blocking`
- `price`
- `state`

[R95] Table 16 shows the attributes that **MUST** be included in the Charge Item based on the `priceType`: [MEF57.2 R86]

priceType	**recurringChargePeriod**	**unitOfMeasure**	**price.dutyFreeAmount**
`recurring`	X		X
`nonRecurring`			X

priceType	recurringChargePeriod	unitOfMeasure	price.dutyFreeAmount
usageBased		X	X

Table 16. Price Type Required Information

6.11.1 Use case 11a: Initiate Charge Associated to Product Order Item

In this case, the Seller detects additional charges or changes in previously communicated charges linked to the fulfillment of the Product Order Item. The model and states have been described earlier. The sequence diagram below presents a Charge use case together with a context of the Use Case 1.

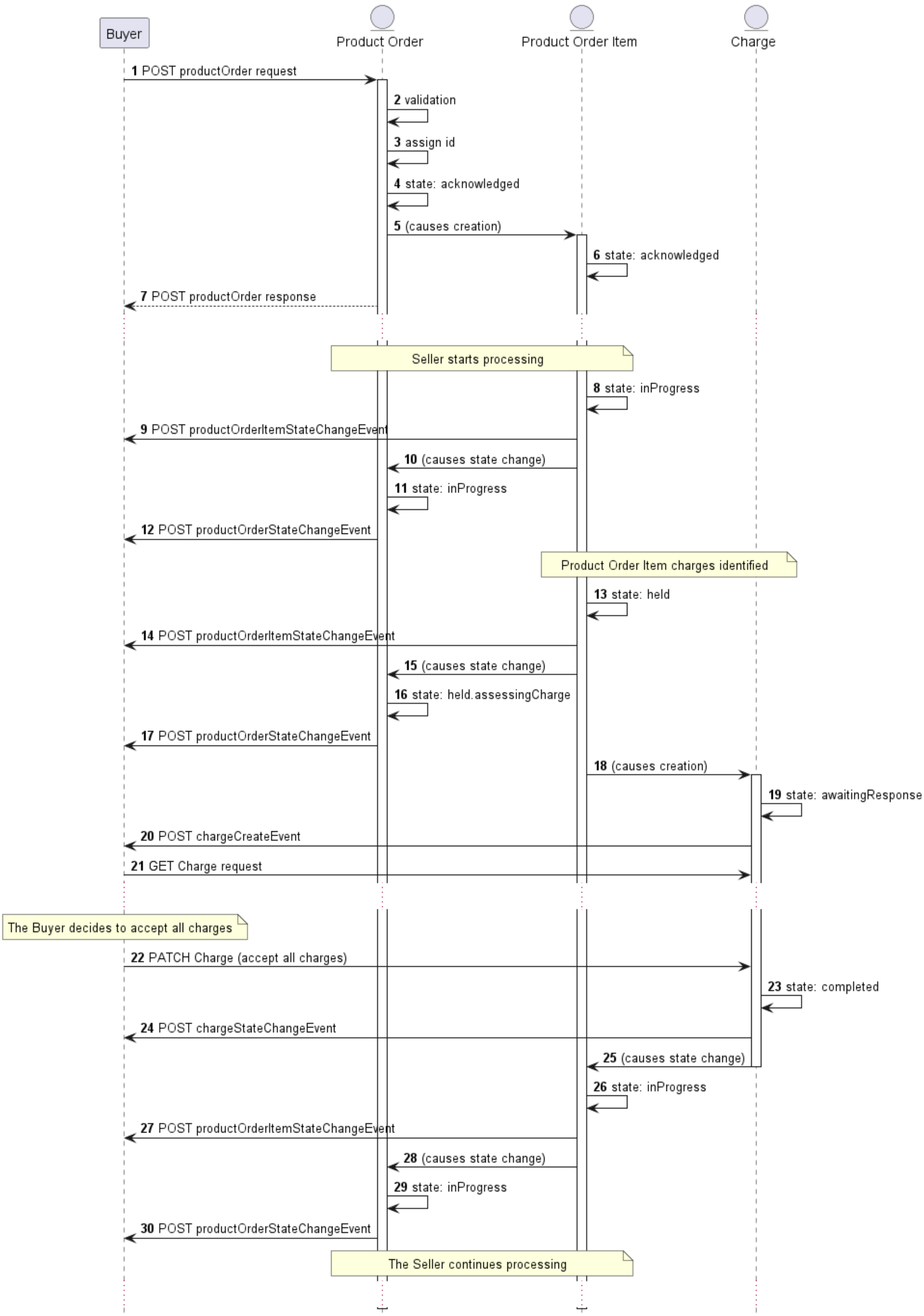

Figure 26. Use case 11a: Initiate Charge Associated to Product Order Item Flow

- The Seller identifies one or more charges associated with a `ProductOrderItem`. The referenced `ProductOrderItem` moves to the `held` state **(13)** and the `ProductOrder` moves to `held.assessingCharge` state **(15,16)**.
- A `Charge` is initiated by the Seller and a `chargeCreateEvent` is sent by the Seller **(18,19,20)**.

- The Buyer queries for the details of the `Charge` **(21)**.
- The Buyer accepts each `ChargeItem` contained within the `Charge` **(22)**.
- The Seller changes the state of the Charge to `completed` **(23)** and changes the referenced `ProductOrderItem` and `ProductOrder` states back to `inProgress` **(25,26,28,29)**.

The snippet below presents what a Charge related to this use case may look like. This exact part will be a body of a response to a Buyer's GET by id request **(21)**.


```json
{
  "id": "00000000-0000-1111-0000-000000000001",
  "href": "{{baseUrl}}/charge/00000000-0000-1111-0000-00000000000",
  "creationDate": "2021-05-25T22:05:48.319Z",
  "productOrderId": "00000000-1111-2222-3333-000000000123",
  "productOrderItemId": "item-001",
  "chargeItem": [
    {
      "id": "item-001",
      "priceType": "nonRecurring",
       "description": "Because of COVID sanitary restrictions there is an additional for the on-site installation
visit",
      "activityType": "new",
      "blocking": true,
      "price": {
        "taxRate": 8,
        "dutyFreeAmount": {
          "unit": "USD",
          "value": 50
        },
        "taxIncludedAmount": {
          "unit": "USD",
          "value": 54
        }
      },
      "state": "awaitingResponse"
    }
  ],
  "responseDueDate": "2021-05-25T22:05:48.319Z",
  "state": "awaitingResponse"
}
```


6.11.2 Use case 11b: Initiate Charge Associated to Modify Product Order Item Requested Delivery Date

In this case, the Charges are identified as a result of a Modify Product Order Item Completion Date request. The model and states have been described earlier. The sequence diagram below presents a Charge use case together with a context of the Use Case 5a: Modify Expedite Indicator - setting the `expediteIndicator` to `true` see section 6.5.

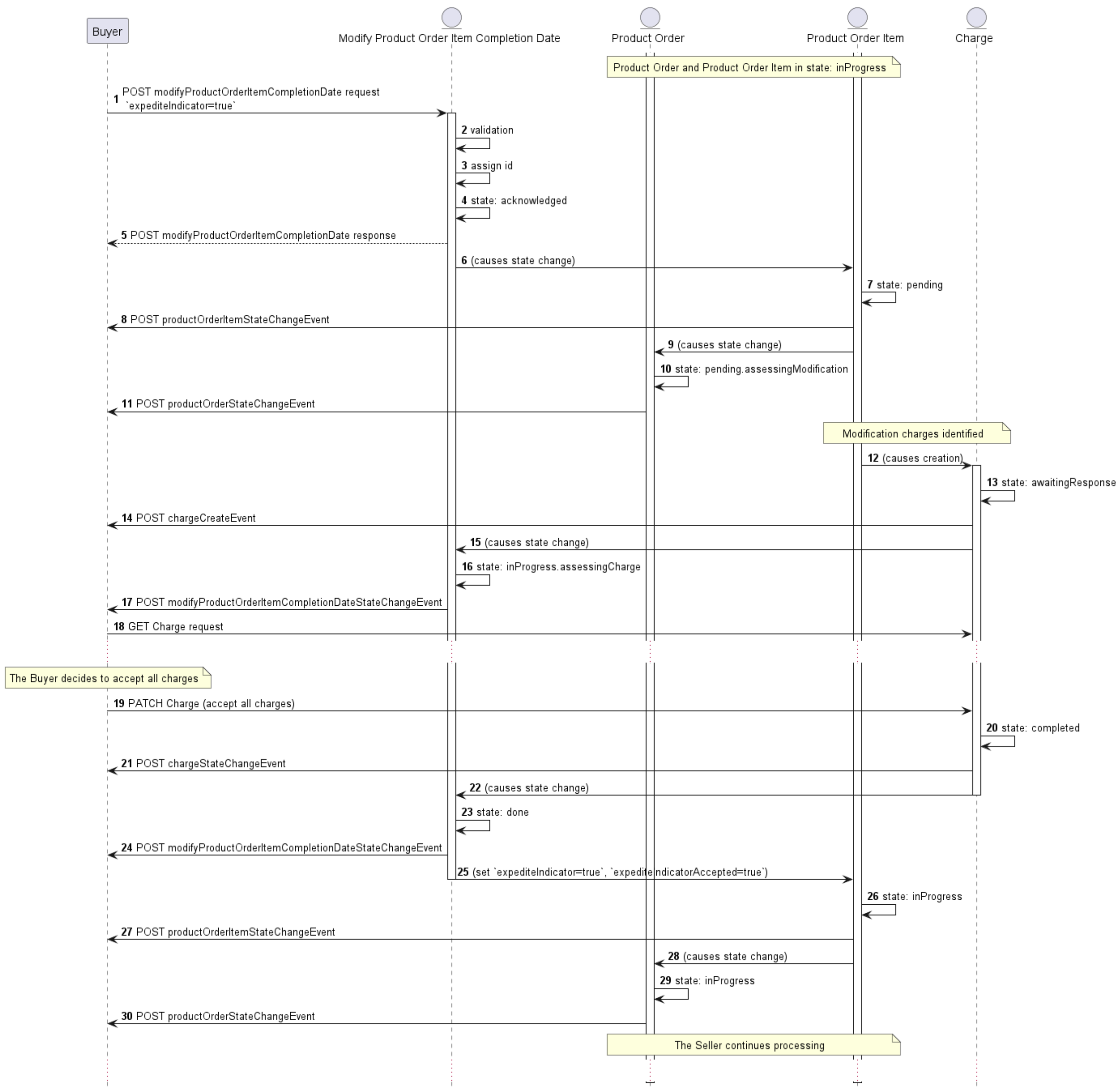

Figure 27. Use case 11b: Initiate Charge Associated to Modify Product Order Item Requested Delivery Date Flow

- The Seller identifies one or more charges associated with a `modifyProductOrderItemRequestedDeliveryDate` request. A `Charge` process is initiated by the Seller **(12,13)** and a `chargeCreateEvent` is sent by the Seller **(14)**.
- The referenced `modifyProductOrderItemRequestedDeliveryDate` moves to the `inProgress.assessingCharge` state **(15,16)** (until a response is received from the Buyer or the `responseDueDate` expires).
- The Buyer queries for the details of the `Charge` **(18)**.
- The Buyer accepts each `ChargeItem` contained within the `Charge` **(19)**.
- The Seller changes the state of the `Charge` to `completed` **(20)** and the referenced `modifyProductOrderItemRequestedDeliveryDate` state to `done` **(22,23)**.
- The Seller updates the `expediteIndicator` and `expediteIndicatorAccepted` **(25)** and changes the `ProductOrderItem` and `ProductOrder` states back to `inProgress` **(26,28,29)**.

6.11.3 Use case 11c: Initiate Charge Associated to Cancel Product Order

In this case, the Charges are identified as a result of a Cancel Product Order request. The model and states have been described earlier. The sequence diagram below presents a Charge use case together with a context of the Use Case 8: Cancel Product Order

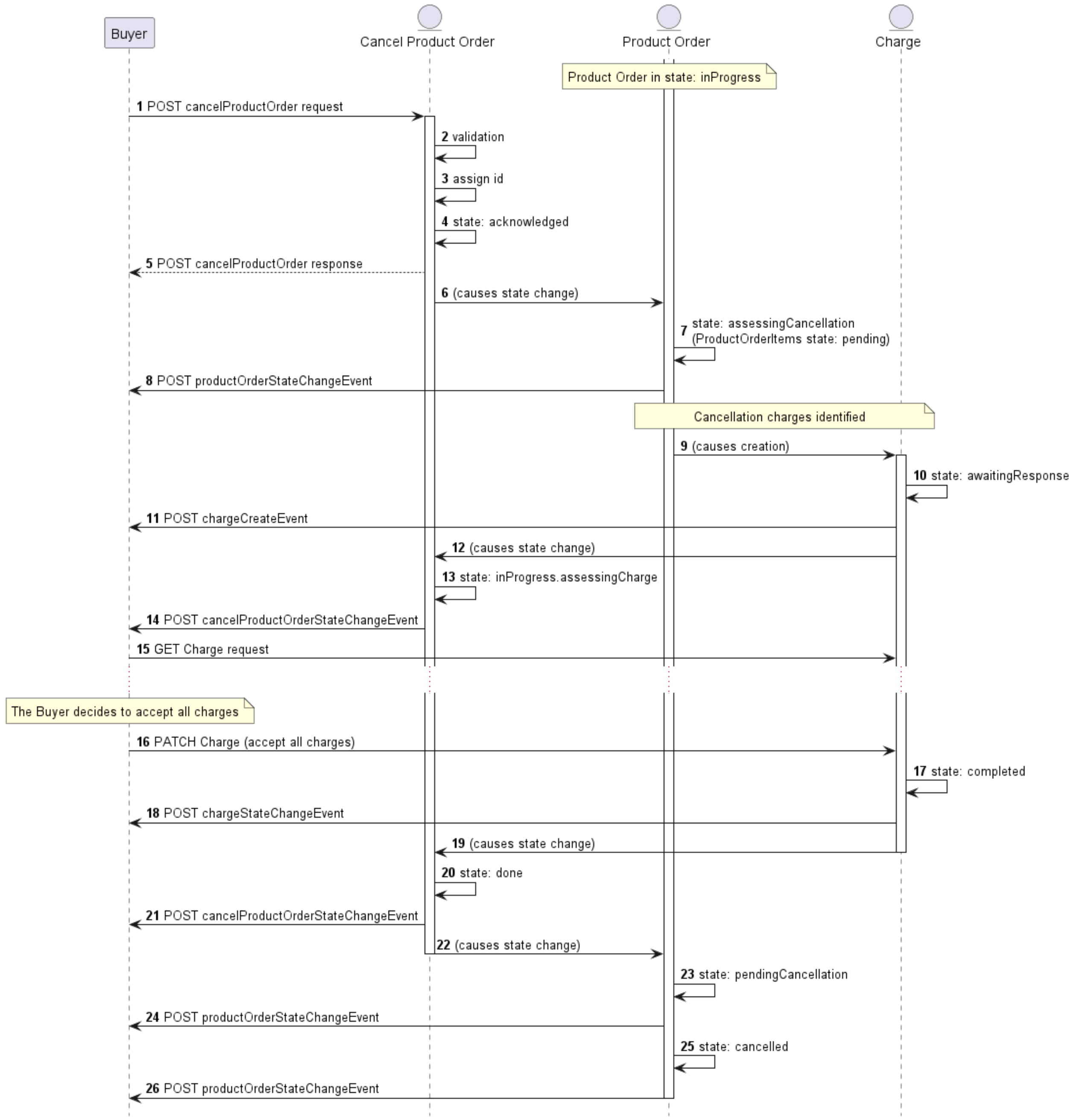

Figure 28. Use case 11c: Initiate Charge Associated to Cancel Product Order Flow

- The Seller identifies one or more charges associated with a `CancelProductOrder` request. A `Charge` process is initiated by the Seller in `awaitingResponse` state **(9,10)** and a `chargeCreateEvent` is sent by the Seller **(11)**.
- The referenced `CancelProductOrder` moves to the `assessingCharge` state **(12,13)** (until a response is received from the Buyer or the `responseDueDate` expires).
- The Buyer queries for the details of the `Charge` **(15)**.
- The Buyer accepts each `ChargeItem` contained within the `Charge` **(16)**.
- The Seller changes the state of the `Charge` to `completed` **(17)** and the referenced `CancelProductOrder` state to `done` **(19,20)**.
- The referenced `ProductOrder.state` is set to `pendingCancellation` **(22,23)**.
- Once the Seller has completed the cancellation process, the referenced `ProductOrder.state` is changed to `cancelled` **(25)**.

6.12. Use case 12: Respond to Charge

The Buyer must respond to a Charge initiated by the Seller with the use of a `PATCH /charge/{{id}}` operation. The model for this case is in Figure 23 section 6.11

The PATCH usage recommendation follows TMF 622 json/merge (https://tools.ietf.org/html/rfc7386).

Below is an example of such a Charge response - PATCH request:

```
{
  "chargeItem": [
    {
      "id": "item-001",
      "acceptanceIndicator": "accepted"
    }
  ]
}
```

[R96] The Buyer's response to the Charge **MUST** update the `acceptanceIndicator` for each and every Charge Item included in the Charge. [MEF57.2 R90]

[R97] The Buyer **MUST** update all Charge Items included in a Charge at once. [MEF57.2 R91]

[O16] The Buyer **MAY** add notes to Charge Items by appending the existing `note` list with a new item. [MEF57.2 O14]

[R98] If there is no response received by the `responseDueDate` is passed the Seller **MUST** treat all Charge Items as `declinedByBuyer` and put the Charge in `timeout` state. [MEF57.2 R92]

[R99] Once a Charge Item has been accepted by the Buyer, the Seller **MUST NOT** withdraw or modify the Charge Item. [MEF57.2 R124]

If in Use Case 11a the Buyer rejects a Charge Item that is identified as Blocking, the Seller changes the state of the Charge to `completed`, changes the referenced Product Order Item state to `failed`, and changes any Product Order Items related to the referenced Product Order Item to `failed`.

If in Use Case 11b the Buyer rejects a Blocking Charge Item, the Seller changes the state of the Charge to `complete` and changes the referenced Modify Product Order Item Requested Delivery Date state to `declined`. No modification to the Product Order Item is Performed.

If in Use Case 11c the Buyer rejects a Blocking Charge Item, the Seller changes the state of the Charge to `complete` and changes the referenced Cancel Product Order state to `declined`, and returns the Product Order to `inProgress`. The Product Order is not cancelled.

6.13. Use case 13: Retrieve List of Charges

The Buyer can retrieve a list of `Charges` by using a `GET /charge` operation with desired filtering criteria.

[O17] The Buyer's request **MAY** contain none or any of the following attributes: [MEF57.2 O20]

- `productOrderId`
- `productOrderItemId`
- `creationDate.gt`
- `creationDate.lt`
- `responseDueDate.gt`
- `responseDueDate.lt`
- `state`

The rules of using pagination and an example request are provided in section 6.3. Please refer to it as the rules also apply to this case.

[R100] The Seller must put the following attributes into the response (if set): [MEF57.2 R116]:

- `id`
- `productOrder` or `productOrderItem`
- `state`
- `creationDate`
- `responseDueDate`

[R101] In case no items matching the criteria are found, the Seller **MUST** return a valid response with an empty list. [MEF57.2 R115]

6.14. Use case 14: Retrieve Charge by Charge Identifier

The Buyer can get detailed information about the Charge communicated by the Seller by using a `GET /charge/{{id}}` operation.

[R102] The Seller's response **MUST** provide the following Charge attributes (if set): [MEF57.2 R119]

- `id`
- `chargeItem`
- `creationDate`
- `responseDueDate`
- `state`
- `productOrder` or `productOrderItem`
- `cancelProductOrder`
- `modifyProductOrderItemRequestedDeliveryDate`

6.15. Use case 15: Register for Notifications

The Seller communicates with the Buyer with Notifications provided that:

- both Seller and Buyer support the notification mechanism
- Buyer has registered to receive notifications from the Seller

To register for notifications the Buyer uses the `registerListener` operation from the API: `POST /hub`. The request model contains only 2 attributes:

- `callback` - mandatory, to provide the callback address the events will be notified to,
- `query` - optional, to provide the required types of event.

The usage of a combination of these attributes fulfills the [MEF57.2 CR7<O21], and [MEF57.2 CR8<O21] requirements.

The figure below shows all entities involved in the Notification use cases.

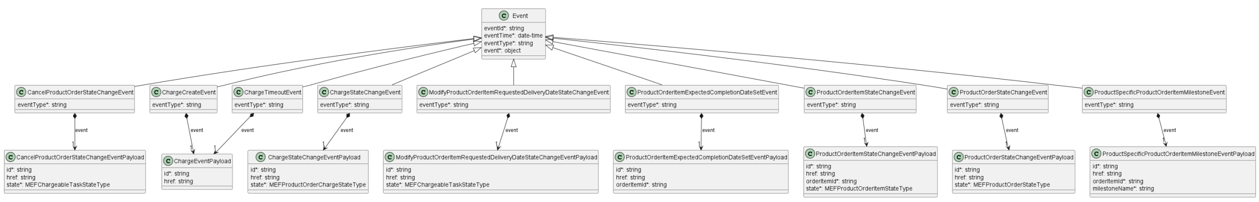

Figure 29. Product Order Management Notification Data Model

By using a simple request:

```
{
  "callback": "https://buyer.mef.com/listenerEndpoint"
}
```

The Buyer subscribes for notification of all types of events. Those are:

- `productOrderStateChangeEvent`
- `productOrderItemStateChangeEvent`
- `productSpecificProductOrderItemMilestoneEvent`
- `productOrderItemExpectedCompletionDateSetEvent`
- `cancelProductOrderStateChangeEvent`
- `chargeCreateEvent`
- `chargeStateChangeEvent`
- `chargeTimeoutEvent`
- `modifyProductOrderItemRequestedDeliveryDateStateChangeEvent`

If the Buyer wishes to receive only notification of a certain type, a `query` must be added:

```
{
  "callback": "https://buyer.mef.com/listenerEndpoint",
  "query": "eventType=productOrderStateChangeEvent"
}
```

If the Buyer wishes to subscribe to 2 different types of events, there are 2 possible syntax variants [TMF630]:

```
eventType=productOrderStateChangeEvent,chargeCreateEvent
```

or

```
eventType=productOrderStateChangeEvent&eventType=chargeCreateEvent
```

The `query` formatting complies with RFC3986 RFC3986. According to it, every attribute defined in the Event model (from notification API) can be used in the `query`. However, this standard requires only `eventType` attribute to be supported.

[O18] The Seller **MAY** support registration for Notifications other than `chargeCreateEvent`. [MEF57.2 O21]

[R103] `eventType` is the only attribute that the Seller **MUST** support in the query.

If any of Charge related use cases are supported, the following 2 requirements apply:

[R104] The Seller **MUST** support `chargeCreateEvent` notifications. [MEF57.2 R120]

[R105] The Buyer **MUST** subscribe to `chargeCreateEvent` notifications. [MEF57.2 R121]

The Seller responds to the subscription request by adding the `id` of the subscription to the message that must be further used for unsubscribing.

```
{
  "id": "00000000-0000-0000-0000-000000000678",
  "callback": "https://buyer.mef.com/listenerEndpoint",
  "query": "eventType=productOrderStateChangeEvent"
}
```

Example of a final address that the Notifications will be sent to (for Sonata, `productOrderStateChangeEvent`):

- `https://buyer.mef.com/listenerEndpoint/mefApi/sonata/productOrderingNotification/v11/listener/productOrderStateChangeEvent`

To stop receiving events, the Buyer has to use the `unregisterListener` operation from the `DELETE /hub/{id}` endpoint. The `id` is the identifier received from the Seller during the listener registration.

[R106] In the `unregisterListener` operation, the Buyer **MUST** provide the `id` of the registered `EventSubscription` that originates from the Seller.

The example below shows an exemplary unregister call sent by the Buyer to the Seller (DELETE operation):

```
http://seller.mef.com:8080/mefApi/sonata/productOrderingManagement/v11/hub/00000000-0000-0000-0000-000000000678
```

[R107] In the successful scenario the Seller **MUST** respond with an empty body and HTTP code `204`.

The Buyer can unregister only the whole `EventSubscription`, regardless of the provided `query`. In the case when the Buyer e.g. resigns from specific types of events (or changes the callback address), the existing `EventSubscription` that includes undesired notification types that need to be removed and replaced by the new `EventSubscription` with an adjusted `query` attribute.

Note: The above note concludes that the Buyer cannot update the existing `EventSubscription`. Every kind of update is done by subscription replacement.

6.16. Use case 16: Send Notification

Notifications are used to asynchronously inform the Buyer about the respective objects and attribute changes. The Seller's synchronous response to a Product Order, Cancel Product Order, and Modify Product Order Item Requested Delivery Date create requests are considered to act as a Create Notification so there is no explicit respective Create Notification type. The next notification must be sent when the `state` changes compared to the previously sent one.

For the sake of readability, all previous flow diagrams presented only cases of using only the `productOrderStateChangeEvent`. Figure 30 presents the end-to-end sequence of communication in Use Case 1 - Create Product Order with Buyer's subscription to both `productOrderStateChangeEvent` and `productOrderItemStateChangeEvent` event types.

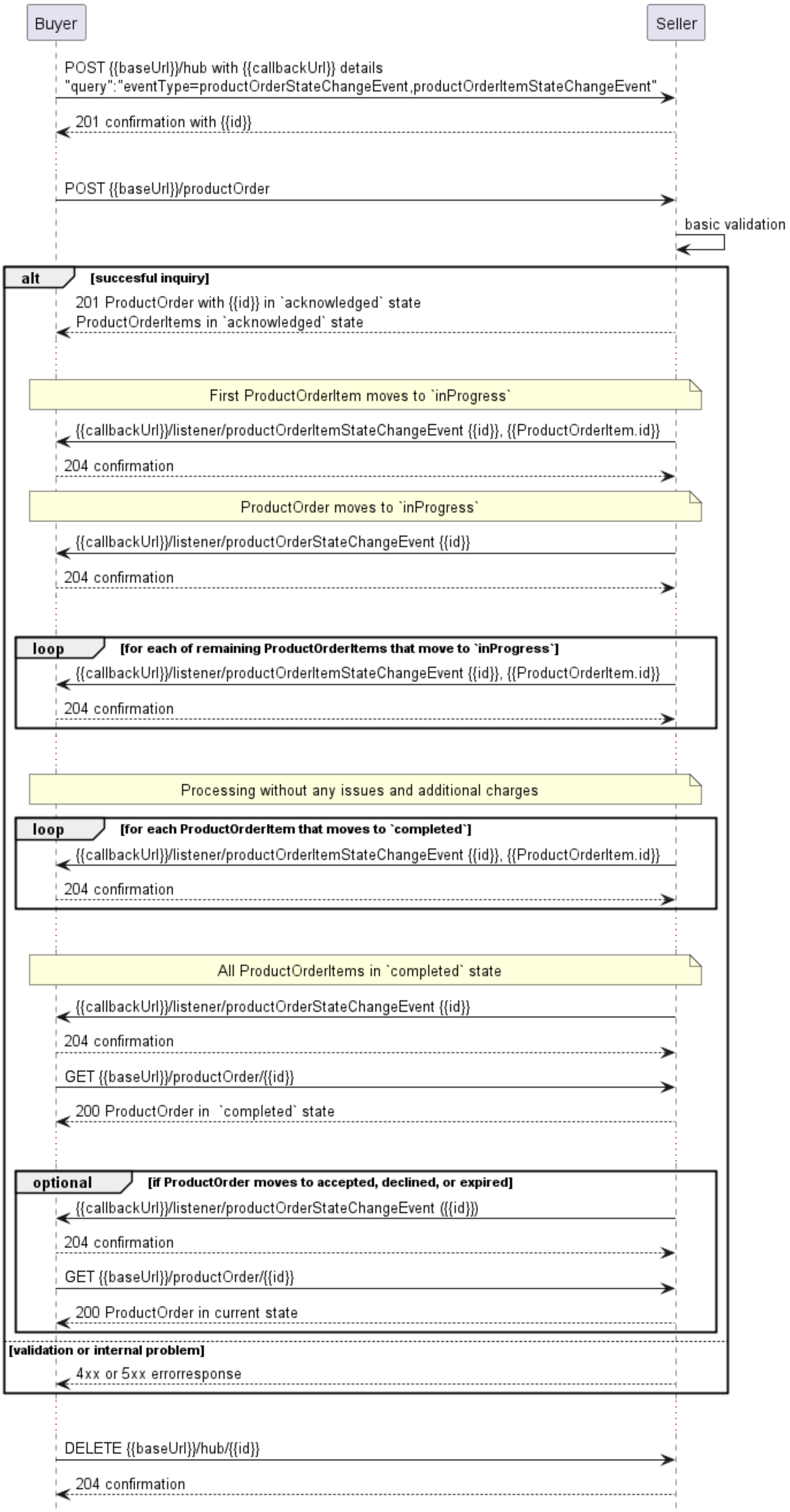

Figure 30. Use Case 1 - Create Product Order with Product Order Item Notifications

After a successful Notification subscription, the Buyer sends a Product Order create request. The Seller responds with a Product Order and all items in an `acknowledged` state. When the first Product Order Item moves to `inProgress`, a `productOrderItemStateChangeEvent` is sent. Immediately the Product Order also changes its state to `inProgress` and the `productOrderStateChangeEvent` is sent. Then the rest (if any) of the Product Order Items are processed. Let's assume that no additional charges were found and the process ends smoothly. When particular items are done processing they reach the `completed` state. Once all are successfully done, the Product Order also changes state to `completed`. The Buyer will likely now ask for the Product Order details.

The events are sent only after a synchronous response to the Product Order create request is provided. Thus there must be no state change notifications set for Product Order and Product Order Items reaching the `acknowledged` state.

[R108] The Seller **MUST NOT** send Notifications to Buyers who have not registered for them. [MEF57.2 R122]

[R109] The Seller **MUST** send Notifications to Buyers who have registered for them. [MEF57.2 R123]

The following snippets present examples of `productOrderStateChangeEvent` and `productOrderItemStateChangeEvent`:

```
{
  "eventId": "event-001",
  "eventType": "productOrderStateChangeEvent",
  "eventTime": "2021-06-02T00:00:00.000Z",
  "event": {
    "id": "00000000-1111-2222-3333-000000000123",
    "state": "inProgress"
  }
}
```

[R110] The Seller **MUST** provide the following attributes of `Event`:

- `event`
- `eventId`
- `eventTime`
- `eventType`

[R111] An event triggered by the Product Order Item (`productOrderItemStateChangeEvent`, `productOrderItemExpectedCompletionDateSetEvent`, `productSpecificProductOrderItemMilestoneEvent`) **MUST** additionally contain the relative `orderItemId`. [MEF57.2 R124]

[R112] A `StateChange` type of event (`productOrderStateChangeEvent`, `productOrderItemStateChangeEvent`, `cancelProductOrderStateChangeEvent`, `chargeStateChangeEvent`, `modifyProductOrderItemRequestedDeliveryDateStateChangeEvent`) **MUST** additionally contain the `state` attribute.

[O19] The Seller **MAY** support Milestones. [MEF57.2 O22]

[CR8]<[O19] If the Seller supports Milestones, the Seller **MUST** support sending `productSpecificProductOrderItemMilestoneEvent` with the `milestoneName` attribute set. [MEF57.2 R124]

[CR9]<[O19] If the Seller supports Milestones, the Seller **MUST** store the Milestone history in the `milestone` attribute of the `ProductOrderItem`. [MEF57.2 CR9<O22]

```
{
  "eventId": "event-002",
  "eventType": "productOrderItemStateChangeEvent",
  "eventTime": "2021-06-02T00:00:00.000Z",
```

```json
    "event": {
      "id": "00000000-1111-2222-3333-000000000123",
      "orderItemId": "item-001",
      "state": "inProgress"
    }
  }
```

Note: the body of the event carries only the source object's `id` and `state` (where applicable). The Buyer needs to query it later by `id` to get details.

7. API Details

7.1. API patterns

7.1.1. Indicating errors

Erroneous situations are indicated by appropriate HTTP responses. An error response is indicated by HTTP status 4xx (for client errors) or 5xx (for server errors) and appropriate response payload. The Product Order API uses the error responses as depicted and described below.

Implementations can use HTTP error codes not specified in this standard in compliance with rules defined in RFC 7231 [RFC7231]. In such a case, the error message body structure might be aligned with the `Error`.

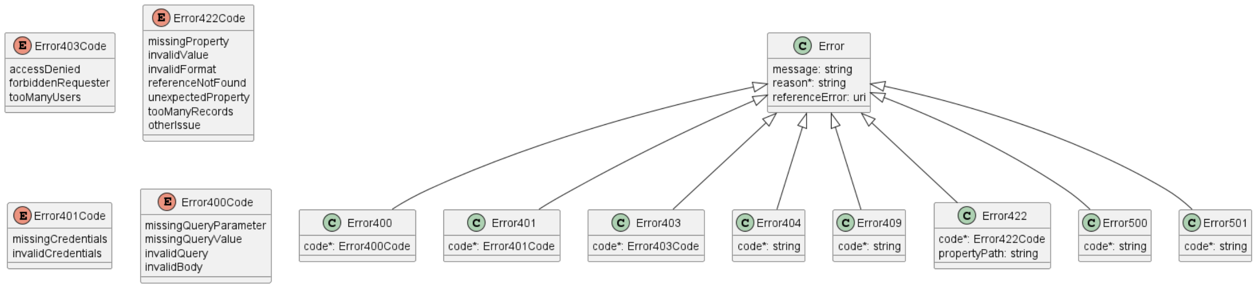

Figure 31. Data model types to represent an erroneous response

7.1.1.1. Type Error

Description: Standard Class used to describe API response error Not intended to be used directly. The `code` in the HTTP header is used as a discriminator for the type of error returned in runtime.

Name	Type	Description
message	string	Text that provides mode details and corrective actions related to the error. This can be shown to a client user.
reason*	string *maxLength = 255*	Text that explains the reason for the error. This can be shown to a client user.
referenceError	uri *format = uri*	URL pointing to documentation describing the error

7.1.1.2. Type Error400

Description: Bad Request. (https://tools.ietf.org/html/rfc7231#section-6.5.1)

Inherits from:

- Error

Name	Type	Description

Name	Type	Description
code*	Error400Code	One of the following error codes: - missingQueryParameter: The URI is missing a required query-string parameter - missingQueryValue: The URI is missing a required query-string parameter value - invalidQuery: The query section of the URI is invalid. - invalidBody: The request has an invalid body

7.1.1.3. enum Error400Code

Description: One of the following error codes:

- missingQueryParameter: The URI is missing a required query-string parameter
- missingQueryValue: The URI is missing a required query-string parameter value
- invalidQuery: The query section of the URI is invalid.
- invalidBody: The request has an invalid body

7.1.1.4. Type Error401

Description: Unauthorized. (https://tools.ietf.org/html/rfc7235#section-3.1)

Inherits from:

- Error

Name	Type	Description
code*	Error401Code	One of the following error codes: - missingCredentials: No credentials provided. - invalidCredentials: Provided credentials are invalid or expired

7.1.1.5. enum Error401Code

Description: One of the following error codes:

- missingCredentials: No credentials provided.
- invalidCredentials: Provided credentials are invalid or expired

7.1.1.6. Type Error403

Description: Forbidden. This code indicates that the server understood the request but refuses to authorize it. (https://tools.ietf.org/html/rfc7231#section-6.5.3)

Inherits from:

- Error

Name	Type	Description

Name	Type	Description
code*	Error403Code	This code indicates that the server understood the request but refuses to authorize it because of one of the following error codes: - accessDenied: Access denied - forbiddenRequester: Forbidden requester - tooManyUsers: Too many users

7.1.1.7. `enum` Error403Code

Description: This code indicates that the server understood the request but refuses to authorize it because of one of the following error codes:

- accessDenied: Access denied
- forbiddenRequester: Forbidden requester
- tooManyUsers: Too many users

7.1.1.8. Type Error404

Description: Resource for the requested path not found. (https://tools.ietf.org/html/rfc7231#section-6.5.4)

Inherits from:

- Error

Name	Type	Description
code*	string	The following error code: - notFound: A current representation for the target resource not found

7.1.1.9. Type Error409

Description: Conflict (https://datatracker.ietf.org/doc/html/rfc7231#section-6.5.8)

Inherits from:

- Error

Name	Type	Description
code*	string	The following error code: - conflict: The client has provided a value whose semantics are not appropriate for the property.

7.1.1.10. Type Error422

The response for HTTP status `422` is a list of elements that are structured using the `Error422` data type. Each list item describes a business validation problem. This type introduces the `propertyPath` attribute which points to the erroneous property of the request, so that the Buyer may fix it easier. It is highly recommended that this property should be used, yet remains optional because it might be hard to implement.

Description: Unprocessable entity due to a business validation problem. (https://tools.ietf.org/html/rfc4918#section-11.2)

Inherits from:

- Error

Name	Type	Description
code*	Error422Code	One of the following error codes: - missingProperty: The property the Seller has expected is not present in the payload - invalidValue: The property has an incorrect value - invalidFormat: The property value does not comply with the expected value format - referenceNotFound: The object referenced by the property cannot be identified in the Seller system - unexpectedProperty: Additional property, not expected by the Seller has been provided - tooManyRecords: the number of records to be provided in the response exceeds the Seller's threshold. - otherIssue: Other problem was identified (detailed information provided in a reason)
propertyPath	string	A pointer to a particular property of the payload that caused the validation issue. It is highly recommended that this property should be used. Defined using JavaScript Object Notation (JSON) Pointer (https://tools.ietf.org/html/rfc6901).

7.1.1.11. `enum` Error422Code

Description: One of the following error codes:

- missingProperty: The property the Seller has expected is not present in the payload
- invalidValue: The property has an incorrect value
- invalidFormat: The property value does not comply with the expected value format
- referenceNotFound: The object referenced by the property cannot be identified in the Seller system
- unexpectedProperty: Additional property, not expected by the Seller has been provided
- tooManyRecords: the number of records to be provided in the response exceeds the Seller's threshold.
- otherIssue: Other problem was identified (detailed information provided in a reason)

7.1.1.12. Type Error500

Description: Internal Server Error. (https://tools.ietf.org/html/rfc7231#section-6.6.1)

Inherits from:

- Error

Name	Type	Description
code*	string	The following error code: - internalError: Internal server error - the server encountered an unexpected condition that prevented it from fulfilling the request.

7.1.1.13. Type Error501

Description: Not Implemented. Used in case Seller is not supporting an optional operation (https://tools.ietf.org/html/rfc7231#section-6.6.2)

Inherits from:

- Error

Name	Type	Description
code*	string	The following error code: - notImplemented: Method not supported by the server

7.2. Management API Data model

Figure 32 presents the whole Product Order Management data model. The data types, requirements related to them and mapping to MEF 57.2 specification are discussed later in this section.

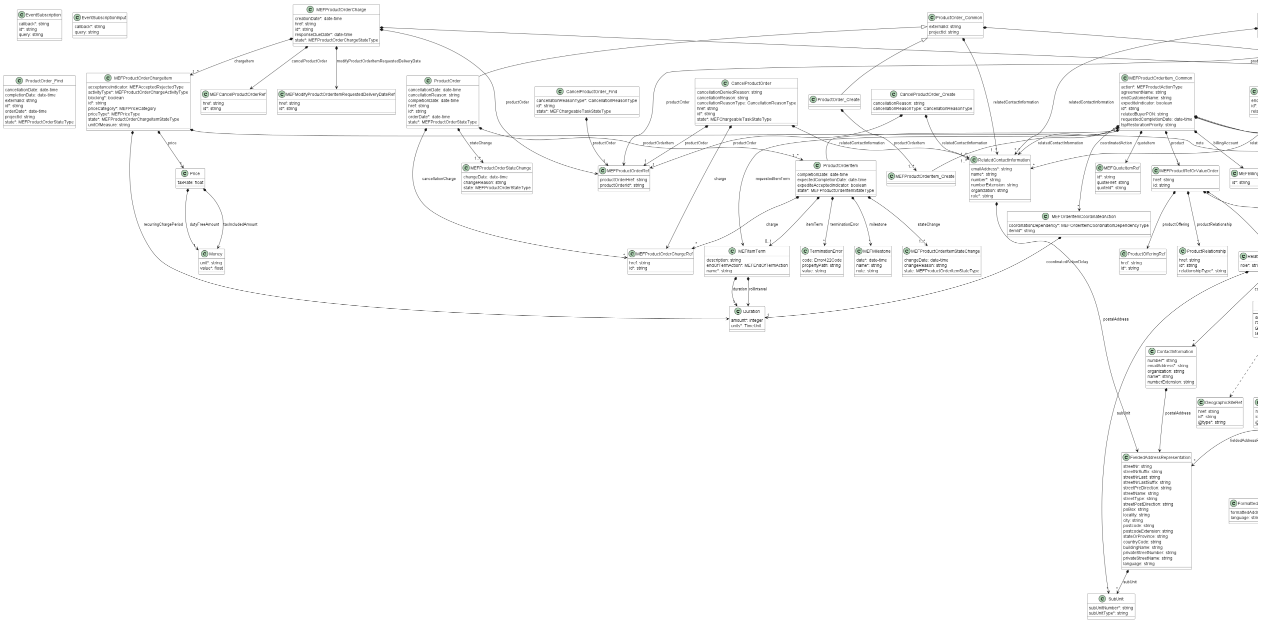

Figure 32. Product Order Management Data Model

7.2.1. ProductOrder

7.2.1.1 Type ProductOrder_Common

Description: A Product Order is a type of order which can be used to place an order between a customer and a service provider or between a service provider and a partner and vice versa,

Name	Type	M/O	Description	MEF 57.2
externalId	string	O	An identifier for this order within the Buyer's enterprise.	

Name	Type	M/O	Description	MEF 57.2
note	Note[]	O	Free form text to clarify or explain the Product Order. Only new notes can be entered. The Buyer and Seller cannot modify an existing Note. The Buyer creates a Note when creating the Product Order or when updating it. The Seller may add notes at any time.	
projectId	string	O	An identifier that is used to group Product Orders that is important to the Buyer. A projectId can be used to relate multiple Product Orders together.	
relatedContactInformation	RelatedContactInformation[] *minItems = 1*	M	Contact information of an individual or organization playing a role in this context. (e.g. Product Order Contact: role=productOrderContact; Seller Contact: role=sellerContact) Providing the Product Order Contact in the request is mandatory.	

7.2.1.2. Type ProductOrder_Create

Description: A Product Order is a type of order which can be used to place an order between a customer and a service provider or between a service provider and a partner and vice versa, Skipped properties: id,href,completionDate,orderDate,state,stateChange,cancellationDate,cancellationReason

Inherits from:

- ProductOrder_Common

Name	Type	M/O	Description	MEF 57.2
productOrderItem	MEFProductOrderItem_Create[] *minItems = 1*	M	Items contained in the Product Order.	

7.2.1.3. Type ProductOrder

Description: A Product Order is a type of order which can be used to place an order between a customer and a service provider or between a service provider and a partner and vice versa

Inherits from:

- ProductOrder_Common

Name	Type	M/O	Description	MEF 57.2
cancellationCharge	MEFProductOrderChargeRef[]	O	Charges associated with cancelling the Product Order	
cancellationDate	date-time *format = date-time*	O	Identifies the date the Seller cancelled the Order. Set by Seller when the Order is moved to the cancelled state.	
cancellationReason	string	O	An optional free-form text field for the Seller to provide additional information regarding the reason for the cancellation. If the Seller cancels the Product Order, the Seller provides the reason. If the Buyer requests the cancellation, the Seller copies the reason provided by the Buyer from the Cancel Product Order request.	
completionDate	date-time *format = date-time*	O	Identifies the date that all Product Order Items within the Order have reached a terminal state. No further action is permitted on the Product Order.	
href	string	O	Hyperlink to access the order	
id	string	M	Unique identifier for the Product Order that is generated by the Seller when the Product Order is initially accepted via an API.	
orderDate	date-time *format = date-time*	M	Date when the Product Order was created in the Seller's system and a Product Order Identifier was assigned	

Name	Type	M/O	Description	MEF 57.2
productOrderItem	ProductOrderItem[] *minItems = 1*	M	Items contained in the Product Order.	
state	MEFProductOrderStateType	M	The states as defined by TMF622 and extended to meet MEF requirements. These states are used to convey the Product Order status during the lifecycle of the Product Order.	
stateChange	MEFProductOrderStateChange[] *minItems = 1*	M	State change for the Product Order	

7.2.1.4. Type ProductOrder_Update

Description: A request initiated by the Buyer to update Product Order and/or Product

Name	Type	M/O	Description	MEF 57.2
externalId	string	O	An identifier for this Product Order within the Buyer's enterprise.	
note	Note[]	O	Free form text to clarify or explain the Product Order. Only new notes can be entered. The Buyer and Seller cannot modify an existing Note. The Buyer creates a Note when creating the Product Order or when updating it. The Seller may add notes at any time.	
productOrderItem	MEFProductOrderItem_Update[]	O	Order Item attributes that may be updated	

Name	Type	M/O	Description	MEF 57.2
projectId	string	O	An identifier that is used to group Product Orders that is important to the Buyer. A projectId can be used to relate multiple Product Orders together.	
relatedContactInformation	RelatedContactInformation[] minItems = 1	O	Contact information of an individual or organization playing a role in this context. Buyer can modify, add, or delete only Buyer-related contacts.	

7.2.1.5. Type ProductOrder_Find

Description: Structure to define GET without id response. A list of productOrder matching request criteria. Provides Product order summary view.

Name	Type	M/O	Description	MEF 57.2
cancellationDate	date-time format = date-time	O	Identifies the date the Seller cancelled the Order. Set by Seller when the Order is moved to the cancelled state.	
completionDate	date-time format = date-time	O	Identifies the date that all Product Order Items within the Order have reached a terminal state. No further action is permitted on the Product Order after this notification.	
externalId	string	O	ID given by the consumer and only understandable by him (to facilitate his searches afterward).	
id	string	M	Unique identifier for the order that is generated by the Seller when the order is initially accepted via an API.	

Name	Type	M/O	Description	MEF 57.2
orderDate	date-time *format = date-time*	M	Date when the Product Order was created	
projectId	string	O	An identifier that is used to group Product Orders that is important to the Buyer. A projectId can be used to relate multiple Product Orders together.	
state	MEFProductOrderStateType	M	The states as defined by TMF622 and extended to meet MEF requirements. These states are used to convey the Product Order status during the lifecycle of the Product Order.	

7.2.1.6. `enum` MEFProductOrderStateType

Description: Possible values for the state of the Product Order The following mapping has been used between `MEFProductOrderStateType` and MEF 57.2:

state	MEF 57.2 name	Description
`acknowledged`	ACKNOWLEDGED	A Product Order has been received by the Seller and has passed basic validation. A `productOrder.id` is assigned in the `acknowledged` state and a response is returned to the Buyer. The Product Order remains in the `acknowledged` state while validations of Product Order and Product Order Item(s) attributes as applicable is completed. If the Product Order and Product Order Item attributes are validated the Product Order moves to the `inProgress` state. If not validated, the Product Order moves to the `rejected` state.

state	MEF 57.2 name	Description
`assessingCancellation`	ASSESSING_CANCELLATION	A Cancel Product Order request has been received by the Seller. The Product Order is being assessed to determine if the Product Order can be cancelled. If there are charges associated with cancelling the Product Order, these are communicated to the Buyer using the Charge process. The Product Order remains in the `assessingCancellation` state until any relevant Charge is completed or withdrawn by the Seller. Once the Buyer's request has been validated and any associated Charges completed, the Product Order moves to the `pendingCancellation` state. If the request is not validated or if any associated Charges are not completed, the Product Order moves to the `inProgress` state and the Product Order is not cancelled.
`held.assessingCharge`	ASSESSING_CHARGE	A Charge has been initiated by the Seller that is not the result of a Modify Product Order Item Requested Delivery Date or Cancel Product Order request and the Seller is awaiting a Buyer response to the Charge. If a blocking or non-blocking charge is accepted by the Buyer, the Product Order moves to `inProgress`. If a non-blocking charge is declined by the Buyer, the Product Order moves to `inProgress`. If a blocking charge is declined by the Buyer and there are no unrelated Product Order Items in the Product Order, the Product Order moves to the `inProgress` and then to the `failed` state. If a blocking charge is declined by the Buyer and there are unrelated Product Order Items in the Product Order, the Product Order moves to the `inProgress` state.

state	MEF 57.2 name	Description
`pending.assessingModification`	ASSESSING_MODIFICATION	A request has been made by the Buyer to modify either the `expediteIndicator` or the `requestedCompletionDate` of a Product Order Item. The Product Order Item is currently being assessed to determine whether the Modify Product Order Item Requested Delivery Date is valid. If there is a charge associated with the Modify Product Order Item Requested Delivery Date, the Product Order remains in the `pending.assessingModification` state until the Charge is completed or withdrawn by the Seller. Once the Buyer's request has been validated and any associated Charges completed, the Product Order returns to the `inProgress` state.
`cancelled`	CANCELLED	The Product Order has been successfully cancelled. This is a terminal state.
`pendingCancellation`	CANCELLING	The Buyer's Cancel Request has been assessed and it has been determined that it is feasible to proceed with the cancellation. This state can also result from a Seller cancelling the Product Order within their systems without a request from the Buyer.
`completed`	COMPLETED	The Product Order has completed fulfillment and the Product is now active. This is a terminal state
`failed`	FAILED	All Product Order Items have failed which results in the entire Product Order failing. This is a terminal state.
`inProgress`	IN_PROGRESS	The Product Order has been successfully validated, and fulfillment has started.
`partial`	PARTIAL	At least one Product Order Item is `failed` or `rejected`, and fulfillment of at least one Product Order Item has been successful. This is a terminal state

state	MEF 57.2 name	Description
`rejected`	REJECTED	A Product Order was submitted, and it has failed at least one of the validation checks the Seller performs after it reached the `acknowledged` state

7.2.1.7. Type MEFProductOrderStateChange

Description: Holds the State notification reasons and associated date the State changed, populated by the server

Name	Type	M/O	Description	MEF 57.2
changeDate	date-time *format = date-time*	O	The date on when the state was reached	
changeReason	string	O	Additional comment related to state change	
state	MEFProductOrderStateType	O	Reached state	

7.2.2. Product Order Item

7.2.2.1 Type MEFProductOrderItem_Common

Description: An identified part of the order. A product order is decomposed into one or more order items. This type holds the attributes common to request and response representation of the Product Order Item.

Name	Type	M/O	Description
action	MEFProductActionType	M	Action to be applie referred by this Produ
agreementName	string	O	The name of the Ag referenced for the Pro
billingAccount	MEFBillingAccountRef	O	Billing account info billing account the E for the Product Order
coordinatedAction	MEFOrderItemCoordinatedAction[]	O	The interval after th one or more relate Items that this Produ be started or complet
endCustomerName	string	O	The name of the End a business name c name depending on t

Name	Type	M/O	Description
expediteIndicator	boolean *default = false*	O	Indicates that exped requested. Set by the set to TRUE, the Requested Completi expedited date. See M 7.3 for a description between the Buyer ar
id	string	M	A Buyer provide identify Product Orde able to relate them This is set by the Bu within the Product O Reference Identifier or A, B, C. The Re can be reused in Orders to identify Item within that Prod
note	Note[]	O	Free form text to clar Product Order Item. can be entered. The cannot modify an ex Buyer creates a No the Product Order updating it. The Sell at any time. This is inform the Seller of Buyer wishes perforn
product	MEFProductRefOrValueOrder	M	The Buyer's existi which the Product requested. Set by t Product Action is mo
productOfferingQualificationItem	ProductOfferingQualificationItemRef	O	The POQ and POQ I this Product Order It may be required by case, this is a mand Seller does not requi reference, then this attribute.
productOrderItemRelationship	OrderItemRelationship[]	O	The relationship b Order Items in the Pr
quoteItem	MEFQuoteItemRef	O	The Quote Item as Product Order Item. reference may be Seller. In that case, th field. If the Seller do Quote, then this attribute.

Name	Type	M/O	Description
relatedBuyerPON	string	O	Identifies the Buyer Number that is relate Order.
relatedContactInformation	RelatedContactInformation[]	O	Contact information or organization playi Order Item. The rul represented attribute is to use the _l pattern e.g. - Buye Item `role=buyerProductO - Buyer Implemen `role=buyerImplemen Buyer Technic `role=buyerTechnical Billing `role=buyerBillingCc Fault `role=buyerFaultCon Fault `role=sellerFaultCont GDPR `role=buyerGDPRCo GDPR `role=sellerGDPRCo
requestedCompletionDate	date-time *format = date-time*	O	Identifies the Buyer's (requested delivery d
requestedItemTerm	MEFItemTerm	O	Requested term of tl Item
tspRestorationPriority	string	O	Within the United Sta provisioning and re: as defined under t Vendor Handbook. ' are defined in ATIS ATIS-0404001.

7.2.2.2. Type MEFProductOrderItem_Create

Description: An identified part of the order. A product order is decomposed into one or more order items. The modelling pattern introduces the `Common` supertype to aggregate attributes that are common to both `ProductOrderItem` and `ProductOrderItem_Create`. The `Create` type has a subset of attributes of the response type and does not introduce any new, thus the `Create` type has an empty definition.

Inherits from:

- MEFProductOrderItem_Common

7.2.2.3. Type ProductOrderItem

Description: An identified part of the order. A product order is decomposed into one or more order items.

Inherits from:

- MEFProductOrderItem_Common

Name	Type	M/O	Description	MEF 57.2
charge	MEFProductOrderChargeRef[]	O	The Charges associated to this Product Order Item. This list contains all completed Charges containing accepted Charge Items initiated by the Seller. Any Charge that is withdrawn or containing all declined Charge Items must not be included in this list.	

Name	Type	M/O	Description	MEF 57.2
completionDate	date-time *format = date-time*	O	Identifies the date the Seller completed the Product Order Item. Set by Seller when all Product Order Items have reached a terminal state. No further action is permitted on the Product Order after this state is reached.	
expectedCompletionDate	date-time *format = date-time*	O	Identifies the date the Seller expects to complete the Product Order Item.	

Name	Type	M/O	Description	MEF 57.2
expediteAcceptedIndicator	boolean *default = false*	O	Indicates if the Seller has accepted the Buyer's Expedite request. See MEF 57.2 section 7.3 for a description of the interaction between the Buyer and Seller. If this is set to true, the Seller provides the costs to expedite the Product Order in the charge attribute	
itemTerm	MEFItemTerm[] *maxItems = 1*	O	Term of the Product Order Item	
milestone	MEFMilestone[]	O	Milestones associated to the Product Order Item. Set by the Seller when a Milestone occurs.	
state	MEFProductOrderItemStateType	M	State of the Product Order Item	
stateChange	MEFProductOrderItemStateChange[] *minItems = 1*	M	State change for the Product Order Item	

Name	Type	M/O	Description	MEF 57.2
terminationError	TerminationError[]	O	When the Seller cannot process the request, the Seller returns a text-based list of reasons here.	

7.2.2.4. Type MEFProductOrderItem_Update

Description: An updatable representation of the Product Order Item.

Name	Type	M/O	Description
endCustomerName	string	O	The name of the End Customer, eithe a business name or an individua name depending on the end customer.
id	string	M	Identifier of the Item. This is t address the Item to be updated withi the Product Order. The id itself cannc be updated.
note	Note[]	O	Free form text to clarify or explain th Product Order Item. Only new note can be entered. The Buyer and Selle cannot modify an existing Note. Th Buyer creates a Note when creatin the Product Order Item or whe updating it. The Seller may add note at any time.
relatedBuyerPON	string	O	This information is not used by th Seller and is maintained for th convenience of the Buyer (e.g. searc purposes).

Name	Type	M/O	Description
relatedContactInformation	RelatedContactInformation[]	O	Contact information of an individual or organization playing a role for this Order Item. Buyer may modify, add or delete only Buyer-related contacts - Buyer Product Order Item Contact `role=buyerProductOrderItemContact` - Buyer Implementation Contact `role=buyerImplementationContact` - Buyer Technical Contact `role=buyerTechnicalContact` - Buyer Fault Contact `role=buyerFaultContact` - Buyer GDPR Contact `role=buyerGDPRContact`

7.2.2.5. enum MEFProductActionType

Description: Action to be performed on the Product that the Order Item refers to.

ProductActionType	MEF 57.2
add	INSTALL
modify	CHANGE
delete	DISCONNECT

7.2.2.6. enum MEFProductOrderItemStateType

Description: Possible values for the state of the Product Order Item The following mapping has been used between `MEFProductOrderItemStateType` and MEF 57.2:

state	MEF 57.2 name	Description
`acknowledged`	ACKNOWLEDGED	A Product Order Item has been received and has passed basic business validations. From the `acknowledged` state the Product Order Item is further validated and depending on the results of the validation and if other Product Order Items in the Product Order are also validated the Product Order Item moves to `inProgress`, `rejected.validated`, or `rejected.unassessed`.
`cancelled`	CANCELLED	The Product Order has moved to the `pendingCancellation` state. All Product Order Items move to `cancelled`.
`completed`	COMPLETED	The Product Order Item has completed provisioning. This is an end state

state	MEF 57.2 name	Description
`failed`	FAILED	The fulfillment of a Product Order Item has failed. A Product Order Item may fail because the Buyer declined a Blocking charge identified via the Charge, the Buyer failed to respond to a Charge Item included in a Charge, or the Seller is unable to fulfill the Product Order Item. A Product Order Item moving to `failed` state results in the Product Order State being `failed` or `partial`. This is a terminal state.
`held`	HELD	The Product Order Item cannot be progressed due to Charge the Seller awaiting a response from the Buyer on a Charge. The Seller stops work on the Product Order Item until the Charge has completed. Upon acceptance by the Buyer of all Blocking charges, the Product Order Item returns to `inProgress` state If the Buyer rejects a Blocking charge, the Product Order Item moves to the `failed` state.
`inProgress`	IN_PROGRESS	The Product Order Item has been successfully validated and fulfillment has started. If the Seller's system links validation between Product Order Items in a Product Order, a Product Order Item in this state also indicates that the other Product Order Items passed validation.
`pending`	PENDING	The Product Order Item cannot be progressed due to the Seller assessing a Cancel Product Order or Modify Product Order Item Requested Delivery Date request. The Seller stops work on the Product Order Item until either the Cancel Product Order has been accepted and the Product Order state moves to `pendingCancellation` and the Product Order Item state moves to `cancelled`, the Cancel Product Order has been rejected and the Product Order Item State moves to `inProgress`, the Modify Product Order Item Requested Delivery Date has been accepted and the Product Order Item State moves to `inProgress`, or the Modify Product Order Item Requested Delivery Date moves to `done.declined` and the Product Order Item state moves to `inProgress` with original delivery dates.
`rejected`	REJECTED	A Product Order Item was submitted, and it has failed at least one validation checks the Seller performs during the `acknowledged` state. Other Product Order Items may continue to be processed.

state	MEF 57.2 name	Description
`rejected.unassessed`	UNASSESSED	A Product Order was submitted and all validation checks the Seller performs during the `acknowledged` state have not been completed, but another Product Order Item in the Product Order has moved to the `rejected` state.
`rejected.validated`	VALIDATED	A Product Order was submitted, and it has passed all validation checks the Seller performs during the `acknowledged` state, but another Product Order Item in the Product Order has moved to the `rejected` state

7.2.2.7. Type MEFProductOrderItemStateChange

Description: Holds the State notification reasons and associated date the State changed, populated by the server

Name	Type	M/O	Description	MEF 57.2
changeDate	date-time *format = date-time*	O	The date on when the state was reached	
changeReason	string	O	Additional comment related to state change.	
state	MEFProductOrderItemStateType	O	Reached state	

7.2.2.8. Type ProductOfferingQualificationItemRef

Description: It's a productOfferingQualification item that has been executed previously.

Name	Type	M/O	Description	MEF 57.2
alternateProductOfferingProposalId	string	O	A unique identifier for this Alternate Product Proposal assigned by the Seller.	
id	string	M	Id of an item of a product offering qualification	
productOfferingQualificationHref	string	O	Reference to a related Product Offering Qualification resource.	
productOfferingQualificationId	string	M	Unique identifier of related Product Offering Qualification resource.	

7.2.2.9. Type ProductOfferingRef

Description: A reference to a Product Offering offered by the Seller to the Buyer. A Product Offering contains the commercial and technical details of a Product sold by a particular Seller. A Product Offering defines all of the commercial terms and, through association with a particular Product Specification, defines all the technical attributes and behaviors of the Product. A Product

Offering may constrain the allowable set of configurable technical attributes and/or behaviors specified in the associated Product Specification.

Name	Type	M/O	Description	MEF 57.2
href	string	O	Hyperlink to a Product Offering in Sellers catalog. In case Seller is not providing a catalog capabilities this field is not used. The catalog API definition is provided by the Seller to the Buyer during onboarding Hyperlink MAY be used by the Seller in responses Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request	
id	string	M	id of a Product Offering. It is assigned by the Seller. The Buyer and the Seller exchange information about offerings' ids during the onboarding process.	

7.2.2.10. Type OrderItemRelationship

Description: The relationship between Product Order Items in the Product Order.

Name	Type	M/O	Description	MEF 57.2
id	string	M	Id of the related Order Item (must be in the same Order).	
relationshipType	string	M	Specifies the nature of the relationship to the related Product Order Item. A string that is one of the relationship types specified in the Product Specification.	

7.2.2.11. Type MEFOrderItemCoordinatedAction

Description: The interval after the completion of one or more related Order Items that this Order Item can be started or completed

Name	Type	M/O	Description	MEF 57.2
coordinatedActionDelay	Duration	M	The period of time for which the coordinated action is delayed.	
coordinationDependency	MEFOrderItemCoordinationDependencyType	M	A dependency between the Order Item and a related Order Item	

Name	Type	M/O	Description	MEF 57.2
itemId	string	M	Specifies Order Item that is to be coordinated with this Order Item.	

7.2.2.12. enum MEFOrderItemCoordinationDependencyType

Description: Possible values of the Order Item Coordination Dependency

OrderItemCoordinationDependencyType	MEF 57.2	Description
startToStart	START_TO_START	Work on the Specified Order Item can only be started after the Coordinated Order Items are started
startToFinish	START_TO_FINISH	The Coordinated Order Items must complete before work on the Specified Order Item begins
finishToStart	FINISH_TO_START	Work on the Related Order Items begins after the completion of the Specified Order Item
finishToFinish	FINISH_TO_FINISH	Work on the Related Order Items completes at the same time as the Specified Order Item

Value	MEF 57.2
startToStart	START_TO_START
startToFinish	START_TO_FINISH
finishToStart	FINISH_TO_START
finishToFinish	FINISH_TO_FINISH

7.2.2.13. Type MEFProductOrderItemRef

Description: It's a ProductOrder item

Name	Type	M/O	Description	MEF 57.2
productOrderHref	string	O	Reference of the related ProductOrder.	
productOrderId	string	M	Unique identifier of a ProductOrder.	
productOrderItemId	string	M	Id of an Item within the Product Order	

7.2.2.14. Type MEFQuoteItemRef

Description: It's a Quote item that has been executed previously.

Name	Type	M/O	Description	MEF 57.2
id	string	M	Id of an Quote Item	
quoteHref	string	O	Reference of the related Quote.	
quoteId	string	M	Unique identifier of a Quote.	

7.2.2.15. Type MEFProductOrderChargeRef

Description: A reference to a Charge instance

Name	Type	M/O	Description	MEF 57.2
href	string	O	Hyperlink to access the Charge	
id	string	M	A unique identifier of the Charge	

7.2.2.16. Type MEFMilestone

Description: Milestones associated to the Product Order Item. Set by the Seller when a Milestone occurs.

Name	Type	M/O	Description	MEF 57.2
date	date-time *format = date-time*	M	The date on when the milestone was reached	
name	string	M	Name of the Milestone.	
note	string	O	Additional comment related to milestone change.	

7.2.3. Product representation

7.2.3.1. Type MEFProductRefOrValueOrder

Description: Used by the Buyer to point to existing and/or describe the desired shape of the product. In case of `add` action - only `productConfiguration` MUST be specified. For `modify` action - both `id` and `productConfiguration` MUST be provided to point which product instance to update and to what state. In `delete` only the `id` must be provided.

Name	Type	M/O	Description	M 57
href	string	O	Hyperlink to the referenced Product. Hyperlink MAY be used by the Seller in responses. Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request.	

Name	Type	M/O	Description	M 57
id	string	O	The unique identifier of an in-service Product that is the ordering subject. This field MUST be populated if an item `action` is either `modify` or `delete`. This field MUST NOT be populated if an item `action` is `add`.	
place	RelatedPlaceRefOrQueryWithSubUnit[]	O	The relationships between this Product Order Item and one or more Places as defined in the Product Specification.	
productConfiguration	MEFProductConfiguration	O	MEFProductConfiguration is used to specify the MEF specific product payload. This field MUST be populated if an item `action` is `add` or `modify`. It MUST NOT be populated when an item `action` is `delete`. The @type is used as a discriminator.	
productOffering	ProductOfferingRef	O	A particular Product Offering defines the technical and commercial attributes and behaviors of a Product.	
productRelationship	ProductRelationship[]	O	A list of references to existing products that are related to the ordered Product.	

7.2.3.2. Type MEFProductConfiguration

Description: MEFProductConfiguration is used as an extension point for MEF specific product/service payload. The `@type` attribute is used as a discriminator

Name	Type	M/O	Description	MEF 57.2
@type	string	M	The name of the type, defined in the JSON schema specified above, for the product that is the subject of the Product Order Request. The named type must be a subclass of MEFProductConfiguration.	

7.2.3.3. Type ProductRelationship

Description: A relationship to an existing Product. The requirements for usage for given Product are described in the Product Specification.

Name	Type	M/O	Description	MEF 57.2
href	string	O	Hyperlink to the product in Seller's inventory that is referenced Hyperlink MAY be used when providing a response by the Seller Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request	
id	string	M	unique identifier of the related Product	
relationshipType	string	M	Specifies the type (nature) of the relationship to the related Product. The nature of required relationships varies for Products of different types. For example, a UNI or ENNI Product may not have any relationships, but an Access E-Line may have two mandatory relationships (related to the UNI on one end and the ENNI on the other). More complex Products such as multipoint IP or Firewall Products may have more complex relationships. As a result, the allowed and mandatory `relationshipType` values are defined in the Product Specification.	

7.2.4. Place representation

7.2.4.1. Type RelatedPlaceRefOrQueryWithSubUnit

Description: Allows pointing to a place by referring a GeographicAddress, GeographicSite, or providing GeographicAddress by value. It also provides additional information like the `role` the place plays for given Product, `subUnit` to provide more detailed information about the precise location of the installation and `contact` needed access to this place.

Name	Type	M/O	Description	MEF 57.2
place	PlaceRefOrQuery	M		
role	string	M	Role of this place. The values that can be specified here are described by Product Specification (e.g. "INSTALL_LOCATION").	
subUnit	SubUnit[]	O	A list of zero or more sub units included within the boundary of the `place` for this POQ Item. This is a list to allow complex sub-unit information such as SUITE 42 ROOM A	
contact	ContactInformation[]	O	The person to call to get access to this place in case such access is required to complete the evaluation of this POQ Item.	

7.2.4.2. Type PlaceRefOrQuery

Description: A place described by reference to Geographic Address, Geographic Site or by Geographic Address Representations.

7.2.4.3. Type GeographicAddress_Query

Description: A list of representations being a subset of Geographic Address entity. This is to be used when providing a list of representations to validate or search for an Geographic Address

Name	Type	M/O	Description	MEF 57.2
fieldedAddressRepresentation	FieldedAddressRepresentation[]	O	A Fielded Address representation	
formattedAddressRepresentation	FormattedAddressRepresentation[]	O	A Formatted Address representation	
geographicPointRepresentation	GeographicPointRepresentation[]	O	A Fielded Address representation	
labelRepresentation	LabelRepresentation[]	O	A Fielded Address representation	

7.2.4.4. Type FieldedAddressRepresentation

Description: A type of Address that has a discrete field and value for each type of boundary or identifier down to the lowest level of detail. For example "street number" is one field, "street name" is another field, etc.

Name	Type	M/O	Description	MEF 150
streetNr	string	O	Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses.	
streetNrSuffix	string	O	The first street number suffix (in a street number range) or the suffix for the street number if there is no range	
streetNrLast	string	O	Last number in a range of street numbers allocated to an Address	
streetNrLastSuffix	string	O	Last street number suffix for a ranged Address	
streetPreDirection	string	O	The direction of the street that appears before the Street Name	
streetName	string	O	Name of the street or other street type	
streetType	string	O	The type of street (e.g., alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf)	

Name	Type	M/O	Description	MEF 150
streetPostDirection	string	O	A modifier denoting a relative direction that appears after the Street Name.	
poBox	string	O	Number identifying a specific location in a post office.	
locality	string	O	An area of defined or undefined boundaries within a local authority or other legislatively defined area.	
city	string	O	City in which the Address is located.	
postcode	string	O	A descriptor for a postal delivery area used to speed and simplify the delivery of mail (also known as zip code)	
postcodeExtension	string	O	The extension used on a postal code. Note: there are different use codes for this attribute depending upon the country.	
stateOrProvince	string	O	The State or Province in which the Address is located.	
countryCode	string *minLength = 2* *maxLength = 2*	O	Country in which the Address is located, defined using two characters as defined in ISO 3166	
subUnit	SubUnit[]	O	The Sub Unit represented as a list. This is a list to allow complex sub-unit information such as SUITE 42 ROOM A	
buildingName	string	O	The well-known name of a building that is located at this Address (e.g., where there is one Address for a campus).	
privateStreetNumber	string	O	Street number on a private street within the Address.	
privateStreetName	string	O	Private streets internal to a property (e.g., a university) may have internal names that are not recorded by the land title office.	
language	string *minLength = 2* *maxLength = 2*	O	The language in which the address is expressed. It MUST use the ISO 639:2023 two letter code 639:2023	

7.2.4.5. Type FormattedAddressRepresentation

Description: A freeform text representation agreed to by the Buyer and Seller.

Name	Type	M/O	Description	MEF 57.2
formattedAddress	string	M	A formatted Address Representation that contains a non-fielded address.	
language	string *minLength = 2* *maxLength = 2*	O	The language in which the address is expressed. Based on ISO 639:2023	

7.2.4.6. Type GeographicPointRepresentation

Description: A GeographicPointRepresentation defines a geographic point through coordinates.

Name	Type	M/O	Description	MEF 57.2
spatialRef	string	M	The spatial reference system used to determine the coordinates. The system used and the value of this field are to be agreed during the onboarding process.	
latitude	string	M	The latitude expressed in the format specified by the `spacialRef`	
longitude	string	M	The longitude expressed in the format specified by the `spacialRef`	
elevation	string	O	The elevation expressed in the format specified by the `spacialRef`	

7.2.4.7. Type LabelRepresentation

Description: A unique identifier controlled by a generally accepted independent administrative authority that specifies a fixed geographical location.

Name	Type	M/O	Description	MEF 57.2
label	string	M	The unique reference to an Geographic Address assigned by the Administrative Authority.	
administrativeAuthority	string	M	The organization or standard from the organization that administers this Geographic Address Label ensuring it is unique within the Administrative Authority.	
language	string *minLength = 2* *maxLength = 2*	O	The language in which the label is expressed. Based on ISO 639:2023	

7.2.4.8. Type GeographicAddressRef

Description: A reference to a Geographic Address resource available through Address Validation API.

Name	Type	M/O	Description	MEF 57.2
href	string	O	Hyperlink to the referenced Address. Hyperlink MAY be used by the Seller in responses. Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request.	

Name	Type	M/O	Description	MEF 57.2
id	string	M	Identifier of the referenced Geographic Address. This identifier is assigned during a successful address validation request (Geographic Address Management API)	
@type	string	M	Used to unambiguously designate the class type when using `oneOf`	

7.2.4.9. Type GeographicSiteRef

Description: A reference to a Geographic Site resource available through Service Site API

Name	Type	M/O	Description	MEF 57.2
href	string	O	Hyperlink to the referenced Site. Hyperlink MAY be used by the Seller in responses. Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request.	
id	string	M	Identifier of the referenced Geographic Site.	
@type	string	M	Used to unambiguously designate the class type when using `oneOf`	

7.2.4.10. Type SubUnit

Description: Allows for sub unit identification

Name	Type	M/O	Description	MEF 57.2
subUnitNumber	string	M	The discriminator used for the subunit, often just a simple number but may also be a range.	
subUnitType	string	M	The type of subunit e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.	

7.2.5. Cancel Product Order

7.2.5.1. Type CancelProductOrder_Create

Description: Request for cancellation an existing product order Skipped properties: id,href,state,effectiveCancellationDate

Name	Type	M/O	Description	M 5
cancellationReason	string	O	An optional attribute that allows the Buyer to provide additional detail to the Seller on their reason for cancelling the Product Order	

Name	Type	M/O	Description
cancellationReasonType	CancellationReasonType	O	Identifies the type of reason, Technical or Commercial, for the Cancellation request
productOrder	MEFProductOrderRef	M	A reference to a Product Order that the buyer wishes to cancel.
relatedContactInformation	RelatedContactInformation[] *minItems = 1*	M	Contact information of an individual or organization playing a role for this Cancel Product Order. The rule for mapping a represented attribute value to a `role` is to use the _lowerCamelCase_ pattern e.g. - Cancel Product Order Contact: `role=cancelProductOrderContact`

7.2.5.2. Type CancelProductOrder

Description: Request for cancellation an existing product order

Name	Type	M/O	Description
cancellationDeniedReason	string	O	If the Cancel Product Order requ denied by the Seller, the Seller pro a reason to the Buyer using attribute.
cancellationReason	string	O	An optional attribute that allow Buyer to provide additional detail t Seller on their reason for cancellin Product Order
cancellationReasonType	CancellationReasonType	O	Identifies the type of reason, Tech or Commercial, for the Cancell request
charge	MEFProductOrderChargeRef	O	The Charge Identifier of any ch that are related to the Cancel Pr Order.
href	string	O	Hyperlink to the cancellation rec Hyperlink MAY be used by the Sel responses Hyperlink MUST be ig by the Seller in case it is provide the Buyer in a request
id	string	M	Unique identifier for the Cancel Pr Order that is generated by the S when the Cancel Product Order re `state` is set to `acknowledged`
productOrder	MEFProductOrderRef	M	A reference to a Product Order th Buyer wishes to cancel.

Name	Type	M/O	Description
relatedContactInformation	RelatedContactInformation[]	M	Contact information of an individu organization playing a role for Cancel Product Order. The rule mapping a represented attribute val a `role` is to use the _lowerCamelC pattern e.g. - Cancel Product C Contact: `role=cancelProductOrderContact` Cancel Product Order Seller Con `role=cancelProductOrderSellerCon
state	MEFChargeableTaskStateType	M	The states as defined by TMF622 extended to meet MEF requirem These states are used to convey Cancel Product Order status durin lifecycle of the Product Order.

7.2.5.3. Type CancelProductOrder_Find

Description: A response to a Buyer's get List of Cancel Product Orders

Name	Type	M/O	Description	MEF 57.2
cancellationReasonType	CancellationReasonType	M	Identifies the type of reason, Technical or Commercial, for the Cancellation request	
id	string	M	Unique identifier for the Cancel Product Order that is generated by the Seller when the Cancel Product Order request `state` is set to `acknowledged`	
productOrder	MEFProductOrderRef	M	A reference to a Product Order that the Buyer wishes to cancel.	
state	MEFChargeableTaskStateType	M	The states as defined by TMF622 and extended to meet MEF requirements. These states are used to convey the Cancel Product Order status during the lifecycle of the Product Order.	

7.2.5.4. enum CancellationReasonType

Description: Identifies the type of reason, Technical or Commercial, for the Cancellation request

Value	MEF 57.2
technical	TECHNICAL
commercial	COMMERCIAL

7.2.5.5. Type MEFProductOrderRef

Description: Holds the MEF Product Order reference

Name	Type	M/O	Description	MEF 57.2
productOrderHref	string	O	Hyperlink to access the order	
productOrderId	string	M	Unique (within the ordering domain) identifier for the order that is generated by the seller when the order is initially accepted.	

7.2.6. Charge

7.2.6.1. Type MEFProductOrderCharge

Description: When non-recurring or updated recurring charges are identified by the Seller during their processing of a Product Order, the Seller must communicate these charges to the Buyer and the Buyer must respond to the Seller informing the Seller if they accept or reject each charge. The Seller indicates for each charge, if the charge is Blocking or non-Blocking. If the Buyer rejects a Blocking Charge, the Seller will cancel that Product Order Item and any related Product Order Items. If the Buyer rejects a non-blocking Charge, the Seller may proceed with fulfillment of the Product Order Item.

Name	Type
cancelProductOrder	MEFCancelProductOrderRef
chargeItem	MEFProductOrderChargeItem[]
creationDate	date-time *format = date-time*

Name	Type
href	string
id	string
modifyProductOrderItemRequestedDeliveryDate	MEFModifyProductOrderItemRequestedDeliveryDa
productOrder	MEFProductOrderRef

Name	Type
productOrderItem	MEFProductOrderItemRef
responseDueDate	date-time *format = date-time*
state	MEFProductOrderChargeStateType

7.2.6.2. Type MEFProductOrderCharge_Update

Description: A subset of MEFProductOrderCharge that is allowed to be updated by the Buyer

Name	Type	M/O	Description	MEF 57.2
chargeItem	MEFProductOrderChargeItem_Update[] *minItems = 1*	M	A list of Charge Items contained in the Charge	

7.2.6.3. Type MEFProductOrderCharge_Find

Description: A response object for Buyer's get Charge List request.

Name	Type	M/O	Description	MEF 57.2
creationDate	date-time *format = date-time*	M	Date that the Charge was created by the Seller.	
id	string	M	A unique identifier of the Charge	
productOrder	MEFProductOrderRef	O	Product Order which the Seller is communicating additional or modified charges to the Buyer. This relation MUST be set when the Charge applies to a Product Order. (Caused by Cancel Product Order request)	
productOrderItem	MEFProductOrderItemRef	O	Product Order Item which the Seller is communicating additional or modified charges to the Buyer. This relation MUST be set when the Charge applies to a Product Order Item. (Identified by Seller or caused by Modify Product Order Item Requested Delivery Date request)	
responseDueDate	date-time *format = date-time*	M	The date by which the Buyer must respond to the Seller's Charge. If there is no response received by the Due Date the Seller will treat all charges as declined and move them to `declinedByBuyer` status and put the Charge to `completed` status.	
state	MEFProductOrderChargeStateType	M	The state of the Charge	

7.2.6.4. `enum` MEFProductOrderChargeActivityType

Description: Possible values for the state of the Charge Activity Type

Value	MEF 57.2
new	NEW
change	CHANGE

7.2.6.5. `enum` MEFProductOrderChargeStateType

Description: Possible values for the state of the Charge

State	Description
`completed`	All Charge Items included in the Charge for a given Product Order Item have moved to either the `accepted` state or the `declined` state.
`awaitingResponse`	A Charge has been initiated by the Buyer. The charge includes one or more charges.
`timeout`	A response has not been received from the Buyer within the `responseDueDate`. This is treated as if the Buyer declined the Charge Items.
`withdrawnBySeller`	The Seller determines that the Charge is incorrect. They withdraw the Charge and initiate a new Charge with the required correction(s).

7.2.6.6. Type MEFProductOrderChargeItem

Description: A single component part of the Charge

Name	Type	M/O	Description	MEF 57.2
acceptanceIndicator	MEFAcceptedRejectedType	O	Indicates if the Buyer has accepted the specified charge.	
activityType	MEFProductOrderChargeActivityType	M	Indicates if this is a new charge or a change to a charge provided in a Quote or in a previous accepted Charge Item.	

Name	Type	M/O	Description	MEF 57.2
blocking	boolean	M	Indicates if rejecting the charge will cause the Seller to cancel the Product Order Item, or close the Cancel Product Order or Modify Product Order Item Requested Delivery Date without action.	
id	string	M	An identifier that is unique among all Charge Items within a Charge	
note	Note[]	O	Free form text to clarify or explain the Charge Item. Only new notes can be entered. The Seller cannot modify an existing Note.	
price	Price	M	The value of the Price associated with the Charge Item	
priceCategory	MEFPriceCategory	M	The category of the price	
priceType	MEFPriceType	M	The type of the price.	

Name	Type	M/O	Description	MEF 57.2
recurringChargePeriod	Duration	O	Used for a Charge Item with a priceType = recurring to indicate the period	
state	MEFProductOrderChargeItemStateType	M	The state of the Charge Item	
unitOfMeasure	string	O	Unit of Measure if price depending on it is usageBased (Gb, SMS volume, etc..)	

7.2.6.7. Type MEFProductOrderChargeItem_Update

Description: A type used to perform Buyer's response to a Charge Item - to accept or reject it.

Name	Type	M/O	Description	MEF 57.2
acceptanceIndicator	MEFAcceptedRejectedType	M	Indicates if the Buyer has accepted the specified charge	
id	string	M	An identifier that is unique among Charge. Used for Charge Item matching, not to be update.	
note	Note[]	O	Free form text to clarify or explain the Charge Item. Only new notes can be entered. The Seller cannot modify an existing Note.	

7.2.6.8. enum MEFProductOrderChargeItemStateType

Description: Possible values for the state of the Charge Item

State	MEF 57.2 Name	Description

State	MEF 57.2 Name	Description
`awaitingResponse`	AWAITING_RESPONSE	A Charge has been initiated by the Buyer. The charge includes one or more charges related to a Product Order or Product Order Item. Buyer has not indicated whether they accept or reject the charges via a Respond to Charge request.
`completed`	COMPLETED	All Charge Items included in the Charge have moved to either the `acceptedByBuyer` state, the `declinedByBuyer` state, or the `withdrawnBySeller` state.
`timeout`	TIMEOUT	A Charge Item has been declined by the Buyer. The referenced Product Order and Product Order Items are updated. If a Blocking charge is declined, the Seller may cancel the referenced Product Order Item and any related Product Order Items, the related Cancel Product Order, or the related Modify Product Order Item Requested Delivery Date.
`withdrawnBySeller`	WITHDRAWN_BY_SELLER	The Seller determines that the Charge Item is incorrect. They withdraw the Charge Item and initiate a new Charge with the required correction(s) if needed.

7.2.6.9. `enum` MEFPriceCategory

Description: A description of the cause of the Charge Item

Value	MEF 57.2
cancellation	CANCELLATION
construction	CONSTRUCTION
connection	CONNECTION
disconnect	DISCONNECT
expedite	EXPEDITE
other	OTHER

7.2.6.10. Type MEFCancelProductOrderRef

Description: A reference to a Cancel Product Order instance

Name	Type	M/O	Description	MEF 57.2
href	string	O	Hyperlink to access the Cancel Product Order	
id	string	M	A unique identifier of the Cancel Product Order	

7.2.6.11. Type MEFModifyProductOrderItemRequestedDeliveryDateRef

Description: a reference to Modify Product Order Item Requested Delivery Date

Name	Type	M/O	Description	MEF 57.2
href	string	O	Hyperlink to access the Modify Product Order Item Requested Delivery Date	
id	string	M	A unique identifier of the Modify Product Order Item Requested Delivery Date	

7.2.7. Modify Product Order Item Requested Delivery Date

7.2.7.1. Type MEFModifyProductOrderItemRequestedDeliveryDate_Create

Description: A request initiated by the Buyer to modify the Requested Requested Delivery Date or the Expedite Indicator of a Product Order Item.

Name	Type	M/O	Description	MEF 57.2
expediteIndicator	boolean *default = false*	O	Indicates that expedited treatment is requested. Set by the Buyer. Default Value = FALSE. If this is set to TRUE, the Buyer sets the Requested Completion Date to the expedited date	
productOrderItem	MEFProductOrderItemRef	M	A reference to the Product Order Item to be modified.	
requestedCompletionDate	date-time *format = date-time*	O	Identifies the Buyer's desired due date (requested delivery date)	

7.2.7.2. Type MEFModifyProductOrderItemRequestedDeliveryDate

Description: A response to a request initiated by the Buyer to modify the Requested Completion Date or the Expedite Indicator of a Product Order Item.

Name	Type	M/O	Description
creationDate	date-time *format = date-time*	M	Date that the Modify Product Ord Delivery Date was created in the S the id was assigned
expediteIndicator	boolean *default = false*	O	Indicates that expedited treatment i the Buyer. Default Value = FALS TRUE, the Buyer sets the Requeste to the expedited date

Name	Type	M/O	Description
href	string	O	Hyperlink to the modification r MAY be used by the Seller in re MUST be ignored by the Seller in by the Buyer in a request
id	string	M	Unique identifier MEFModifyProductOrderItemRequ that is generated by the S MEFModifyProductOrderItemRequ request is moved to the 'acknowledg
productOrderItem	MEFProductOrderItemRef	M	A reference to the Product Order Ite
requestedCompletionDate	date-time *format = date-time*	O	Identifies the Buyer's desired du delivery date)
state	MEFChargeableTaskStateType	M	The state of the Modify Product Orc Delivery Date request

7.2.8. Notification registration

Notification registration and management are done through `/hub` API endpoint. The below sections describe data models related to this endpoint.

7.2.8.1. Type EventSubscriptionInput

Description: This class is used to register for Notifications.

Name	Type	M/O	Description
callback	string	M	This callback value must be set to *host* property from (productOrderNotification.api.yaml). This property is appended with the base API to construct an URL to which notification is sent. E.g. for "callback": `ht order state change event notification `https://buyer.mef.com/listenerEndpoint/mefApi/sonata/productOrderingNotifi
query	string	O	This attribute is used to define to which type of events to productOrderStateChangeEvent". To subscribe for more than one even `eventType=productOrderStateChangeEvent,productOrderItemStateChangeEv 'ProductOrderEventType', `CancelProductOrderEventType` in productOrderN specifying no filters - ending in subscription for all event types.

7.2.8.2. Type EventSubscription

Description: This resource is used to respond to notification subscriptions.

Name	Type	M/O	Description	MEF 57.2
callback	string	M	The value provided by the Buyer in `EventSubscriptionInput` during notification registration	

Name	Type	M/O	Description	MEF 57.2
id	string	M	An identifier of this Event Subscription assigned by the Seller when a resource is created.	
query	string	O	The value provided by the Buyer in `EventSubscriptionInput` during notification registration	

7.2.9. Common

Types described in this subsection are shared among two or more Cantata and Sonata APIs.

7.2.9.1. Type Duration

Description: A Duration in a given unit of time e.g. 3 hours, or 5 days.

Name	Type	M/O	Description	MEF 57.2
amount	integer *minimum = 0*	M	Duration (number of seconds, minutes, hours, etc.)	
units	TimeUnit	M	Time unit enumerated	

7.2.9.2. enum MEFAcceptedRejectedType

Description: Indicator of acceptance

Value	MEF 57.2
accepted	ACCEPTED
rejected	REJECTED

7.2.9.3. Type MEFBillingAccountRef

Description: A reference to the Buyer's Billing Account

Name	Type	M/O	Description	MEF 57.2
id	string	M	Identifies the buyer's billing account to which the recurring and non-recurring charges for this order or order item will be billed. Required if the Buyer has more than one Billing Account with the Seller and for all new Product Orders.	

7.2.9.4. enum MEFBuyerSellerType

Description: Indicates if the note is from Buyer or Seller.

Value	MEF 57.2
buyer	BUYER
seller	SELLER

7.2.9.5. `enum` MEFChargeableTaskStateType

Description: The states as defined by TMF622 and extended to meet MEF requirements.

Name	MEF 57.2 Name	Description
inProgress.assessingCharge	ACCESSING_CHARGE	The Modify Product Order Item Requested Delivery Date request results in a Charge being initiated by the Seller. The Modify Product Order Item Requested Delivery Date remains in this state until the Charge is completed or withdrawn by the Seller. All charges within a Charge that was initiated due to a Modify Product Order Item Requested Delivery Date are considered Blocking charges. If any charge is not accepted by the Buyer, the Modify Product Order Item Requested Delivery Date moves from the `inProgress.assessingCharge` state to the `done.declined` state.
acknowledged	ACKNOWLEDGED	A Modify Product Order Item Requested Delivery Date request has been received and has passed basic validation. The Modify Product Order Item Requested Delivery Date Identifier is assigned in the `acknowledged` state. Validation of Modify Product Order Item Requested Delivery Date attributes as applicable is completed in the `acknowledged` state.
done	ACCEPTED	A Modify Product Order Item Requested Delivery Date request has been received, passed all validations, if a Charge is associated all Charge Items have been accepted by the Buyer, and the Product Order Item Completion Date has been updated as requested.
done.declined	DECLINED	Blocking charges associated with a Modify Product Order Item Requested Delivery Date have been declined by the Buyer. No updates are made to the Product Order Item.

Name	MEF 57.2 Name	Description
rejected	REJECTED	A Modify Product Order Item Requested Delivery Date request was submitted by the Buyer, and it has failed any validation checks the Seller performs during the `acknowledged` state. No updates are made to the referenced Product Order Item.

Value	MEF 57.2
acknowledged	ACKNOWLEDGED
done	DONE
done.declined	DONE.DECLINED
inProgress.assessingCharge	IN_PROGRESS.ASSESSING_CHARGE
rejected	REJECTED

7.2.9.6. `enum` MEFEndOfTermAction

Description: The action the Seller will take once the term expires. Roll indicates that the Product's contract will continue on a rolling basis for the duration of the Roll Interval at the end of the Term.
Auto-disconnect indicates that the Product will be disconnected at the end of the Term. Auto-renew indicates that the Product's contract will be automatically renewed for the Term Duration at the end of the Term.

Value	MEF 57.2
roll	ROLL
autoDisconnect	AUTO_DISCONNECT
autoRenew	AUTO_RENEW

7.2.9.7. Type MEFItemTerm

Description: The term of the Item

Name	Type	M/O	Description	MEF 57.2
description	string	O	Description of the term	
duration	Duration	M	Duration of the term	
endOfTermAction	MEFEndOfTermAction	M	The action that needs to be taken by the Seller once the term expires	
name	string	M	Name of the term	
rollInterval	Duration	O	The recurring period that the Buyer is willing to pay for the Product after the original term has expired.	

7.2.9.8. `enum` MEFPriceType

Description: Indicates if the price is for recurring or non-recurring charges.

Value	MEF 57.2
recurring	RECURRING
nonRecurring	NON_RECURRING
usageBased	USAGE_BASED

7.2.9.9. Type Money

Description: A base/value business entity used to represent money

Name	Type	M/O	Description	MEF 57.2
unit	string	M	Currency (ISO4217 norm uses 3 letters to define the currency)	
value	float *format = float*	M	A positive floating point number	

7.2.9.10. Type Note

Description: Extra information about a given entity. Only useful in processes involving human interaction. Not applicable for the automated process.

Name	Type	M/O	Description	MEF 57.2
author	string	M	Author of the note	
date	date-time *format = date-time*	M	Date the Note was created	
id	string	M	Identifier of the note within its containing entity (may or may not be globally unique, depending on provider implementation)	
source	MEFBuyerSellerType	M	Indicates if the note is from Buyer or Seller	
text	string	M	Text of the note	

7.2.9.11. Type Price

Description: Provides all amounts (tax included, duty-free, tax rate) and used currency of a Price

Name	Type	M/O	Description	MEF 57.2
dutyFreeAmount	Money	M	All taxes excluded amount (expressed in the given currency)	
taxIncludedAmount	Money	O	All taxes included amount (expressed in the given currency)	
taxRate	float *format = float*	O	Price Tax Rate. Unit: [%]. E.g. value 16 stand for 16% tax.	

7.2.9.12. Type ContactInformation

Description: Contact data for a person or organization that is involved in the product offering qualification. In a given context it is always specified by the Seller (e.g. Seller Contact Information) or by the Buyer.

Name	Type	M/O	Description	MEF 57.2
number	string	M	Phone number	
emailAddress	string	M	Email address	
postalAddress	FieldedAddressRepresentation	O	Identifies the postal address of the person or office to be contacted.	
organization	string	O	The organization or company that the contact belongs to	
name	string	M	Name of the contact	
numberExtension	string	O	Phone number extension	

7.2.9.13. Type RelatedContactInformation

Description: Contact information of an individual or organization playing a role for this Order Item. The rule for mapping a represented attribute value to a `role` is to use the *lowerCamelCase* pattern e.g.

- Buyer Order Item Contact: `role=buyerOrderItemContact`
- Buyer Implementation Contact: `role=buyerImplementationContact`
- Buyer Technical Contact: `role=buyerTechnicalContact`

Name	Type	M/O	Description	MEF 57.2
emailAddress	string	M	Email address	
name	string	M	Name of the contact	
number	string	M	Phone number	
numberExtension	string	O	Phone number extension	
organization	string	O	The organization or company that the contact belongs to	
postalAddress	FieldedAddressRepresentation	O	Identifies the postal address of the person or office to be contacted.	
role	string	M	A role the party plays in a given context.	

The `role` attribute is used to provide a reason the particular party information is used. It can result from MEF 57.2 requirements (e.g. Seller Contact Information) or from the Product Specification requirements.

The rule for mapping a represented attribute value to a `role` is to use the *lowerCamelCase* pattern e.g.

- Seller Contact: `role` equal to `sellerContact`
- Buyer Contact Information: `role` equal to `buyerContactInformation`

7.2.9.14. Type TerminationError

Description: This indicates an error that caused an Item to be terminated. The code and propertyPath should be used like in Error422.

Name	Type	Description
code	Error422Code	One of the following error codes: - missingProperty: The property the Seller has expected is not present in the payload - invalidValue: The property has an incorrect value - invalidFormat: The property value does not comply with the expected value format - referenceNotFound: The object referenced by the property cannot be identified in the Seller system - unexpectedProperty: Additional property, not expected by the Seller has been provided - tooManyRecords: the number of records to be provided in the response exceeds the Seller's threshold. - otherIssue: Other problem was identified (detailed information provided in a reason)
propertyPath	string	A pointer to a particular property of the payload that caused the validation issue. It is highly recommended that this property should be used. Defined using JavaScript Object Notation (JSON) Pointer (https://tools.ietf.org/html/rfc6901).
value	string	Text to describe the reason of the termination.

7.2.9.15. `enum` TimeUnit

Description: Represents a unit of time.

Value	MEF 57.2
seconds	SECONDS
minutes	MINUTES
businessHours	BUSINESS_HOURS
calendarHours	CALENDAR_HOURS
businessDays	BUSINESS_DAYS
calendarDays	CALENDAR_DAYS
months	MONTHS
years	YEARS

7.3. Notification API Data model

Figure 33 presents the Product Order Management Notification data model. The data types, requirements related to them and mapping to MEF 57.2 are discussed later in this section.

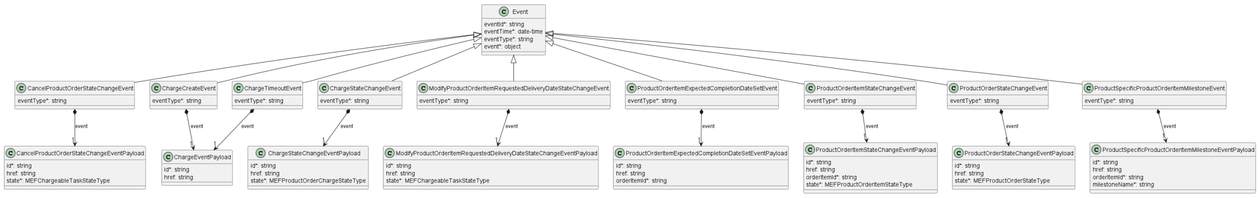

Figure 33. Product Order Management Notification Data Model

This data model is used to construct requests and responses of the API endpoints described in Section 5.2.2.

7.3.1. Type Event

Description: Event class is used to describe information structure used for notification.

Name	Type	M/O	Description	MEF 57.2
eventId	string	M	Id of the event	Not represented in MEF 57.2
eventTime	date-time *format = date-time*	M	Date-time when the event occurred	Not represented in MEF 57.2
eventType	string	M	The type of the notification.	
event	object	M	The event linked to the involved resource object	

7.3.2. Type ProductOrderItemExpectedCompletionDateSetEvent

Description: productOrderItemExpectedCompletionDateSetEvent structure

Inherits from:

- Event

Name	Type	M/O	Description	MEF 57.2
eventType	string	M	Indicates the type of the event.	
event	ProductOrderItemExpectedCompletionDateSetEventPayload	M	A reference to the Product Order that is source of the notification.	

7.3.3. Type ProductOrderItemExpectedCompletionDateSetEventPayload

Description: A reference to the Product Order Item that is the source of the notification.

Name	Type	M/O	Description	MEF 57.2
id	string	M	The Product Order unique identifier.	

Name	Type	M/O	Description	MEF 57.2
href	string	O	Link to the ProductOrder	
orderItemId	string	M	ID of the Product Order Item (within the Product Order) which state change triggered the event	

7.3.4. Type ProductOrderItemStateChangeEvent

Description: ProductOrderItemStateChangeEvent structure

Inherits from:

- Event

Name	Type	M/O	Description	MEF 57.2
eventType	string	M	Indicates the type of the event.	
event	ProductOrderItemStateChangeEventPayload	M	A reference to the Product Order that is source of the notification.	

7.3.5. Type ProductOrderItemStateChangeEventPayload

Description: A reference to the Product Order Item that is the source of the notification.

Name	Type	M/O	Description	MEF 57.2
id	string	M	The Product Order unique identifier.	
href	string	O	Link to the ProductOrder	
orderItemId	string	M	ID of the Product Order Item (within the Product Order) which state change triggered the event	
state	MEFProductOrderItemStateType	M	The state reached	

7.3.6. Type ProductOrderStateChangeEvent

Description: ProductOrderStateChangeEvent structure

Inherits from:

- Event

Name	Type	M/O	Description	MEF 57.2
eventType	string	M	Indicates the type of product order event.	

Name	Type	M/O	Description	MEF 57.2
event	ProductOrderStateChangeEventPayload	M	A reference to the Product Order that is source of the notification.	

7.3.7. Type ProductOrderStateChangeEventPayload

Description: A reference to the Product Order that is the source of the notification.

Name	Type	M/O	Description	MEF 57.2
id	string	M	The Product Order unique identifier.	
href	string	O	Link to the Product Order	
state	MEFProductOrderStateType	M	The state reached	

7.3.8. Type ProductSpecificProductOrderItemMilestoneEvent

Description: ProductSpecificProductOrderItemMilestoneEvent structure

Inherits from:

- Event

Name	Type	M/O	Description	MEF 57.2
eventType	string	M	Indicates the type of the event.	
event	ProductSpecificProductOrderItemMilestoneEventPayload	M	A reference to the Product Order that is source of the notification.	

7.3.9. Type ProductSpecificProductOrderItemMilestoneEventPayload

Description: A reference to the Product Order Item that is the source of the notification.

Name	Type	M/O	Description	MEF 57.2
id	string	M	The Product Order unique identifier.	
href	string	O	Link to the ProductOrder	
orderItemId	string	M	ID of the Product Order Item (within the Product Order) which state change triggered the event	

Name	Type	M/O	Description	MEF 57.2
milestoneName	string	M	The name of the Milestone that was reached by give Product Order or Product Order Item. Mandatory for Product Specific Milestone reached events.	

7.3.10. `enum` MEFProductOrderStateType

Description: Possible values for the state of the Product Order The following mapping has been used between `MEFProductOrderStateType` and MEF 57.2:

state	MEF 57.2 name	Description
`acknowledged`	ACKNOWLEDGED	A Product Order has been received by the Seller and has passed basic validation. A `productOrder.id` is assigned in the `acknowledged` state and a response is returned to the Buyer. The Product Order remains in the `acknowledged` state while validations of Product Order and Product Order Item(s) attributes as applicable is completed. If the Product Order and Product Order Item attributes are validated the Product Order moves to the `inProgress` state. If not validated, the Product Order moves to the `rejected` state.

state	MEF 57.2 name	Description
`assessingCancellation`	ASSESSING_CANCELLATION	A Cancel Product Order request has been received by the Seller. The Product Order is being assessed to determine if the Product Order can be cancelled. If there are charges associated with cancelling the Product Order, these are communicated to the Buyer using the Charge process. The Product Order remains in the `assessingCancellation` state until any relevant Charge is completed or withdrawn by the Seller. Once the Buyer's request has been validated and any associated Charges completed, the Product Order moves to the `pendingCancellation` state. If the request is not validated or if any associated Charges are not completed, the Product Order moves to the `inProgress` state and the Product Order is not cancelled.
`held.assessingCharge`	ASSESSING_CHARGE	A Charge has been initiated by the Seller that is not the result of a Modify Product Order Item Requested Delivery Date or Cancel Product Order request and the Seller is awaiting a Buyer response to the Charge. If a blocking or non-blocking charge is accepted by the Buyer, the Product Order moves to `inProgress`. If a non-blocking charge is declined by the Buyer, the Product Order moves to `inProgress`. If a blocking charge is declined by the Buyer and there are no unrelated Product Order Items in the Product Order, the Product Order moves to the `inProgress` and then to the `failed` state. If a blocking charge is declined by the Buyer and there are unrelated Product Order Items in the Product Order, the Product Order moves to the `inProgress` state.

state	MEF 57.2 name	Description
`pending.assessingModification`	ASSESSING_MODIFICATION	A request has been made by the Buyer to modify either the `expediteIndicator` or the `requestedCompletionDate` of a Product Order Item. The Product Order Item is currently being assessed to determine whether the Modify Product Order Item Requested Delivery Date is valid. If there is a charge associated with the Modify Product Order Item Requested Delivery Date, the Product Order remains in the `pending.assessingModification` state until the Charge is completed or withdrawn by the Seller. Once the Buyer's request has been validated and any associated Charges completed, the Product Order returns to the `inProgress` state.
`cancelled`	CANCELLED	The Product Order has been successfully cancelled. This is a terminal state.
`pendingCancellation`	CANCELLING	The Buyer's Cancel Request has been assessed and it has been determined that it is feasible to proceed with the cancellation. This state can also result from a Seller cancelling the Product Order within their systems without a request from the Buyer.
`completed`	COMPLETED	The Product Order has completed fulfillment and the Product is now active. This is a terminal state
`failed`	FAILED	All Product Order Items have failed which results in the entire Product Order failing. This is a terminal state.
`inProgress`	IN_PROGRESS	The Product Order has been successfully validated, and fulfillment has started.
`partial`	PARTIAL	At least one Product Order Item is `failed` or `rejected`, and fulfillment of at least one Product Order Item has been successful. This is a terminal state.

state	MEF 57.2 name	Description
`rejected`	REJECTED	A Product Order was submitted, and it has failed at least one of the validation checks the Seller performs after it reached the `acknowledged` state

Value	MEF 57.2
acknowledged	ACKNOWLEDGED
assessingCancellation	ASSESSING_CANCELLATION
cancelled	CANCELLED
completed	COMPLETED
failed	FAILED
held.assessingCharge	HELD.ASSESSING_CHARGE
inProgress	IN_PROGRESS
partial	PARTIAL
pending.assessingModification	PENDING.ASSESSING_MODIFICATION
pendingCancellation	PENDING_CANCELLATION
rejected	REJECTED

7.3.11. `enum` MEFProductOrderItemStateType

Description: Possible values for the state of the Product Order Item The following mapping has been used between `MEFProductOrderItemStateType` and MEF 57.2:

state	MEF 57.2 name	Description
`acknowledged`	ACKNOWLEDGED	A Product Order Item has been received and has passed basic business validations. From the `acknowledged` state the Product Order Item is further validated and depending on the results of the validation and if other Product Order Items in the Product Order are also validated the Product Order Item moves to `inProgress`, `rejected.validated`, or `rejected.unassessed`.
`cancelled`	CANCELLED	The Product Order has moved to the `pendingCancellation` state. All Product Order Items move to `cancelled`.
`completed`	COMPLETED	The Product Order Item has completed provisioning. This is an end state
`failed`	FAILED	The fulfillment of a Product Order Item has failed. A Product Order Item may fail because the Buyer declined a Blocking charge identified via the Charge, the Buyer failed to respond to a Charge Item included in a Charge, or the Seller is unable to fulfill the Product Order Item. A Product Order Item moving to `failed` state results in the Product Order State being `failed` or `partial`. This is a terminal state.

state	MEF 57.2 name	Description
held	HELD	The Product Order Item cannot be progressed due to Charge the Seller awaiting a response from the Buyer on a Charge. The Seller stops work on the Product Order Item until the Charge has completed. Upon acceptance by the Buyer of all Blocking charges, the Product Order Item returns to inProgress state If the Buyer rejects a Blocking charge, the Product Order Item moves to the failed state.
inProgress	IN_PROGRESS	The Product Order Item has been successfully validated and fulfillment has started. If the Seller's system links validation between Product Order Items in a Product Order, a Product Order Item in this state also indicates that the other Product Order Items passed validation.
pending	PENDING	The Product Order Item cannot be progressed due to the Seller assessing a Cancel Product Order or Modify Product Order Item Requested Delivery Date request. The Seller stops work on the Product Order Item until either the Cancel Product Order has been accepted and the Product Order state moves to pendingCancellation and the Product Order Item state moves to cancelled, the Cancel Product Order has been rejected and the Product Order Item State moves to inProgress, the Modify Product Order Item Requested Delivery Date has been accepted and the Product Order Item State moves to inProgress, or the Modify Product Order Item Requested Delivery Date moves to done.declined and the Product Order Item state moves to inProgress with original delivery dates.
rejected	REJECTED	A Product Order Item was submitted, and it has failed at least one validation checks the Seller performs during the acknowledged state. Other Product Order Items may continue to be processed.
rejected.unassessed	UNASSESSED	A Product Order was submitted and all validation checks the Seller performs during the acknowledged state have not been completed, but another Product Order Item in the Product Order has moved to the rejected state.
rejected.validated	VALIDATED	A Product Order was submitted, and it has passed all validation checks the Seller performs during the acknowledged state, but another Product Order Item in the Product Order has moved to the rejected state

Value	MEF 57.2
acknowledged	ACKNOWLEDGED

Value	MEF 57.2
cancelled	CANCELLED
completed	COMPLETED
failed	FAILED
held	HELD
inProgress	IN_PROGRESS
pending	PENDING
rejected	REJECTED
rejected.validated	REJECTED.VALIDATED
rejected.unassessed	REJECTED.UNASSESSED

7.3.12. Type CancelProductOrderStateChangeEvent

Description:

Inherits from:

- Event

Name	Type	M/O	Description	MEF 57.2
eventType	string	M	Indicates the type of the event.	
event	CancelProductOrderStateChangeEventPayload	M	A reference to the object that is source of the notification.	

7.3.13. Type CancelProductOrderStateChangeEventPayload

Description: The identifier of the Cancel Product Order being subject of this event.

Name	Type	M/O	Description	MEF 57.2
id	string	M	ID of the Cancel Product Order	
href	string	O	Hyperlink to access the Cancel Product Order	
state	MEFChargeableTaskStateType	M	The state reached	

7.3.14. Type ModifyProductOrderItemRequestedDeliveryDateStateChangeEvent

Description:

Inherits from:

- Event

Name	Type	M/O	Descri

Name	Type	M/O	Descri
eventType	string	M	Indicat the ty the eve
event	ModifyProductOrderItemRequestedDeliveryDateStateChangeEventPayload	M	A refe to the that source the notific

7.3.15. Type ModifyProductOrderItemRequestedDeliveryDateStateChangeEventPayload

Description: The identifier of the Modify Product Order Item Requested Delivery Date being subject of this event.

Name	Type	M/O	Description	MEF 57.2
id	string	M	ID of the Modify Product Order Item Requested Delivery Date	
href	string	O	Hyperlink to access the Modify Product Order Item Requested Delivery Date	
state	MEFChargeableTaskStateType	M	The state reached	

7.3.16. `enum` MEFChargeableTaskStateType

Description: The states as defined by TMF622 and extended to meet MEF requirements.

Name	MEF 57.2 Name	Description
inProgress.assessingCharge	ACCESSING_CHARGE	The Modify Product Order Item Requested Delivery Date request results in a Charge being initiated by the Seller. The Modify Product Order Item Requested Delivery Date remains in this state until the Charge is completed or withdrawn by the Seller. All charges within a Charge that was initiated due to a Modify Product Order Item Requested Delivery Date are considered Blocking charges. If any charge is not accepted by the Buyer, the Modify Product Order Item Requested Delivery Date moves from the `inProgress.assessingCharge` state to the `done.declined` state.

Name	MEF 57.2 Name	Description
acknowledged	ACKNOWLEDGED	A Modify Product Order Item Requested Delivery Date request has been received and has passed basic validation. The Modify Product Order Item Requested Delivery Date Identifier is assigned in the `acknowledged` state. Validation of Modify Product Order Item Requested Delivery Date attributes as applicable is completed in the `acknowledged` state.
done	ACCEPTED	A Modify Product Order Item Requested Delivery Date request has been received, passed all validations, if a Charge is associated all Charge Items have been accepted by the Buyer, and the Product Order Item Completion Date has been updated as requested.
done.declined	DECLINED	Blocking charges associated with a Modify Product Order Item Requested Delivery Date have been declined by the Buyer. No updates are made to the Product Order Item.
rejected	REJECTED	A Modify Product Order Item Requested Delivery Date request was submitted by the Buyer, and it has failed any validation checks the Seller performs during the `acknowledged` state. No updates are made to the referenced Product Order Item.

Value	MEF 57.2
acknowledged	ACKNOWLEDGED
done	DONE
done.declined	DONE.DECLINED
inProgress.assessingCharge	IN_PROGRESS.ASSESSING_CHARGE
rejected	REJECTED

7.3.17. Type ChargeCreateEvent

Description:

Inherits from:

- Event

Name	Type	M/O	Description	MEF 57.2

Name	Type	M/O	Description	MEF 57.2
eventType	string	M	Indicates the type of the event.	
event	ChargeEventPayload	M	A reference to the object that is source of the notification.	

7.3.18. Type ChargeTimeoutEvent

Description:

Inherits from:

- Event

Name	Type	M/O	Description	MEF 57.2
eventType	string	M	Indicates the type of the event.	
event	ChargeEventPayload	M	A reference to the object that is source of the notification.	

7.3.19. Type ChargeEventPayload

Description: The identifier of the Charge being subject of this event.

Name	Type	M/O	Description	MEF 57.2
id	string	M	ID of the Charge	Not represented in MEF 57.2
href	string	O	Hyperlink to access the Charge	Not represented in MEF 57.2

7.3.20. Type ChargeStateChangeEvent

Description:

Inherits from:

- Event

Name	Type	M/O	Description	MEF 57.2
eventType	string	M	Indicates the type of the event.	
event	ChargeStateChangeEventPayload	M	A reference to the object that is source of the notification.	

7.3.21. Type ChargeStateChangeEventPayload

Description: The identifier of the Charge being subject of this event.

Name	Type	M/O	Description	MEF 57.2
id	string	M	ID of the Charge	

Name	Type	M/O	Description	MEF 57.2
href	string	O	Hyperlink to access the Charge	
state	MEFProductOrderChargeStateType	M	The state reached	

7.3.22. `enum` MEFProductOrderChargeStateType

Description: Possible values for the state of the Charge

State	Description
`completed`	All Charge Items included in the Charge for a given Product Order Item have moved to either the `accepted` state or the `declined` state.
`awaitingResponse`	A Charge has been initiated by the Buyer. The charge includes one or more charges.
`timeout`	A response has not been received from the Buyer within the `responseDueDate`. This is treated as if the Buyer declined the Charge Items.
`withdrawnBySeller`	The Seller determines that the Charge is incorrect. They withdraw the Charge and initiate a new Charge with the required correction(s).

Value	MEF 57.2
awaitingResponse	AWAITING_RESPONSE
completed	COMPLETED
timeout	TIMEOUT
withdrawnBySeller	WITHDRAWN_BY_SELLER

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